

Wage theft, renegeing and liquidity constraints in informal labor markets *

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July 11, 2026

Abstract

Informal labor markets in low- and middle-income countries exhibit high unemployment and job turnover despite high labor demand. We conduct two field experiments to study how wage theft, worker renegeing, and liquidity constraints generate a mismatch between workers' labor supply and firms' labor demand. In the first experiment, 1,360 workers receive real job offers on either flat/daily-pay contracts, which do not withhold wages and therefore reduce wage theft concerns and workers' liquidity constraints, or back-loaded contracts, which withhold wages and impose these constraints. Labor supply is around three times larger for flat/daily-pay contracts. This preference is driven by workers' fears of wage theft, liquidity constraints, and contractual lock-in costs—the expected losses from forfeiting accrued wages when workers exit back-loaded contracts in response to shocks or excess work extraction by firms. These frictions are costly and 72% of workers who rejected job offers went on to earn less during the period than the offers would have paid. In the second experiment, we make real offers to hire workers on either flat/daily-pay or back-loaded contracts to 349 firms. Firms' labor demand almost doubles for back-loaded contracts and this is driven by liquidity constraints, and the costs of worker renegeing under flat/daily-pay contracts. When we match the firms and workers who accepted contracts in the first two experiments, workers are 69% more likely to complete back-loaded contracts than flat/daily-pay contracts, which have a 34% completion rate, thereby imposing significant costs on firms. Firms extract longer working hours under back-loaded contracts, consistent with workers' stated concerns.

Keywords: Labor markets, Informality, Labor Coercion, Absenteeism, Liquidity constraints.

JEL Classification: O12, O17, D86, D83, D22, D52, J22, J23, J41, J46, J47, J63, J64, J83

*I am incredibly thankful for support and advice from Michael Kremer, Christina Brown, Anne Karing and James Robinson. I thank Arun Chandrasekhar, Michael Dinerstein, Chinmaya Kumar, Erin Kelley, Nick Swanson, Frank Schilbach, and seminar audiences at the University of Chicago, APU, MIT, MWIEDC (UIUC), LSE, NEUDC, NBER SI (Development), CEPR Development Economics Symposium, G²LM | LIC/path2dev/BREAD Conference, and PEDL for helpful comments and feedback. Jay Prasad provided excellent research assistance. I am deeply indebted to the field team in Patna led by Gufran Ahmad and Neeraj Kumar. This work would not have been possible without helpful discussions with Gufran Ahmad, Raman Singh Chhina, Karan Jain, Sabahat Ali Khan, Neeraj Kumar, Randhir Kumar, Avinash Kumar, Rakesh Kumar, Madan Paswan, Jay Prasad and Manoj Yadav. The project was funded by the Development Economics Research Fund at the University of Chicago, the Weiss Fund, and the Pearson Institute. The IRB approval for this project was provided by the University of Chicago. The experiment was registered at AEA RCT registry (AEARCTR-0013100).

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1 Introduction

The informal sector is a large component of the economy in low- and middle-income countries. In India, it accounts for roughly one-half of the GDP and employs approximately 84% of the labor force (Murthy, 2019). Informal labor markets are characterized by small, liquidity-constrained firms with high worker-search costs and by poor workers who rely on day-to-day earnings, face substantial job-search costs, and experience significant unemployment (Farazi, 2014; Abebe et al., 2021b; Breza et al., 2021; Caria et al., 2024; Breza and Kaur, 2025). Written contracts are uncommon, enforcement is weak, and both sides face limited-commitment concerns: workers fear wage theft and being asked to work excess hours, while firms are concerned about worker renegeing and absenteeism (Breza et al., 2018; Krishnaswamy, 2019; Boudreau et al., 2024).

These institutional features can have important implications for the contractual preferences of firms and workers. When workers can exit at low cost and firms face high costs of finding workers, firms may have incentives to withhold wages to reduce renegeing by workers (Shapiro and Stiglitz, 1984; Becker and Stigler, 1974), and ease their liquidity constraints. Conversely, if low-income workers believe that firms may not pay wages, they may be wary of deferred wages, as withholding increases the potential losses from non-payment and exacerbates their existing liquidity constraints. Together, non-payment risk, worker renegeing, and liquidity constraints can shape labor supply and demand for contractual payment structures, often creating a wedge between firms' and workers' preferred contracts. Agreement between firms and workers over the contractual payment structure is a prerequisite for a match in the labor market, and mismatch can have significant consequences for efficiency, welfare and firm growth.

This paper studies how concerns about wage theft by firms, renegeing by workers, and liquidity constraints jointly shape labor supply and labor demand in informal labor markets in LMICs. Using two field experiments with construction-sector firms and low-income workers, this paper addresses three questions. First, we study how concerns about wage theft (non-payment) and liquidity constraints affect workers' labor supply and their preferences for different contractual payment structures. We also examine whether these workers' preferences are driven by concerns around contractual lock-in costs — the expected losses from forfeiting accrued wages when shocks lead workers to exit back-loaded contracts before completion. Second, we examine how firms' labor demand responds to worker renegeing, and liquidity constraints. Building on this, the third question asks whether firms and workers in fact renege on contracts when institutional enforcement is absent and, importantly, how renegeing varies across different types of contracts. For workers, we treat failure to complete the contract as renegeing, while for firms we treat non-payment of agreed wages and requiring workers to work more than eight hours as renegeing. Answering these three questions helps shed light on the simultaneous existence of unemployment, unmet labor demand, and high rates of worker renegeing in informal labor markets of LMICs (Banerjee and Sequeira (2023), Breza and Kaur (2025)).

We proceed as follows. First, we use descriptive data from surveys with firms and workers to establish key facts about the incidence of worker renegeing, non-payment by firms, and liquidity

constraints, and the costs they impose on low-income workers and firms. We then incorporate these facts into a simple model which analyzes the preferences and actions of firms and workers along a single margin: the payment structure of the contract, which captures how much workers are paid and when. It generates predictions about contractual preferences and on-the-job actions as the severity of liquidity constraints and the risk of reneging by the other party vary.

We test the model using two experiments in the construction labor market in Patna, India, a sector that employs approximately 57.7 million unskilled workers (Bajjal and Awasthi, 2023). To isolate the effects of different frictions, we design three contracts of either 3 or 7 days in length, developed through surveys with firms and workers, that represent a subset of agreements in this market. In *daily pay* contracts, firms pay due wages at the end of each day. In *steep back-loaded* contracts, firms pay a portion of wages on the first day and the remainder on the last day.¹ In *smooth back-loaded* contracts, firms pay a fraction of wages daily, with the withheld portion paid at the end of the contract.

The model highlights four forces that shape preferences over these contracts, and the experiment helps in empirically isolating them. First, worker reneging: firms prefer back-loaded contracts because withheld wages deter workers from quitting (Becker and Stigler, 1974; Lazear, 1981; Thomas and Worrall, 1988). Both steep and smooth contracts reduce reneging relative to daily pay.² Second, wage theft: firms may fail to pay wages with positive probability, and the expected loss from non-payment increases with the intensity of back-loading—the total wages that remain unpaid at the end of a working day. This loss is greatest under steep contracts, intermediate under smooth, and zero under daily pay. Back-loading also enables firms to extract longer working hours at the same wage, since workers have lower bargaining power when wages are withheld.³ Third, liquidity constraints: workers require some wages daily for consumption, reinforcing their preference for daily pay, while firms must borrow to pay wages in excess of their liquid assets, reinforcing their preference for back-loading.⁴ The smooth contract is designed so that the daily payment is at least as large as the daily consumption of 99% of workers in our sample, isolating wage theft concerns from liquidity constraints. Fourth, transaction costs: paying wages requires the firm representative’s physical presence at the worksite, and these costs are higher under daily pay, which requires more frequent payments. Additionally, back-loaded contracts impose contractual lock-in costs: because workers anticipate shocks, outside opportunities, or excess work extraction that may lead them to exit before completion, they face losses from forfeiting accrued wages. These lock-in costs are distinct from wage theft risk and liquidity constraints, and further

¹Throughout, we use *daily pay* and *flat* contracts interchangeably.

²By holding back wages, firms can reduce the probability that workers renege. From the worker’s point of view, back-loading allows firms to extract work and then renege on the last day, resulting in losses larger than in the case where the firms would renege on a daily pay contract.

³In our setting, it is common knowledge among workers that total working hours for a commonly known daily wage is 8 hours. As has been shown in other settings (for instance Boudreau et al. (2024)), firms ask workers to work overtime without pay.

⁴Precautionary saving motives could push workers to prefer back-loaded contracts especially when preferences are time inconsistent (Casaburi and Macchiavello, 2019). However, these forces are less relevant for workers who consume a large portion of their daily income (Carroll et al., 2021).

suppress labor supply for back-loaded contracts. The three contracts are cross-randomized with other treatments, as described below.

In the first experiment, 1,360 workers seeking employment at labor stands or spot markets receive job offers to work at construction sites on randomly chosen contracts at the market wage. The job itself is identical across all offers and only the contract terms vary. Workers are randomly assigned one of three contract types (daily pay, smooth, or steep), with contract lengths of either 3 or 7 days also randomly assigned. To formally test whether labor supply is driven by concerns about wage theft, we cross-randomize these contracts with a zero-cost insurance treatment. The insurance treatment arm guarantees the workers that their due wages will be paid by the field team if the firm reneges.⁵

In the second experiment, we offer to hire workers for 349 construction firms on the three payment structures for either 3 or 7 days. As the steep contract does not fully relax firms' liquidity constraints, we offer a *credit* treatment, which provides credit to firms to pay workers. The credit contract also shuts down the transaction costs of paying workers as we offer to pay the worker on firm's behalf.⁶ Additionally, we offer a *guarantor* treatment that provides compensation if the worker reneges on a daily pay contract. All contracts are offered to each firm, and one is chosen randomly at the end and the firm's choice is implemented. The worker and firm experiments give an incentive-compatible measure of worker and firm preferences. At the end of these two experiments, we obtain a set of firms and workers which have agreed to specific contracts. We then match firms and workers within steep and daily-pay contracts (henceforth, the *matching stage*).⁷ For each matched pair, we collect data on contract completion by the worker, firm non-payment (wage theft), hours worked and productivity. These matched outcomes allow us to study whether and how often firms and workers renege on contracts under different payment structures. To separate the effect of selection into contracts from the effect of contracts themselves we use a quasi-experimental sample for whom contract assignment in the matching stage was random.

The experiments have the following findings. First, workers' labor supply, measured by the acceptance rate, is much lower for uninsured steep contracts (28%) than for uninsured daily pay contracts (82%). Uninsured smooth contracts, which relax liquidity constraints but do not alleviate wage theft concerns, have an acceptance rate of 40%. Second, workers' acceptance of jobs increases when contracts are insured against wage theft. The insurance induced increase in acceptance is larger for back-loaded contracts (steep and smooth) than for daily pay contracts, and the resulting difference-in-differences estimate is statistically significant at 5%.

To isolate liquidity constraints, we compare smooth insured contracts, which relax liquidity constraints and reduce wage theft concerns, to steep insured contracts, which reduce wage theft concerns but leave liquidity constraints largely unchanged. Smooth insured contracts are accepted at 54.4%, compared to 42.2% for steep insured contracts (a 12.2 p.p. difference). The gap is large

⁵Firms were not informed of this treatment to avoid moral hazard.

⁶These costs arise because representatives of firms have to be present at the work site at the end of the day to pay workers which can be costly.

⁷Firms (workers) on smooth contracts were upgraded to steep (daily pay) contracts to increase power.

and statistically significant for 7-day contracts but close to zero for 3-day contracts, suggesting liquidity constraints matter less over shorter horizons. Within 7-day contracts, the effect is larger for workers who report higher daily consumption ex ante. Comparing smooth uninsured and smooth insured contracts shows that removing wage-theft risk raises take-up by about 15 p.p. Consistent with this interpretation, workers who experienced non-payment in the month prior to the survey respond more strongly to insurance on smooth contracts suggesting that the insurance arm operates through wage-theft concerns.

The smooth insured contract relaxes both liquidity constraints and wage-theft concerns, so any remaining acceptance gap between this contract and the insured daily-pay contract is a residual which we define as contractual lock-in costs — the losses workers expect from forfeiting accrued wages if they exit before completion. This gap is large and insured (uninsured) daily-pay contracts are accepted at 33.3 (27) p.p. higher rates than smooth insured contracts. Workers highlight three circumstances that make early exit likely: (i) health shocks or social obligations ([Goraya et al., 2025a](#)); (ii) pressure from firms to work longer hours when wages are withheld; and (iii) positive economic shocks, such as a better work opportunity.

Third, in the subsample of workers who rejected the job offer, 72% earned less over the relevant contract period than they would have earned had they accepted the offer.⁸ On average, earnings of workers who rejected jobs are 27.27% lower than the earnings they would have received under the offered contract. This suggests that were these frictions absent, workers may have earned more income.

Fourth, on the firm side, demand for steep contracts, which reduce liquidity constraints and the risk of worker renegeing, is 81%, compared with 43% for daily pay contracts. The demand for smooth contracts, which impose liquidity constraints but reduce the risk of renegeing, is 67%. Of the 38 p.p. gap between daily pay and steep contracts, we interpret roughly 24 p.p. (about 63%)—the daily pay–smooth difference—as primarily reflecting higher losses from worker renegeing under daily pay, while the remaining 14 p.p. (about 37%)—the smooth–steep difference—largely reflects liquidity constraints and transaction costs. Consistent with this decomposition, guarantor contracts, which provide compensation against worker renegeing, have a take-up rate of 69%, statistically similar to that of smooth contracts. Credit contracts, which fully alleviate firms’ liquidity constraints and transaction costs, have a take-up rate of 89%. Using reasons provided by the firms for accepting or rejecting contracts, we argue that alleviating liquidity constraints alone would increase the demand to 82.5%.

Lastly, in the matching stage, workers are 69% more likely to complete back-loaded contracts than daily-pay contracts, for which the completion rate is 34%. Relative to daily-pay contracts, workers on back-loaded contracts work longer hours and are more likely to report being required to work beyond the initially agreed-upon hours. We do not find differences in non-payment across contracts and overall rates of non-payment are around 3%. Because workers and firms select into contracts, it is important to determine whether these results are driven by the contract

⁸829 workers rejected the offers. We were able to collect data on the earnings for a sub-sample of workers.

structure or by selection. When we restrict analysis to workers for whom contract assignment was quasi-random, the estimated effects remain quantitatively similar. Taken together, the results from matching indicate that back-loaded contracts effectively increase worker retention but also allow firms to extract more work hours, which reinforces the contractual lock-in costs workers anticipate under back-loaded contracts.

The frictions highlighted here can generate a mismatch between firms' and workers' preferred payment structures, with consequences for market efficiency. In our setting, firms may be willing to hire workers but not on daily pay contracts, while workers may be willing to work but reluctant to accept back-loaded contracts. This disagreement can delay the start of production and discourage hiring when commitment concerns prevent firms from employing unfamiliar but capable workers (Chandrasekhar et al., 2020). Consistent with this mechanism, in a survey of a sub-sample of firms, 54% reported waiting for a trusted worker to start work, and 57% reported facing delays in hiring new workers.⁹ These patterns align with evidence that firms often cannot replace absent workers and incur losses due to absenteeism (Krishnaswamy, 2019). These findings underscore the importance of labor institutions that can enforce contracts, since reputational forces alone may be insufficient to discipline fraudulent firms.¹⁰

Contributions to the literature This paper speaks to work on frictions in labor markets in LMICs, including search frictions, information frictions, and contractual rigidities (Heath, 2018; Carranza et al., 2022; Fernando et al., 2023; Caria et al., 2024; Barker et al., 2025). It is connected to work on incomplete and relational contracts (Aghion and Holden, 2011; Macchiavello, 2022) and payment timing (Casaburi and Macchiavello, 2019). It also relates to evidence on partial enforcement and compliance with labor regulation in developing countries (Boudreau, 2024; Bhorat et al., 2019; Kanbur and Ronconi, 2018) and to work on liquidity constraints among the poor and small firms (Casaburi and Willis, 2018; Brune and Kerwin, 2019; Brune et al., 2021; Sharma et al., 2024).

First, we contribute to the empirical literature on contract enforcement in settings with weak formal institutions. Existing work has largely focused either on historical coercive enforcement (Naidu and Yuchtman, 2013) or on relational and reputational mechanisms in product markets and global value chains (Fernando and Singh, 2025; Macchiavello and Morjaria, 2015). In contrast, we provide experimental evidence from a large urban informal labor market in which contracts are not externally enforceable. We quantify how the risk of wage theft and firms' limited commitment affect workers' willingness to accept contracts, and show that concerns about nonpayment are an important determinant of labor supply. To our knowledge, this study provides one of the first experimental evaluations of how wage-theft risk affects workers' labor supply in informal labor markets. In related work, Earle and Sabirianova Peter (2009), Danzer (2019) and Musa and Sonnenstuhl (2025) study the consequences of wage arrears. Our paper complements the empiri-

⁹In a similar setting, Sharma et al. (2024) report that 40% of firms face difficulties in hiring workers.

¹⁰While reputation effects could drive defaulting firms out of the market or induce them to improve their behavior (Fernando and Singh, 2025), the arrival of new workers and incomplete diffusion of information can allow such firms to continue finding workers.

cal literature on hiring frictions, which emphasizes firms' information about worker productivity and moral hazard on the worker side ([Abebe et al., 2021a](#); [Caria and Falco, 2024](#)), by showing that firms' concerns about worker renegeing can also reduce labor demand—a margin that has not been systematically quantified in LMIC settings.

Second, we add to the literature on how liquidity constraints and payment timing affect labor market behavior of low-income workers. [Casaburi and Macchiavello \(2019\)](#) show that Kenyan dairy farmers willingly accept lower prices to receive infrequent lump-sum payments as a commitment device to save for lumpy expenditures, and [Brune et al. \(2021\)](#) likewise find that workers value deferred wages as a savings tool. These results obtain in environments where the counterparty is sufficiently credible. We provide a contrasting case. In our setting, employers are spot-market contractors operating without formal enforcement, making wage theft a salient risk. Workers face binding day-to-day liquidity constraints for consumption and commuting. Back-loaded contracts also impose lock-in costs: workers who accept them forfeit the flexibility to respond to health shocks, outside job offers, or exploitative working conditions. Together, these frictions overwhelm any potential commitment benefit.

Third, we document new facts on separation, limited commitment, and working conditions in informal labor markets. We show that separation rates are very high under daily pay contracts. This suggests that search and adjustment costs for firms are large ([Barker et al., 2025](#)). Back-loaded contracts reduce turnover relative to daily pay and provide novel evidence, in an LMIC setting, of firm-side costs from limited commitment. We further show that back-loaded contracts are associated with longer work hours and more frequent reports that firms require workers to work beyond the initially agreed schedule, linking the findings to research on worker rights, coercion, and working conditions ([Boudreau, 2024](#); [Boudreau et al., 2024](#); [Naidu and Yuchtman, 2013](#); [Fernando and Singh, 2025](#)).

Lastly, the paper sheds light on why high unemployment, unmet labor demand, and high absenteeism can coexist in low-income labor markets ([Breza and Kaur, 2025](#); [Adhvaryu et al., 2024](#); [Banerjee and Sequeira, 2023](#); [Abebe et al., 2021b](#)). Using a 10-day panel of all workers arriving at the labor stand, we document unemployment rates as high as 50%. On the firm side, our survey indicates that firms often face difficulties and delays in filling positions. The firm-side preference for back-loaded contracts and workers' strong preference for daily-pay contracts generate contractual mismatch: firms may be willing to hire but not on daily pay, while workers may be willing to work but not on back-loaded terms. Even when matches form on daily pay, only 34% are completed because workers face health shocks, family emergencies, and better outside options, and the daily-pay structure allows them to exit without forfeiting wages. This further raises search and replacement costs for firms and sends workers back into the pool of job seekers, helping account for the coexistence of unemployment, unmet labor demand, and high turnover in these markets.

The shocks that drive renegeing in our setting are pervasive among low-income workers. [Goraya et al. \(2025a\)](#) show that casual workers in rural manufacturing forgo around 8% of potential earnings for contracts that allow spontaneous absences, reflecting the importance of unpredictable dis-

ruptions such as illness and social obligations. In our setting, the anticipation of such disruptions leads workers to reject back-loaded contracts because these contracts impose contractual lock-in costs.

The rest of the paper proceeds as follows. Section 2 describes the data we collect and the setting in detail. Section 3 lays out the theoretical framework which generates predictions that will be tested in the experiment. Section 4 lays out the details and results from the worker-side experiment. Section 5 shows the results from the firm side experiment. Section 6 includes results from the matching stage. Section 7 concludes.

2 Data and Setting

This section introduces the main data sources and provides a brief overview of the experimental design. We then use descriptive statistics to highlight the frictions faced by firms and workers in the labor market in our setting.

The setting is the market for construction in Patna, the capital of the state of Bihar, India. Bihar’s per capita gross state domestic product (GSDP) is approximately US \$800 ([Government of Bihar, Finance Department, 2025](#)), significantly lower than the national average of around US \$2,700 ([World Bank, 2025](#)), and comparable to income levels in several Sub-Saharan African countries. At the same time, construction is a large and economically important sector. Nationally, it employs around 57.7 million workers and contributes about 8% to India’s GDP. Focusing on construction workers and firms in Bihar therefore allows us to study contracting and enforcement in a poor, informal labor market that is embedded in a major sector of the Indian economy.

2.1 Data collection

Our data collection proceeds in four steps. First, we survey a subsample of construction firms to identify the most common contract types in the local labor market. Second, we conduct an experiment in which workers at labor stands are offered jobs under different payment structures corresponding to these contract types. Third, in a separate experiment, we offer firms the opportunity to hire a worker under different contract types that mirror those offered to workers, and elicit whether they are willing to hire under each set of terms. Fourth, we match workers and firms that accepted the same contract type, and observe them for the duration of the contract, tracking contractual compliance, worker attendance, and productivity. A summary of the experiment design is shown in Figure 1. We describe each aspect of the design in the forthcoming sections.

2.2 Worker side

Workers in construction often seek work at labor stands in urban areas of Patna. They arrive at the stand early in the morning, and firms hire them from the stand and take them to the work site. Firms and workers are aware of the locations of the labor stands in the city, and this allows a thick

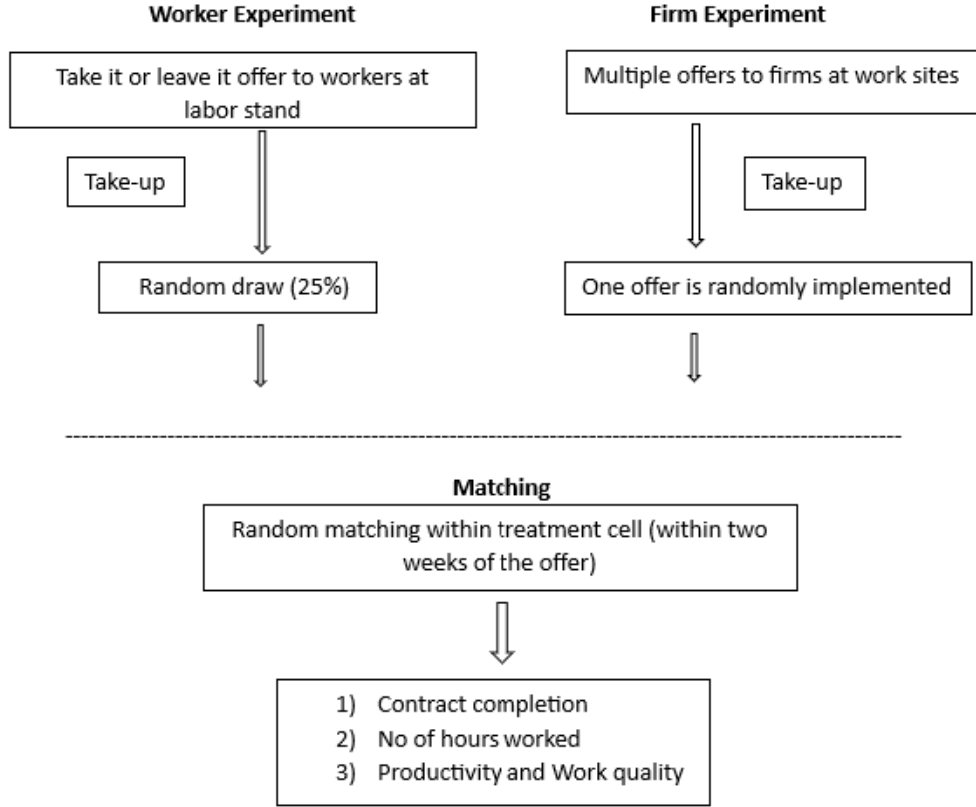


Figure 1: Experiment Design

spot market to flourish at these stands. Recent work by Cefala et al. (2025) likewise underscores the central role of these stands in the job search of low-income workers.

We recruited workers for our experiment from the labor stands. There are around 40 labor stands in the city of Patna. On average, about 180 workers search for work at each stand each day.¹¹ A back-of-the-envelope calculation suggests that approximately 41,000 unique workers seek work at the labor stands of Patna each month.¹²

Out of the 40 labor stands in Patna, we selected 26 for the experiment. Stand selection and survey order were determined by worker footfall, proximity to firms, and logistical considerations. Smaller stands with limited daily attendance were excluded to allocate resources efficiently toward stands with higher worker footfall. The number of job offers at each stand was scaled in proportion to the number of workers present. To capture variation at the city level, two survey teams operated simultaneously across different geographic regions of Patna. Table 1 shows the summary statistics of workers who were part of our experiment.¹³

¹¹These figures are based on data collected at each of the 40 labor stands over a period of two days.

¹²Based on data collected over ten days at four labor stands. Around 10% of workers present on a given day are new, and the remaining workers come to the stand roughly once every three days. Across India, the number of workers seeking work at similar labor stands is expected to be in the millions. Most Indian urban centres and towns have similar labor stands. Patna is the 19th largest city in India by population. Assuming that cities larger than Patna have a similar footfall of workers per month, at least 760,000 workers search for work at these labor stands each month.

¹³Both men and women work in construction, though women make up a small fraction of the workforce. We found almost no women searching for work at labor stands, possibly because they use different search strategy or because

2.2.1 Key facts

High non-employment and search costs at the labor stand. Workers face significant search costs in our setting. On average, a worker takes 55 minutes to reach the stand. They spend nearly 8% of their daily wage on transportation to and from the stand. The frequency of the search relative to the number of days spent working is much higher than in formal jobs.

Despite spending these resources, workers often do not find work. Appendix Figure B-10 plots unemployment levels at four labor stands over a period of 7–9 days. This share exceeded 50% on almost all days. Because these workers incurred the cost of traveling to the stand, this is a revealed-preference measure of labor supply.¹⁴ These data were collected in July and August, which tend to be lean seasons in construction due to the monsoons. In our experimental sample, collected during the peak season, workers worked an average of only 2.5 out of 4 days before the survey.

Workers have concerns about non-payment (wage theft). Non-payment or wage arrears is a significant issue in the construction industry, as previously reported in Praveen (2024), Kumar and Fernandez (2015), and Wells (2023). Non-payment can manifest in various ways. One form occurs when firms refuse to pay workers for work performed. Another arises when firms are unable to pay due to liquidity constraints, and ask workers to return later for payment. The former is a more severe form of theft, in which the firm exploits the worker’s vulnerability.

In our experimental sample, 12% of workers had not been paid their due wages at least once in the month before the survey. The average amount due was 1,209 INR, which equals wages for 2.159 days and approximately 14.4% of workers’ monthly earnings.^{15,16} This suggests that non-payment by firms is an important concern for workers in this setting.

Workers face liquidity constraints for consumption. The daily wage in these markets at the time of our experiment was largely fixed between INR 500–550. In our experimental sample, the average daily consumption of workers was approximately 190 INR, around 35–40% of the going wage.¹⁷ Workers in similar settings have very limited liquid savings — Kaur et al. (2025) find that 64% of workers in their sample would not have 1,000 INR on hand. Under a steep back-loaded contract lasting more than three days, most workers would therefore need to borrow to maintain consumption.

their families search for work as a unit. Among the women in construction, migrants form a large majority.

¹⁴Breza et al. (2021) document severe labor rationing in similar Indian labor markets, showing that a substantial share of non-employment among casual workers is involuntary. Section 4.2.2 provides further discussion of how the frictions we study contribute to unemployment at the stand.

¹⁵Monthly earnings are calculated by extrapolating earnings over a four-day period.

¹⁶Note that this is an equilibrium figure that reflects the precautions workers take to avoid non-payment.

¹⁷This figure only includes household spending on day-to-day activities and does not include spending on education or health.

Workers renege on contracts. Reneging on contracts is common among workers. In our sample, 53% of workers know of others who have broken an oral contract without informing the firm. Workers believe this happens for several reasons: 33% think it occurs because firms ask the worker to do too much work; 18% attribute it to mistreatment by the firm; 60% think workers renege when they fear the firm will not pay them for the work they do; and 10% state that workers renege when they are either ill or simply do not feel like going to work.¹⁸ This suggests that workers may place high value on being able to exit contracts without forfeiting substantial back-loaded wages.

Table 1: Summary stats of workers

	Mean	SD	N
Age	32.34	9.53	1,378
Years of education	4.17	4.34	1,376
Backward caste	0.71	0.45	1,378
Scheduled caste/tribe	0.24	0.43	1,378
two sided travel cost (INR)	41.40	36.76	1,332
Time to market (hrs)	0.90	0.70	1,345
Not paid at least once (in last month)	0.12	0.32	1,318
Days not paid (in last month)	2.19	1.84	155
Non payment amount (in last month)	1,209.50	1,108.67	154
Non payment amount (in last 6 months)	2,597.10	2,687.59	230
Daily consumption cost	189.20	75.83	1,330
Know workers who have reneged on work	0.53	0.50	1,307
Fraction days worked	0.63	0.56	1,378
Total earnings (previous 4 days) (INR)	1,198.17	874.64	1,378

Notes: The table shows summary stats of the workers who were part of the experiment. All questions were asked as part of the same survey. Number of observations may vary due to incomplete responses by workers. Monetary amounts are in INR. *Not paid at least once* refers to any non-payment in the month prior to the survey. *Days not paid* and non-payment amounts are conditional on reporting any non-payment in the relevant recall period. *Fraction days worked* is number of days worked in the four days prior to the survey / 4. *Others have reneged on work* captures a yes or no response to the question: *Do you know workers who did not turn up for work on a day without informing the firm.*

2.3 Firm side

Table 2 presents summary statistics of firms in our sample. The firms in our setting are small, with an average of 10 workers operating across three construction sites, and nearly 14 years of industry experience. Most build houses or small office complexes. Only 12% are registered with the state; the remainder are informal, unregistered, and do not pay taxes. This firm size is representative of the predominantly small-scale manufacturing sector in India, where 98% of firms employ 9 or fewer workers (Goldar et al., 2025). Only 10% have a manager in addition to the firm owner. 75% of firms hire most of their workers from the labor stands, while the remainder are more likely to hire from villages. We recruited firms using snowball sampling, starting with firms encountered at the labor stands who then provided contacts for additional firms.

¹⁸Figures not shown in the table. Workers could provide multiple answers, so the percentages do not sum to 100.

2.3.1 Key facts

Firms face significant renegeing by workers. Seventy-nine percent of surveyed firms experienced at least one instance of a worker failing to show up without notice in the two months preceding the survey, with an average of 5.2 such instances. This pattern of renegeing influences firms' beliefs about worker reliability. Only 24% of firms believe that a worker would complete a three-day contract with daily pay, whereas 70% believe that a worker would complete a three-day contract with steep back-loaded pay.

Firms face high costs due to worker renegeing. When a worker reneges on a contract without notifying the firm, the firm must hire a new worker from the labor stand. This can be costly for three reasons. First, the firm spends money traveling to the labor stands. Second, work starts late, resulting in lost productive hours for other workers who still need to be paid. Third, in some cases, work may need to stop for a day or two until a new worker is hired. Overall, firms bear costs of around 100 INR on average if a worker does not turn up as agreed. This is around 18% of the daily wage paid to the worker and roughly equals the profit a firm makes from hiring one worker.¹⁹ The 99th percentile of this distribution is around 200 INR and we offer this amount to firms to remove concerns about this worker renegeing in the *guarantor* contract. This suggests that search costs due to worker separation are large.

Firms face liquidity constraints. About 30% of clients who hire construction firms fail to pay on time, creating difficulties for firms in paying their workers. In addition, firms face contractual fraud, with many reporting at least one instance of outright non-payment by clients in the six months prior to the survey. The average amount of non-payment was 46,000 INR—roughly 40% of the average monthly earnings of firms in the sample. The firms are typically paid by clients on a weekly or less frequent basis. This payment structure creates periods when firms are cash-strapped, as they must simultaneously cover worker wages and household consumption expenses. As a result, paying workers on time becomes difficult, and liquidity constraints are likely to play an important role in shaping firms' equilibrium behavior.

Firms face transaction costs in paying workers. In our setting, transaction costs arise from paying workers at the end of each workday, which requires a firm representative to be present at the site. These costs are more salient for firm owners who manage multiple sites and cannot always be present. In the experiment 79 offers refused by firms stated transaction costs as a reason. Among these 68 refusals (around 86%) were from firms which had above median sites (see Appendix Figure B-14). Thus, although these costs are not inherently tied to daily payments, contracts that require daily payment are more likely to expose firms to them.

The facts about firms and workers inform the model and experiment design. Specifically, the model aims to capture the constraints faced by low-income workers and informal firms in our

¹⁹Surveys with firms show that they make around 20% profit per day on every worker they hire.

Table 2: Summary stats of firms

	Mean	SD	N
Age	41.27	9.72	349
Years of education	7.17	4.30	345
Backward caste	0.72	0.45	349
Scheduled caste/tribe	0.19	0.39	349
Years working as a contractor	14.52	9.12	349
Worked as a mason before	0.85	0.36	349
Firm is registered	0.12	0.32	323
Hire most laborers from stand	0.75	0.43	253
No of workers in the firm	10.64	10.06	349
No of operational construction sites	3.23	2.34	349
Firm has a manager	0.10	0.33	331
Worker did not turn up (in last two months)	0.79	0.68	331
Instances of worker not turning up (in last two months)	5.18	4.58	222
Instances of Employer reneging on contract	1.10	1.73	331
Amount not paid by employer	46,366.65	35,349.65	141
Costs borne if worker does not turn up (INR)	93.64	56.01	330
Income over last month	126,455.59	244,079.92	349
Pct. Employers do not pay on time	29.45	27.83	165

Notes: The table shows summary stats of the firms that were part of the experiment. The survey was conducted with the respondent who was the head of the firm. Individual level data corresponds to the respondent. *Worker did not turn up* and related counts refer to the two months prior to the survey (conditional questions are asked only when applicable). *Amount not paid by employer* refers to reported unpaid/withheld wages over the stated recall period (conditional on any non-payment) by the firm's client. Number of observations may vary due to non-responses or because the question was asked to only a sub-sample of firms.

setting. The experiment is designed to highlight the importance of these constraints in the labor supply and demand decisions of workers and firms, respectively.

2.4 Contract design

The jobs offered to workers in the experiment are similar to the construction jobs they would otherwise take in the market. For the contracts, we focus on two lengths longer than one day, since back-loading is not meaningful for single-day jobs.

Contract length. To inform the choice of contract lengths, we use a ten-day panel of job outcomes for all workers seeking employment at four labor stands, collected between July and September 2023. Appendix Figure B-1a shows that approximately 70% of realized jobs at labor stands last a single day. The high share of one-day jobs reflects two factors: first, many tasks at labor stands—such as unloading materials, breaking rocks, or casting—are inherently short-term and completed within a single day and second, firms occasionally hire workers for one day to manage temporary increases in workload at sites where other workers are already present. Since back-loading is not meaningful for single-day jobs, the relevant comparison for our experiment is the distribution of multi-day contracts. Among these, 3-day contracts account for roughly 18% and contracts of 7 or more days account for about 25% (Appendix Figure B-1b; see Appendix B.1 for details on data collection). We therefore chose a three-day duration for the shorter contract, representing

the minimum period for which worker-side concerns such as wage theft and liquidity constraints would be relevant. For the longer contract, we needed a duration that differed from three days yet remained feasible within our budget and time constraints. Given that about 25% of all contracts (excluding one-day contracts) offered at the labor stands last seven or more days, we settled on seven days for the longer contract. The wage paid per day was set at the prevailing market rate, which was either 500 or 550 INR.²⁰

Payment structure. To design the payment structure for the main experiment, we surveyed a sub-sample of construction firms in May 2023. Firms were asked to specify the total payment and daily payment structure they usually use when hiring a worker for a seven-day job that pays the firm 5,000 INR. Ninety-five percent of firms stated a daily wage of either 400, 450, or 500 INR and multiplied it by seven to arrive at the total payment.²¹ The stated payment structures, plotted in Appendix Figures B-2a–B-2c, show that firms commonly back-load wages toward the end of the contract. While approximately 18% of firms pay a flat daily wage, the majority withhold some or all wages until the final days, and about 14% pay the entire wage on the last day (see Appendix B.1 for details). The steep and smooth contracts used in our experiment reflect payment schedules that firms actually employ.

A survey with workers indicated that the 99th percentile of daily consumption was 350 INR. We therefore set a daily wage of 350 INR for all days except the last, with the remainder paid on the final day, in the smooth back-loading treatment. This ensures that liquidity constraints are covered for almost all workers. For the steep back-loading contract, we set the wage at 350 INR on the first day, with the remainder paid on the final day.

While the panel data on jobs collected at the labor stands provide information about job duration, they do not reveal the payment structure of contracts that workers have worked on in the past. A potential concern is that workers are unfamiliar with back-loaded contracts and may therefore reject them at higher rates. To assess this, we surveyed 108 randomly chosen workers from four labor stands. We asked workers an open-ended question: *We want to understand the different payment structures used by contractors and firms in the industry. Please tell us all the payment arrangements you know are offered in the industry.* In response, 98% described daily pay contracts (106 out of 108), and 95% described at least one back-loaded payment structure in which wages are partially or fully withheld until the end of the contract (103 out of 108). The latter includes steep, smooth, and other types of back-loaded contracts (such as those that pay all wages at the end). Specifically, 90% described a payment structure resembling the smooth back-loaded contract used in our experiment (partial daily payments with a lump sum at the end), and 76% described one resembling the steep contract (most wages paid on the final day). That nearly all workers independently described both daily-pay and back-loaded structures in response to the same open-ended question

²⁰500 INR was used in first 15% of the job offers and 550 INR was used for the remaining 85%. The increase was driven by the change in the prevailing market wage. Both these wages exceeded the minimum wage for unskilled work in Patna, which was 392 INR during the experiment.

²¹The remaining 5% misunderstood the nature of the job and provided wages applicable to another position.

indicates that workers are familiar with both types of contracts.

When directly asked whether back-loaded contracts are offered at the labor stands, 92% confirmed that they are. More importantly, 62% reported having personally worked under a back-loaded contract. The gap between awareness (95%) and personal experience (62%) of back-loaded contracts may reflect an equilibrium response to the frictions we seek to test: workers learn about the risks of back-loaded contracts from colleagues at the stand and select away from them. We further asked workers to report how a firm hiring workers for a 7-day job would pay on each day of work: 74% reported a wage structure in which some or all wages are withheld until after the first day of work, and 26% reported a flat daily-pay structure. Together, these responses indicate that workers are familiar with back-loaded contracts from direct experience, and that the experimental results are not driven by unfamiliarity with these contract types.²²

3 Conceptual Framework

In this section, we outline a framework to help understand how non-payment (wage theft) concerns, reneging by workers and liquidity constraints shape the preferences and actions of firms and workers under contracts with different payment structures. The model is partial-equilibrium and takes the experimental menu of contracts and wages as given.

3.1 Worker side

Worker i , has a type given by θ_i . Type captures the productivity of the worker. A firm k can hire a worker i on a contract j . A contract j for L periods is represented by $\mathcal{C}_j^L \equiv \{\omega_{\tau j}\}_{\tau=1}^L$, where $\omega_{\tau j}$ are the wages paid on day τ . The set of contracts is given by $\mathcal{J} \equiv \{0, 1, \dots, J\}$, where $j = 0$ is the opportunity to seek work at the labor stand, and $j = J$ represents the opportunity of staying at home. The average daily wage for the jobs we offer in the experiment is given by $\tilde{\omega} = \frac{\sum_{\tau=1}^L \omega_{\tau j}}{L}$ and is fixed across all contracts $j \in \{1, \dots, J-1\}$. Wages in the markets we study do not adjust in response to frictions. Wages are already above the competitive level, and wage floors further limit adjustment (Breza et al., 2025; Varun, 2025).

Workers agree to work for H hours on each day.²³ Work beyond H hours is costly for workers, but the firm can ask the worker to do so. e_{ijt} equals the hours in excess of H for which a worker works. However, they cannot increase hours beyond a limit, $\bar{H}$.²⁴ The cost of effort is given by $\lambda_e e_{ijt}^2$, where λ_e is the quadratic cost parameter.

A worker has liquid assets given by A_{it} . As workers in our setting are extremely poor, they require liquidity for consumption purposes. When they run out of liquid assets (A_{it} falls below

²²Furthermore, if unfamiliarity rather than wage-theft concerns were driving rejection of back-loaded contracts, we would not expect insurance against non-payment to differentially increase take-up of back-loaded contracts relative to daily-pay contracts. Yet this is precisely what we find (Section 4, Figure 5b).

²³ H takes the value of 8 hours per day in our setting.

²⁴Above this threshold, the worker would not find it worthwhile to turn up to work the next day which would be costly for firms.

0) they have to borrow and face a borrowing cost given by λ_B .²⁵ Each worker starts with initial assets A_{i0} which have a distribution $\mathcal{A}$ with support $[0, \mathbf{A}]$. Workers do not want to pay the cost of borrowing, and hence they would prefer contracts that do not push their liquid assets below 0. We normalize consumption choices so that workers always consume c_i per day. The utility that a worker i gets on day t from working on a contract j is given by:

$$U_{ijt} = c_i - \lambda_B \mathbb{1}_{A_{it} < 0} - \lambda_e e_{ijt}^2 \quad (3.1)$$

Workers believe that the firm they work with will pay their dues at the end of a day with probability q . We model q as an ex-ante likelihood of wage theft which is not strategically chosen by the firms for each contract. This is done for tractability. The worker's expected asset evolves as:

$$\mathbb{E}[A_{i,t+1}] = A_{it} - c_i - m_{ij} + q\omega_{jt}, \quad (3.2)$$

In addition to consumption, workers incur commuting costs m_{ij} whenever they leave home to work. We assume that commuting costs are the same across all contracts that involve working and $m_{iJ} = 0$.

We assume workers do not update q upward over the course of a contract, though our predictions hold under bounded updating. If a firm fails to pay the agreed amount on any day, the worker sets the continuation value to zero and does not return. Ex ante, q is common across firms.

We make the following assumptions about workers:

- ASSUMPTION. 1. *Workers require a minimum amount of daily consumption $\underline{c}_i$, with $\underline{c}_i < \tilde{\omega}$, and they always consume at this level (so $c_i = \underline{c}_i$).*
2. *The probability of finding work at the labor stand, $p(\theta_i)$, is (weakly) increasing in type θ_i .*
3. *If the worker is not paid the agreed amount at the end of any day $\tau < L$, they break the contract on that day and do not work for that firm again.*

Consumption and continuation value The worker's intertemporal objective under contract j is

$$\mathbb{E} \left[\sum_{t=1}^L \rho^{t-1} U_{ijt} + \rho^L \Phi A_{iL} \right],$$

where $\rho \in (0, 1)$ is the discount factor and $\Phi > 0$ measures the value of terminal assets. We assume $\Phi > 1/\rho^L$, so that the marginal value of saving exceeds the marginal value of current consumption. Workers therefore optimally consume $\underline{c}_i$ each day, and we treat $c_i = \underline{c}_i$ as fixed.

Outside option On any day t , the worker can seek work at the labor stand on a one-day contract or stay at home (Figure 2). Worker i finds work at the stand with probability $p(\theta_i)$ at expected

²⁵ Assets cannot fall below $-\underline{A}$. Since L is small, workers borrow at most once during the contract.

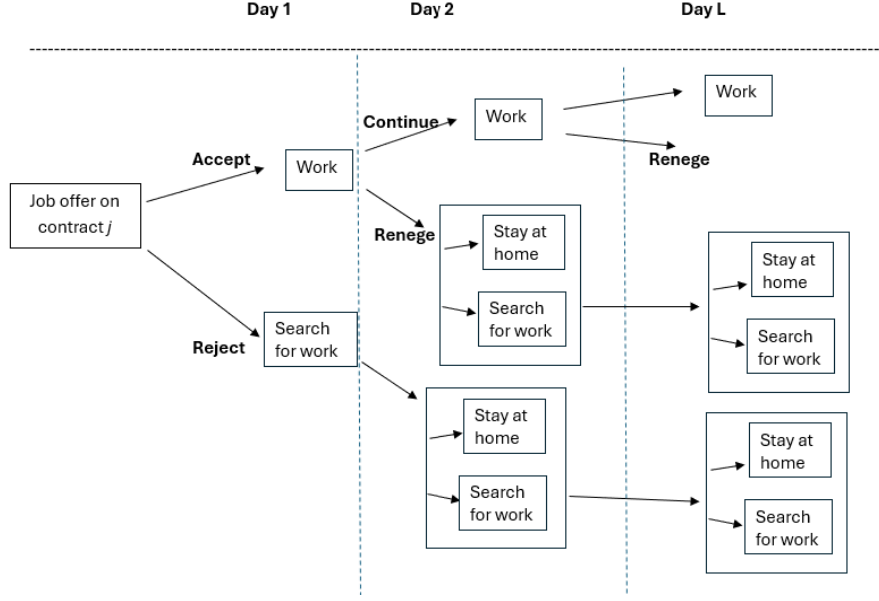


Figure 2: Worker choice

Notes: The figure shows the choices which the worker faces on each day.

wage $\omega^s > \tilde{\omega}$. The expected value of searching is:

$$U_{i0t} = \underline{c}_i - \lambda_B \mathbb{1}_{A_{it} < 0} - p(\theta_i) \lambda_e e_{i0t}^2 \quad (3.3)$$

where e_{i0t} is the effort required for the modal job at the stand, and assets evolve as:

$$\mathbb{E}[A_{it+1}] = A_{it} - \underline{c}_i - m_{i0} + p(\theta_i) q_0 \omega^s \quad (3.4)$$

where q_0 is the probability of being paid. The utility of staying at home is:

$$U_{iJt} = \underline{c}_i - \lambda_B \mathbb{1}_{A_{it} < 0} + \mu_J \quad (3.5)$$

where μ_J captures the value of staying at home.

Reneging on contracts Workers can renege after each day. Let $d_{it} \in \{0, 1\}$ denote worker i 's continuation decision at the end of day t , where $d_{it} = 1$ means continue and $d_{it} = 0$ means renege. At the end of each day, the worker is either paid or not. If not paid, the worker breaks the contract immediately and does not return (Assumption 3). If paid, the worker compares the remaining value of the contract to the outside option and reneges if the outside option dominates. A worker who reneges at t chooses between searching at the stand or staying at home in each remaining period $t + 1, \dots, L$. Under a back-loaded contract, a worker who reneges forfeits accrued but unpaid wages. We refer to the expected loss in continuation value from this forfeiture — net of wage-theft risk and borrowing costs — as *contractual lock-in costs*.

Value of accepting a contract Given the primitives above, the worker's value of accepting contract j at $t = 0$, denoted $V_{ij}^0(A_{i0}; e_j)$, aggregates the expected stream of per-period utilities U_{ijt} , discounted at rate ρ , accounting for: (i) the probability q of being paid on each day, (ii) the optimal continuation decisions $\{d_{it}\}$ that determine when the worker reneges, and (iii) the value of the outside option upon reneging. Similarly, the value of rejecting, $V_{i0}^0(A_{i0})$, is the expected payoff from searching at the labor stand on day 0 followed by optimal choices over search and staying home in each subsequent period. The formal value functions are defined in Appendix A.3.

3.2 Firm Side

Firm k hires workers on contracts $j \in \{1, \dots, J - 1\}$ without observing worker type θ_i . Firms face search costs λ_s when hiring a new worker, reflecting the cost of traveling to the labor stand and the time lost in the process. Firms believe that workers on contract j renege with probability r_j . Initial firm liquidity is drawn from $\mathcal{B}$. The firm's only choice variable is $e_j \in [0, \bar{H} - H]$, the number of excess hours beyond H that a worker is required to work under contract j . Firms do not fire workers.

States and output. Let B_{kt} denote cash-on-hand at the *beginning* of day t and let A_{kt}^f denote outstanding accounts receivable from the employer (completed work not yet paid) at the beginning of day t . For a worker hired under contract j , per-day output (booked as receivables) is

$$y_t = \nu \hat{\theta} (H + e_{kj}), \quad (3.6)$$

where $\hat{\theta}$ is the mean worker type and $e_{kj} \in [0, \bar{H} - H]$ is the excess-hours worked.

Flow profit. We denote the one-period flow profit by f_{kjt} and write

$$f_{kjt} = y_t - \omega_{jt} - \kappa_{kjt} \lambda_{pf} \mathbb{1}\{\omega_{jt} > 0\}, \quad (3.7)$$

where $\lambda_{pf} \geq 0$ is the per-payment administrative cost and κ_{kjt} is a firm- and day-specific shock capturing whether making the payment on day t actually entails a transaction cost. For firms where the owner is already on site to oversee operations, κ_{kjt} is often zero and daily payments are not very costly, whereas for owners who are not always present or who manage multiple sites, κ_{kjt} is more likely to be one. Contracts that require daily payments therefore expose these firms to transaction costs more frequently. Firms receive payments from their employers stochastically over the contract period, which we model through an arrival process

State transitions. Firm cash-on-hand B_{kt} evolves as a function of wage payments ω_{jt} , borrowing costs λ_f on overdrafts, and stochastic settlement of accounts receivable from the employer. The formal law of motion is in Appendix A.3.

Firm value and outside option. The firm's value of hiring under contract j , denoted $\Pi_j^0(B_{k0}, A_{k0}^f; e_j^k, r_j)$, aggregates the expected stream of flow profits f_{kjt} , discounted at rate ρ , accounting for the probability r_j that the worker reneges on each day and the search cost λ_s incurred upon separation. The firm's outside option Π_{0k}^0 is a firm-specific value that depends on the firm's resources and hiring networks. The formal value functions are defined in Appendix A.3.

Non-payment by firms. For tractability, we do not model defaulting firms. Instead, we restrict attention to honest firms that always pay the scheduled wage. Appendix Section A.1 sketches an extension with strategic non-payment.

3.3 Equilibrium

Firms choose excess hours e_j^k and workers choose continuation decisions d_{it}^j that determine renegeing. Higher e_j^k raises workers' effort costs and increases renegeing, creating a trade-off for firms.

Worker's decision problem. A worker makes a choice to accept a contract j over their outside option. This acceptance decision is evaluated contract-by-contract. Given a wage schedule ω for contract j and an average excess-hours requirement e_j^* , worker i 's problem when offered contract j at $t = 0$ is

$$W_i^j(\omega) \equiv \max_{a_i^j \in \{0,1\}} \left\{ a_i^j V_{ij}^0(A_{i0}; e_j^*) + (1 - a_i^j) V_{i0}^0(A_{i0}) \right\}, \quad (3.8)$$

$$\text{with } e_j^* = \int e_j^k d\mathcal{B}(k). \quad (3.9)$$

$V_{ij}^0(A_{i0}; e_j^*)$ is the expected discounted value of accepting contract j , accounting for the probability q of being paid each day, borrowing costs when assets fall below zero, effort costs from excess hours e_j^* , and the worker's optimal renegeing decisions. $V_{i0}^0(A_{i0})$ is the value of rejecting and searching at the labor stand. The value of acceptance is lower when the contract back-loads wages, for three reasons: (i) expected losses from non-payment are larger because more wages remain unpaid at any point, (ii) the worker is more likely to exhaust liquid assets and incur borrowing costs, and (iii) renegeing entails forfeiting a larger stock of accrued but unpaid wages.

Firm's decision problem. Firm k hires under contract j if the expected value of hiring exceeds the outside option:

$$T_k^j(\omega) \equiv \max_{b_k^j \in \{0,1\}, e_j^k \in [0, \bar{H} - H]} b_k^j \cdot \Pi_j^0(B_{k0}, A_{k0}^f; e_j^k, r_j) + (1 - b_k^j) \cdot \Pi_{0k}^0, \quad (3.10)$$

Here $b_k^j = 1$ denotes that firm k chooses to hire a worker under contract j , and $b_k^j = 0$ denotes that it takes its outside option. The function $\Pi_j^0(B_{k0}, A_{k0}^f; e_j^k, r_j)$ is the value of hiring under contract j starting from (B_{k0}, A_{k0}^f) , as defined by the firm's Bellman equation, and Π_{0k}^0 is the value

of the firm's outside option. The term $d_i^j(A_i, \theta_i; e_j^*)$ summarizes worker i 's equilibrium reneging behavior under contract j and aggregate excess-hours e_j^* ; it can be interpreted as the per-period probability that a worker who is active at the beginning of a day reneges before the next period. The expectation $\mathbb{E}_{A, \theta}[\cdot]$ is taken over the joint distribution of worker initial assets and types, $(A_{i0}, \theta_i) \sim (\mathcal{A}, \Theta)$.

Equilibrium An equilibrium in contract j is a vector $\{e_j^*, \{b_k^j\}_k, r_j, \{a_i^j\}_i\}$ such that: (i) each worker's acceptance a_i^j solves (3.8) given e_j^* ; (ii) each firm's hiring and hours (b_k^j, e_j^k) solve (3.10) given r_j ; and (iii) beliefs are consistent with behavior:

$$r_j = \mathbb{E}_{A, \theta} \left[d_i^j(A_i, \theta_i; e_j^*) \right], \quad (3.11)$$

$$e_j^* = \int e_j^k d\mathcal{B}(k). \quad (3.12)$$

3.4 Predictions from the model

Definition. 1. A daily pay contract denoted by $\mathcal{C}^D$ is one which pays the worker the same amount daily: $\{\omega\}_{\tau=1}^L$

2. A smooth back-loaded contract denoted by $\mathcal{C}^{Sm}$ is one which pays the worker the same amount for all days except the last day, and $\omega_i < \omega_L, \forall i < L$.

3. A steep back-loaded contract denoted by $\mathcal{C}^S$ is one which pays the worker only on the first and the last day, and $\omega_1 < \omega_L$.

The total payment made across the three contracts of the same length is the same. The probability that workers accept a contract j which pays an average daily wage of $\tilde{\omega}$ is given by $P(j)$, where

$$P(j) = \mathbb{E}_{\Theta, \mathcal{A}} \left[\mathbb{1} \{ V_{ij}^0(A_{i0}; e_j^*) \geq V_{i0}^0(A_{i0}) \} \right], \quad (3.13)$$

and the expectation is taken over the joint distribution of worker types and initial assets $(\Theta, \mathcal{A})$.

Worker side

Prediction. W.1: The probability of accepting a contract of length L with same average daily wage $\tilde{\omega}$, where $\tilde{\omega} > \omega_1^{\mathcal{C}^{Sm}} = \omega_1^{\mathcal{C}^S} > \underline{c}_i$, weakly declines in the steepness of back-loading, that is, $P(\mathcal{C}^D) \geq P(\mathcal{C}^{Sm}) \geq P(\mathcal{C}^S)$

A daily pay contract is *worker optimal* (among the three contracts we described) as it does not impose liquidity constraints and poses the least wage loss risk to the worker which imposes no lock-in costs. Among all contracts, the optimal contract for firms (assuming that workers face no constraints) would be one that pushes payment of all wages to the last day.

Prediction. W.2: If q , the worker's ex-ante belief that a firm is honest, increases, then probability of accepting a contract (keeping L and $\tilde{\omega}$ the same) increases, that is, $P(j_{q1}) \geq P(j_{q2})$, where $q_1 > q_2$.

Prediction. W.2.1: For a smooth contract, providing insurance, that is shifting q to 1, increases the take-up.

Prediction. W.3: If $q = 1$, that is workers do not have concerns about non-payment, workers are more likely to accept smooth contracts compared to steep contracts due to liquidity constraints.

Prediction. W.4: Workers are more likely to accept daily pay insured contracts than smooth insured contracts, due to contractual lock-in costs.

This prediction implies that workers are more likely to accept daily pay contracts which are insured against non-payment, compared to smooth contracts which are insured against non-payment. This difference captures the fact that exiting a daily pay contract does not impose forfeiture of accrued wages, whereas exiting a back-loaded contract does.

Prediction. WE.1: In equilibrium, workers renege at a higher rate in daily payment contracts (C^D) than steep contracts (C^S) with same length L and average daily wage $\tilde{\omega}$.

Firm side The probability that firms are willing to hire a worker on a contract j which pays an average daily wage of $\tilde{\omega}$ is given by $P_f(j)$, where

$$P_f(j) = \mathbb{E}_{\mathcal{B}} \left[\mathbb{1} \{ \Pi_{kj}^0(B_{k0}, A_{k0}^f) \geq \Pi_{0k}^0 \} \right], \quad (3.14)$$

and the expectation is taken over the distribution of firm initial states induced by $\mathcal{B}$.

Prediction. F.1: The probability of hiring a worker on a contract of length L with same average daily wage $\tilde{\omega}$ weakly increases in the steepness of backloading, that is, $P_f(C^D) \leq P_f(C^{Sm}) \leq P_f(C^S)$.

The steep back-loaded contract is *firm optimal* (among the three contracts described) as it does not impose liquidity constraints and has the least probability of worker reneging, thereby reducing separation costs for the firm.

Prediction. F.2: Alleviating liquidity constraints of firms increases their probability of hiring workers.

Prediction. F.3: Providing compensation to firms if worker reneges on a contract increases their probability of hiring a worker.

Prediction. F.4: Alleviating transaction costs of firms increases their probability of hiring workers.

Prediction. FE.1: In equilibrium, firms ask workers to work for longer hours on steep contracts compared to daily contracts with same length L and average daily wage $\tilde{\omega}$.

The contractual lock-in costs borne by workers under back-loaded contracts are thus the source of firms' retention and extraction benefits as the same withheld wages that make exit costly for workers give firms leverage to reduce reneging and increase hours. All proofs are in Appendix Section A.4. It is possible to estimate the welfare maximizing contract using this setup. We briefly discuss what such a contract might look like in Appendix A.2.

4 Worker experiment

This section studies how non-payment (wage theft) concerns, liquidity constraints, and contractual lock-in costs shape workers' willingness to accept contracts with different payment structures. The experiment described in the section was conducted at labor stands in Patna between April and August 2024.

4.1 Recruitment and design

Worker recruitment We surveyed workers at labor stands and made them job offers whose contract terms (payment schedule, contract length, and insurance) were randomized using a pre-generated list of unique IDs. Enumerators, who were randomly assigned these IDs, approached workers at the stand and delivered the corresponding job offer. Because each ID mapped one-to-one to a treatment cell, treatment assignment was random at the worker level. Enumerators were unaware of the treatment associated with each ID prior to the survey.

Workers were eligible if they were searching for work or if their existing employment was expected to end within two weeks.²⁶ Each eligible worker was offered a job at a construction site near the stand, with contract terms determined by the randomized assignment. To ensure consistency across workers and contract types, we did not disclose the exact site location at the time of the offer. Workers were instead told that the site would be close to the stand and that they would be escorted there on the first day of work.²⁷ All offers involved the same underlying task at the same prevailing daily wage, and only the contract terms varied by treatment. The prevailing wage at the stand for these jobs is relatively fixed.²⁸ The task involved assisting masons with bricklaying, the most common task for which firms hire workers in construction.

Each worker was offered two jobs, one with a three-day contract and one with a seven-day contract, with the order of offers randomized.²⁹ For each job offer that a worker accepted, implementation occurred with 25% probability based on a random draw. This implementation probability was disclosed before eliciting choices. Because any accepted job had a positive chance of being realized, the elicited choices are incentive compatible.³⁰ Realization was determined independently for each accepted offer, with a 25% probability for each. Workers were informed of the outcomes of the random draw at the end of the experiment, and those whose accepted offers were selected were told that the job would start within two weeks of the offer. The firm-side experiment

²⁶ Approximately 14% of the 1,413 workers surveyed were engaged in another job at the time of the survey. Among these, 2.3% (N=33) were employed in jobs expected to last more than two weeks and were therefore not surveyed further.

²⁷ This procedure mimics the behavior of firms hiring workers from the labor stand.

²⁸ We document the existence of wage floors for the jobs offered in our experiment in other work (see Varun (2025)), consistent with recent work by Breza et al. (2025). More broadly, Kaur (2019) provides evidence of downward nominal wage rigidity in Indian casual labor markets, consistent with the limited wage adjustment we observe.

²⁹ In a pre-registered initial cohort of the first 300 workers, we offered only a three-day job. All remaining workers were offered two jobs, one of each contract length.

³⁰ The randomization was introduced to increase the sample size of the experiment without having to increase the number of firms (which was a constraint) or the time period between the survey and job offer (which might make choices less realistic).

was conducted concurrently by a separate field team. In addition to the contract choice module, we administered a short survey.

To limit information spillovers across workers at the stand, enumerators were instructed to make job offers a short distance away from the stand. Job offers were made between 7:30 a.m. and 10:30 a.m. over two to four consecutive days at a labor stand, with the number of consecutive days depending on stand size. In some cases, particularly at larger stands, the field team returned after a one-week gap. To avoid making duplicate job offers to the same worker on different days, we uniquely identified workers using their names, age, and villages. Enumerators were also instructed to confirm with the worker whether they had previously been offered a job. Further details on the implementation can be found in Appendix E.1.

Experiment Design The experiment was designed to tease apart the effect of non-payment concerns, liquidity constraints and contractual lock-in costs on workers' labor supply.

Table 3: Payment schedule by contract type

<i>Panel A: 3-day contract</i>								
	Day 1	Day 2	Day 3	Total				
Daily pay	W	W	W	$3W$				
Smooth	350	350	$3W - 700$	$3W$				
Steep	350	0	$3W - 350$	$3W$				

<i>Panel B: 7-day contract</i>								
	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Total
Daily pay	W	W	W	W	W	W	W	$7W$
Smooth	350	350	350	350	350	350	$7W - 2,100$	$7W$
Steep	350	0	0	0	0	0	$7W - 350$	$7W$

Notes: W denotes the prevailing daily wage at the labor stand (either 500 or 550 INR). All contracts pay the same total wage within each panel.

The experiment varies three contract dimensions—payment structure, insurance, and contract length—using random assignment at the contract-offer level.

1. **Payment structure of Contract:** We offer workers contracts with three payment structures. Total compensation per day of work is identical across all three: every contract pays the prevailing daily wage W per day worked, so that a 7-day contract pays $7W$ and a 3-day contract pays $3W$ regardless of payment structure.^{31, 32}

³¹ W denotes the going wage at the labor stand. This was either 550 or 500 INR at the time of the experiment.

³²There might be concerns that time discounting leads workers to value back-loaded contracts less. To check this, we ran a small pilot in which we offered workers two back-loaded contracts with the same payment structure and same daily wage, except that we paid a small additional amount on the last day in one contract accounting for a daily discount factor of 0.96. The difference in take-up of the two contracts was close to zero, suggesting that discounting is not relevant given the short length of the contracts.

- Steep back-loading: 350 INR being paid on Day 1 and the remaining payment ($3 * W - 350$ INR for a 3-day contract, and $7 * W - 350$ for a 7-day contract) being made on the last day.
 - Smooth back-loading: 350 INR is paid on each day except the last day, and the remaining payment is made on the last day.
 - Daily Payment: Workers are paid INR W at the end of each day.
2. **Insurance vs No insurance:** A random sub-sample of workers was offered zero-cost insurance against non-payment. Under the insured contract, workers were told that the survey firm would pay any wages due to them if the employer failed to pay, and that this guarantee would be provided in writing and bear the signature of the survey firm. Workers were also given a card with the survey firm's name, address, and phone number.³³
 3. **Length of contract:** We offered contracts of two lengths—3 days and 7 days and each worker was offered one contract of each length, with the order of the two offers randomized.

The insurance treatment shifts some of the relevant trust from the employer to the survey firm. We assumed that providing a written insurance guarantee—something firms do not typically offer—would increase workers' trust in the insured contracts. Because some workers may still mistrust the survey firm, the difference in take-up between insured and uninsured contracts provides a lower bound on the role of wage-theft concerns in job acceptance. To gauge workers' trust in the survey firm, we asked them why they accepted or rejected each contract offer and used these responses to identify cases in which insured contracts were rejected primarily due to mistrust in the guarantee rather than other factors.

In total, job offers were made to 1,378 workers, of whom 18 refused. Our main worker sample therefore consists of 1,360 workers. The sample size for each treatment cell is shown in Table 4. Cell sizes vary across treatment arms by design. They were pre-registered and chosen to maximize statistical power based on take-up rates observed during piloting. Appendix Table C-2 reports balance in observable characteristics across the six main treatment cells (pooling 3-day and 7-day versions).³⁴ Out of 165 pairwise comparisons, 9 are significant at the 10% level and 2 at the 5% level, fewer than would be expected by chance.

Separating effects of non-payment, liquidity constraints and contractual lock-in: Our experimental design allows us to separate the effects of different frictions which influence the labor supply for contracts with different payment structures. Qualitative survey with workers suggests that they dislike contracts which withhold wages even if non-payment concerns and liquidity constraints are alleviated. We refer to this friction as *contractual lock-in costs*: the expected losses workers face from forfeiting accrued wages if they exit a back-loaded contract for any reason.

³³The exact text of the offer is shown in Appendix Section E.1

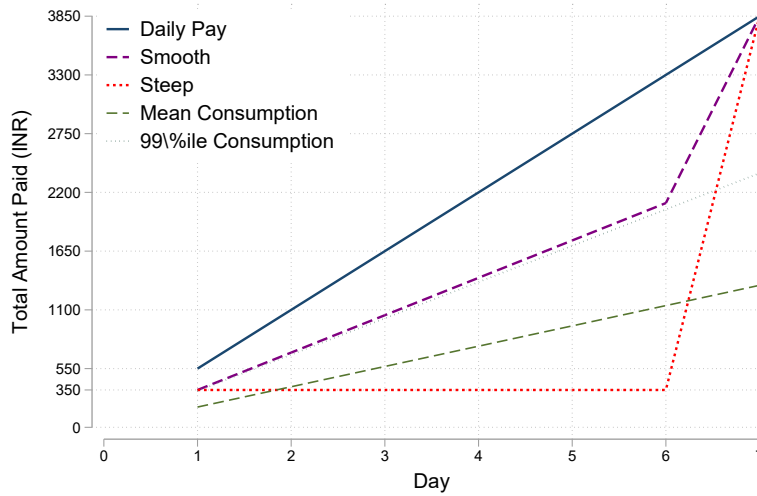
³⁴Since 78% of the sample was offered one 3-day and one 7-day contract, it is highly likely that we also achieve balance on observables by contract length.

Table 4: Treatment Cells and Sample size

Cell	3 Days	7 Days	Cell	3 Days	7 Days
No insurance, no backloading	9%	6%	Insurance, no backloading	6%	4%
No insurance, smooth backloading	12%	8%	Insurance, smooth backloading	12%	8%
No insurance, steep backloading	9%	6%	Insurance, steep backloading	12%	8%

Notes: The total sample size is 1360. Out of these the first 300 workers were offered just one job for 3-day contracts. The remaining workers were offered two jobs. Cell proportions were pre-registered and chosen to maximize power based on piloting.

Figure 3: Constraints faced by worker under different contracts

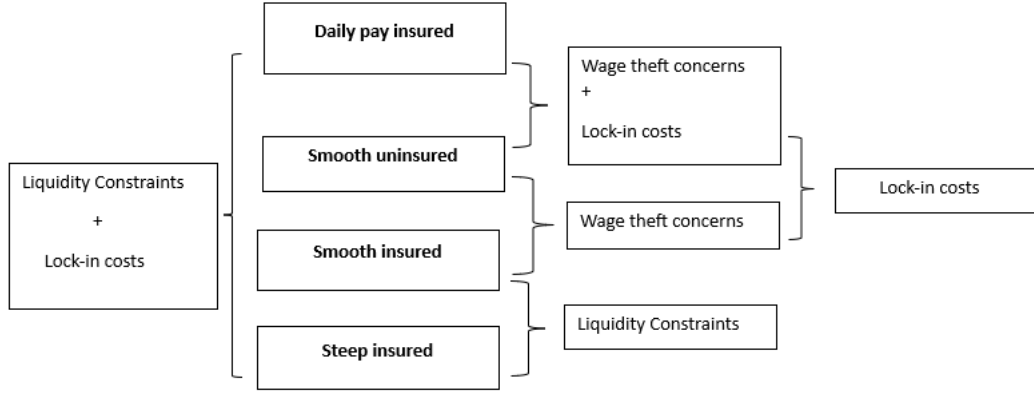


Notes: The figure shows the amounts paid to workers under different contracts. The figure illustrates the case where $W = 550$. Total mean consumption indicates the average total daily consumption up to a given day for workers in the sample. The 99th percentile consumption series shows the corresponding values for workers at the 99th percentile of the consumption distribution in the sample.

Figure 3 illustrates the constraints imposed by each payment structure. A steep back-loaded contract imposes all three constraints: workers are exposed to non-payment, receive little cash until the end of the contract, and lose a large share of their earnings if they renege early. A smooth back-loaded contract relaxes liquidity constraints by paying 350 INR each day, which exceeds the 99th percentile of reported daily consumption in our survey. Thus daily consumption needs are met for almost all workers. An insured contract contractually removes wage-theft risk, shifting any remaining concerns to mistrust in the survey firm. A daily-pay contract relaxes liquidity constraints and eliminates lock-in costs, although at the end of each day one day's wage remains at risk. An insured daily-pay contract relaxes all three constraints up to workers' trust in the survey firm. Figure 4 summarizes how we use these contracts to disentangle the three effects.

To separate the effect of each friction, we compare take-up across pairs of contracts that differ only along one dimension. First, holding contract length fixed, we compare smooth back-loaded contracts with and without insurance. As described above, liquidity constraints and lock-in costs are the same under these two contracts, while wage-theft risk is present only in the uninsured con-

Figure 4: Worker side identification



Notes: The figure shows the constraints faced by the worker under different contracts.

tract. The difference in take-up between smooth insured and smooth uninsured contracts therefore identifies the effect of wage-theft concerns on labor supply ([Prediction W.2.1](#)).

Second, we compare smooth insured and steep insured contracts. Both contracts are insured, so wage-theft concerns are removed. Interpreting the difference in take-up between these two contracts as the effect of liquidity constraints requires assuming that (i) the insurance contract fully relaxes perceived wage-theft risk and (ii) contractual lock-in costs do not vary systematically with the steepness of back-loading once daily liquidity is held constant. Under these identifying assumptions, the difference in take-up between smooth insured and steep insured contracts therefore captures the effect of liquidity constraints on contract acceptance ([Prediction W.3](#)).

Third, we compare insured smooth contracts with insured daily-pay contracts. With insurance, wage-theft risk is removed in both cases, and, as discussed above, liquidity constraints are not present under either contract. The key remaining difference is lock-in: under daily pay, workers can exit without forfeiting accumulated earnings, whereas under smooth back-loading they lose accrued pay if they leave before the last day. The difference in take-up between insured daily-pay and insured smooth contracts therefore captures contractual lock-in costs ([Prediction W.4](#)).

4.2 Results

We use the following specification for analysis.

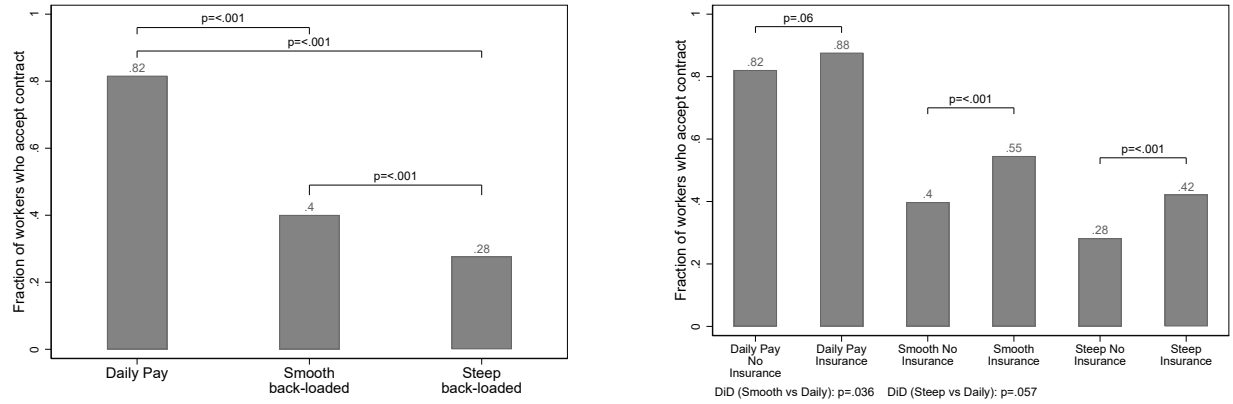
$$Y_{isl} = \alpha + \beta_1 \text{Daily Pay}_{isl} + \beta_2 \text{Smooth}_{isl} + \beta_3 \text{Insurance}_{isl} + \beta_4 \text{Daily Pay}_{isl} \times \text{Insurance}_{isl} + \beta_5 \text{Smooth}_{isl} \times \text{Insurance}_{isl} + X_i' \gamma + \delta_s + \delta_\ell + \epsilon_{isl}. \quad (4.1)$$

Here, Y_{isl} is an indicator equal to 1 if worker i at labor stand s accepts the offered contract of length ℓ . Daily Pay_{isl} and Smooth_{isl} are indicators for the contract's payment structure, with uninsured steep back-loaded contracts as the omitted category, and Insurance_{isl} is an indicator for an insured contract. X_i denotes a vector of controls (worker age, education, and half-hour-of-day

for the survey time), and δ_s and δ_ℓ are fixed effects for the labor stand and contract length respectively. Our analysis is robust to using alternate control variables including those selected using post-double-selection Lasso (Belloni et al., 2014).

Figure 5a summarizes how contract acceptance varies across uninsured payment structures and corresponds to Prediction W.1 from the model. Overall, 28% of workers accept an uninsured steep back-loaded contract and 40% accept an uninsured smooth contract. Appendix Figure B-4a shows the decomposition by contract length: steep uninsured contracts have acceptance rates of 36% and 17% for 3- and 7-day contracts, respectively, while uninsured smooth contracts have a 40% take-up rate for both durations. These rates are substantially lower than the 82% take-up of uninsured daily-pay contracts (80% for 3 days and 84% for 7 days). Pairwise differences across contract types are statistically significant (p-values reported in Figure 5a, based on specification 4.1).

Figure 5: Take-up of daily pay and the effect of insurance on back-loaded contracts



(a) Take-up of uninsured daily pay vs back-loaded contracts

(b) Effect of insurance on take-up of daily pay and back-loaded contracts

Notes: Panel A shows results from equation 4.1 for uninsured contracts. Panel B shows results from equation 4.1 comparing insured and uninsured contracts. In both panels, each bar is the sum of the control mean and the relevant regression coefficients. We use fixed effects for the labor stand and length of contract, and control for respondent age, education, and half hour of survey time. Each respondent (except the first 300) accounts for 2 observations (one for three-day contracts and one for seven-day contracts). Standard errors are clustered at the respondent level. p-values are shown for different pairs of treatments for a two-sided test.

Figure 5b examines how wage-theft concerns affect contract acceptance. Adding insurance increases take-up of daily-pay contracts by 6 percentage points and take-up of back-loaded contracts by around 14 p.p. The difference-in-differences in take-up between insured and uninsured daily-pay and back-loaded contracts is statistically significant (p-value = 0.036 for daily-smooth and 0.057 for daily-steep), suggesting that wage-theft concerns are more salient for back-loaded contracts, where the amount due to the worker at the end of any day is larger (see Figure 3). These patterns are consistent with Prediction W.2: insurance raises acceptance overall and has a larger effect for back-loaded than for daily-pay contracts. The insured contract thus acts as a third-party

guarantee of payment, mimicking the role that a labor court could play.³⁵

4.2.1 Decomposing the effect of different frictions

Until now, we have documented how workers' labor supply varies across payment structures and how insurance shifts these choices. We now decompose these differences in take-up to quantify the contribution of three frictions—concerns about non-payment, liquidity constraints, and contractual lock-in—to workers' contract choices. We use the comparisons described in Section 4.1 and summarized in Figure 4.

Wage Theft Table 5 reports the estimated effects. Comparing smooth insured and smooth uninsured contracts, insurance raises take-up by 15.1 percentage points, which corresponds to a 37.5 percent increase relative to the uninsured mean of 40%. This positive and statistically significant effect shows that removing wage-theft risk substantially increases workers' willingness to accept back-loaded contracts, in line with Prediction W.2.1. Similar magnitudes for 3- and 7-day contracts indicate that this effect is present at both horizons.

A natural alternative interpretation of the insurance effect is that it increases take-up because the offer comes from a research team that is perceived to be more reliable than typical employers, rather than because workers believe that wage-theft risk on the contract is lower. We refer to this as a “generic experimenter-trust” channel. If this were the main mechanism, we would expect similar insurance responses for workers with and without prior exposure to non-payment. In contrast, Appendix Figure B-5 shows that workers who experienced non-payment in the month prior to the survey display a substantially larger response to insurance on smooth contracts (by about 11.6 p.p.), even though the estimate is imprecise in our sample. This pattern is more naturally interpreted as insurance working through concerns about non-payment, rather than through a purely generic increase in trust in the experiment.

Liquidity constraints To isolate liquidity constraints, we compare take-up of smooth insured and steep insured contracts. For 3-day contracts, the difference in take-up is small and statistically insignificant, and for 7-day contracts, the smooth insured contract is accepted 22.1 percentage points more often than the steep insured contract, and this difference is highly significant, suggesting that liquidity constraints become important over longer contract lengths. Interpreting this difference as reflecting liquidity constraints rather than differential lock-in costs requires assuming that lock-in costs are similar across smooth and steep contracts. Appendix Figure B-6 provides evidence consistent with this interpretation by looking at heterogeneity in response by daily consumption (which should increase liquidity constraints). For 7-day contracts, the smooth-steep gap is much larger for workers with above-median daily consumption than for those below the median, whereas for 3-day contracts the gap is similar across the two groups. If the difference

³⁵When offering the insured contract, workers were informed that shirking at work would disqualify them from coverage. This condition was intended to limit moral hazard under third-party insurance.

between smooth and steep reflected differential lock-in costs, we would not expect it to be concentrated among workers with higher day-to-day consumption needs. The observed heterogeneity is instead exactly what we would expect if the 7-day smooth–steep difference is primarily driven by liquidity constraints. Finally, workers’ stated reasons for accepting smooth 7-day contracts are also consistent with this view: among those who accept these contracts, 67% report that they value receiving some wages daily because it helps them meet their day-to-day consumption needs, while the remaining workers cite either insurance or the assurance of continued work as their main reason.

Contractual Lock-in The striking result from Table 5, which confirms [Prediction W.4](#), is that contractual lock-in costs are quantitatively large: take-up of contracts that impose lock-in but relax non-payment and liquidity concerns is 33.3 percentage points lower than contracts without lock-in but with the same characteristics. For lock-in to be costly, workers must expect to exit the contracts before completion. Survey responses from workers highlight three reasons why exit may be likely: (i) health shocks or social obligations ([Goraya et al., 2025a](#)); (ii) pressure from firms to work longer hours when wages are withheld; and (iii) positive economic shocks, such as a better work opportunity. In the matching stage, we show that worker–firm separation before contract completion is indeed higher under daily-pay contracts, and that the reported reasons for renegeing align with these three channels.

To further unpack the residual daily-insured versus smooth-insured gap, we examine heterogeneity by worker type. Following our pre-registered measure, we classify workers in the top decile of earnings over the four days prior to the survey as high type. This measure is intended to capture stronger outside options and higher opportunity costs of remaining in a back-loaded contract. Appendix Figure B-9 shows that the gap between take-up of daily-insured and smooth-insured contracts is substantially larger for high-type workers than for low-type workers and the difference-in-differences is statistically significant (p -value = 0.023). This pattern is consistent with the view that an important component of contractual lock-in cost is the value of preserving outside options: workers with stronger outside opportunities are especially reluctant to accept back-loaded contracts that make exit costly by forcing them to forfeit accrued wages. Consistent with this, Appendix Figure B-7 shows that the estimated lock-in cost is 27 p.p. for workers with below-median travel time and 38 p.p. for workers with above-median travel time (DiD = 11 p.p., $p = 0.118$), suggesting that workers who commute from more distant, peri-urban areas and are therefore more exposed to delays and shocks, place especially high value on the option to exit without forfeiting accrued wages.

While the effect size is larger for 7 days as theory would predict, the gap is small. One potential explanation for this smaller than expected gap is that there are discontinuities in the distribution of lock-in costs. To further examine how contractual lock-in cost varies with contract length, we exploit within-worker comparisons for the subset of workers who were offered two jobs in random order (1,000 out of 1,360 workers). We restrict attention to workers who received either (i) two

smooth insured offers—one 3-day and one 7-day—or (ii) two daily-pay offers—one 3-day and one 7-day—where daily-pay offers may be insured or uninsured.³⁶ Appendix Figure B-8 plots predicted take-up from specification (4.1) for daily-pay and smooth insured contracts of 3 and 7 days for this subsample. In this subsample, the difference in take-up between daily-pay and smooth insured contracts is significantly larger for 7-day than for 3-day contracts: the difference-in-differences in predicted take-up is about 18 percentage points (p-value = 0.008). This indicates that contractual lock-in costs increase with contract length, as expected given that longer contracts accumulate more unpaid wages at risk of forfeiture.

Table 5: Wage Theft concerns, Liquidity constraints and Contractual Lock-in Costs

	Wage Theft Concerns			Liquidity constraints			Lock-in Costs		
<i>Smooth insured vs uninsured</i>	0.151*** (0.033)	0.167*** (0.043)	0.134*** (0.049)						
<i>Smooth vs Steep (Insured)</i>				0.122*** (0.033)	0.048 (0.043)	0.221*** (0.046)			
<i>Daily insured vs Smooth insured</i>							0.333*** (0.034)	0.321*** (0.045)	0.351*** (0.050)
Observations	973	549	420	964	540	424	716	399	316
Control group mean	0.40	0.39	0.41	0.42	0.50	0.32	0.55	0.55	0.54
Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Complete	3 days	7 days	Complete	3 days	7 days	Complete	3 days	7 days

Notes: The dependent variable is an indicator for whether the worker was willing to accept the contract offered. The first row reports coefficients from regressions of the dependent variable on the insurance indicator for the sample of smooth contracts, estimated separately by day and for the pooled sample. The second row reports coefficients from regressions of the dependent variable on the smooth-contract indicator for the sample of steep and smooth insured contracts, again estimated separately by day and for the pooled sample. The third row reports coefficients from regressions of the dependent variable on the daily-pay contract indicator for the sample of daily-pay and smooth insured contracts, estimated separately by day and for the pooled sample. All specifications include labor-stand fixed effects and controls for respondent age, education, and half-hour survey-time. Each respondent contributes two observations, except for the first 300 observations. Standard errors are clustered at the individual level. * p < 0.10, ** p < 0.05, *** p < 0.01.

A natural alternative explanation is that workers reject back-loaded contracts because they heavily discount future income. To assess this, we collected data on the potential earnings of workers who rejected the experimental contracts (see Section 4.2.1 for details). Using these data, we compute the average daily earnings of workers who rejected each type of contract and ask what constant daily discount factor would make a worker indifferent between accepting the experimental contract and earning this outside option each day. The implied daily discount factors are 0.21 for smooth 3-day contracts, 0.43 for smooth 7-day contracts, 0.51 for steep 3-day contracts, and 0.83 for steep 7-day contracts. For comparison, a monthly interest rate of 25%—a high rate charged by informal moneylenders—corresponds to a daily discount factor of about 0.9917.³⁷ The implied discount factors are therefore orders of magnitude lower than what even very high market interest rates would suggest, and are not a plausible explanation for workers' rejection of back-loaded contracts. This makes it more likely that the observed preferences are driven by

³⁶We include uninsured daily-pay contracts, in addition to insured daily-pay contracts, to increase power. As shown above, insurance has only a small effect on the take-up of daily-pay contracts.

³⁷Using survey data from moneylenders, Surendra (2025) reports annual interest rates as high as 120%.

frictions associated with back-loaded contracts, rather than by pure time discounting.³⁸

Workers forego significant earnings by rejecting jobs The results indicate that workers' labor supply is substantially lower for contracts that back-load wages. Workers' outside options consist of highly similar jobs offered by other firms at the labor stand, typically at the same or lower wages, and we have documented that they find jobs on only 66% of days before our experimental offer and spend significant amounts on search. Given these limited outside options, high rejection rates of back-loaded contracts suggest that these frictions weigh heavily in workers' decisions. At the same time, some participants may have received comparable offers during the experiment, either at the labor stand or elsewhere, in which case rejections of back-loaded contracts could partly reflect the availability of other jobs rather than the frictions we study. To assess this possibility, we collected seven days of post-offer earnings data. Follow-up was conducted primarily by phone. Because many workers did not have cell phones, we reached only a subsample ($N = 521$, approximately 38% of the full sample).³⁹

Figure 6 plots the CDF of earnings for workers who rejected the experimental offer. Approximately 72% of rejecters earned less per day, averaged over the seven days after the offer, than they would have earned had they accepted. On average, workers who rejected earned 27% less than the offered pay. These losses are concentrated among those who turned down back-loaded contracts: a 33% loss for back-loaded rejections versus a 3% loss for daily-pay rejections, and a 25.4% loss for workers who rejected insured back-loaded contracts. These comparisons do not account for income from self-employment or home production on days when workers were not searching for wage work. To address this, we restrict the comparison to days when the worker was actively searching in the labor market. By this measure, 52% of rejecters still earned less than the offered contract wage.

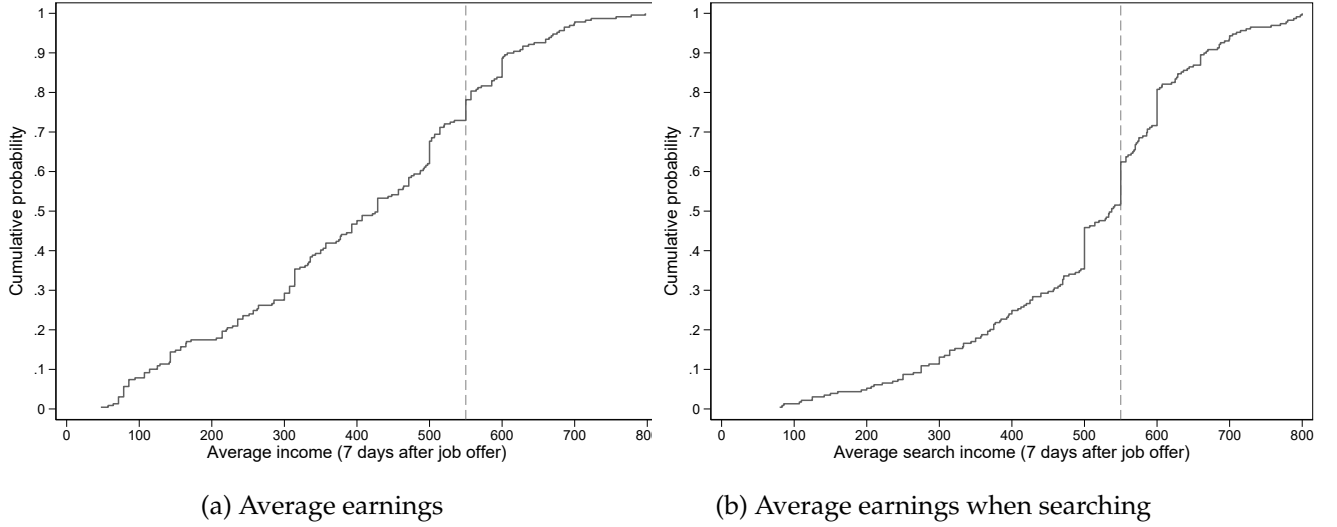
We view these figures as descriptive evidence that the frictions we study are quantitatively important, rather than as complete measures of welfare losses. In particular, a worker who rationally rejects a back-loaded contract to avoid its associated risks may earn less in the realized state while still having maximized ex ante expected utility. However, it is precisely the frictions we study that generate the risks making rejection rational. In a counterfactual where these frictions are absent, more workers would accept and earn more.

Robustness checks Our main results are robust to a variety of specification checks. They are unchanged when we include additional covariates (Appendix D.2), when we control for covariates selected via the post-double-selection Lasso procedure of Belloni et al. (2014) (Appendix D.1), and when we add surveyor fixed effects (Appendix D.2.2). Appendix Figure B-11 examines how worker characteristics correlate with acceptance of daily-pay and back-loaded contracts.

³⁸In Appendix Section B.2.1, we repeat this exercise allowing for hyperbolic discounting.

³⁹Workers with a cell phone reported earning 3% more income in the four days prior to the job offer, so the post-offer mean is likely an underestimate.

Figure 6: Earnings of workers who rejected job offers



Notes: The figure shows the distribution of earnings of workers who rejected the job offers. The figure restricts the sample to workers who rejected any one of the two job offers and who we were able to survey. In Panel A, we calculate average earnings as the mean of total earnings over the 7-day period following the day of the job offer. In Panel B, we calculate average income as the mean of total earnings on days when the worker is searching for a job during the same 7-day period. After winsorizing at 1% the total observations are 229. This is approximately 27.6% of the sample which rejected the job offer. The black line indicates the daily wage which was offered in the experiment. These data were collected by phone or through an in-person meeting with the worker at the labor stand. In some cases the earnings were recorded for 3 days instead of 7 days. In such cases, we calculate average earnings for the 3 day period.

A related concern is that some workers may not fully trust the field team to honor the insurance guarantee. To address this, we asked workers who rejected a job offer why they did so. If a worker reported that they did not trust the survey team, we classify them as having accepted the insurance treatment arm under full trust and recompute take-up using this adjusted measure. Re-estimating our main specifications with this adjusted outcome increases non-payment concerns by 4 p.p. and reduces liquidity constraints and lock-in costs by 1 p.p. (Appendix D.2.1). A second concern is that insurance could act as a generic signal of higher quality or different firm type, thereby affecting acceptance for reasons beyond wage-theft risk. Because insurance is randomly assigned conditional on contract type and introduced only after the contract is fully described, any such signalling effect should raise acceptance of all insured contracts similarly and is therefore differenced out in our cross-contract comparisons. Finally, in the matched sample (see Section 6) we do not observe systematic differences in completion, or hours between insured and uninsured workers, suggesting that the insurance does not materially distort on-the-job behavior (see Appendix D-12).

An additional concern is that the payment schedule may convey information about the employer. If workers interpret back-loaded offers as coming from less reliable firms, then differences in acceptance across payment structures could partly reflect belief updating about firm type rather than only the direct cash-flow implications of payment timing. Our design mitigates this concern by withholding firm identity and the exact site location at the offer stage, and by holding fixed the task, prevailing wage, and proximity of the site across treatments. In addition, the insurance treat-

ment directly reduces exposure to non-payment without changing the payment schedule, which helps distinguish wage-theft concerns from other dimensions of perceived job quality. Consistent with this interpretation, insurance increases take-up more for back-loaded contracts than for daily-pay contracts.

4.2.2 Interpreting rejection of contracts due to frictions as an explanation for unemployment

In our experiment, all job offers are made at the same daily wage; only the payment structure and insurance vary. Random assignment of contract types ensures that workers receiving different offers are, on average, comparable, so there is no systematic reason for their reservation wages for this job to differ across treatment arms. When a worker rejects a back-loaded job at this wage, it therefore reflects a higher effective reservation value for that type of contract rather than an absence of job opportunities: their non-employment at that wage is driven by the frictions embedded in the contract. At the same time, acceptance of the same job at the same wage rises when insurance is provided, indicating that more credible protection against non-payment can move workers into these jobs. Thus the frictions documented here contribute to observed unemployment even when jobs are available. Taken together, these patterns suggest that reducing frictions related to non-payment, liquidity, and contractual lock-in can potentially increase match formation at a given wage and may translate into higher employment in equilibrium.

Rejection rates in equilibrium Our job-offer protocol is designed to mimic how firms hire at the labor stand, but there are two important differences. First, firms typically make one job offer at a time, whereas in our sample the majority of workers receive two offers. Second, in the market firms are surrounded by many workers, so an offer rejected by one worker may be accepted by another after some bargaining (as documented in [Varun \(2025\)](#)), while in our experiment workers receive offers in isolation. To gauge whether these procedural differences affect the interpretation of our results, we compare rejection rates in our experiment with those from equilibrium firm-worker negotiations. In our experiment, 47% of the 2,420 job offers we made were rejected. In [Varun \(2025\)](#), real-time data show that 16% of jobs ($N = 1,588$) end with no agreement. Because firms can make offers to multiple workers before reaching agreement or no agreement, this implies more than one offer per job. Under a simple calibration in which firms make on average 1.5 offers per job, the implied offer-level rejection rate is about 45%, very close to the 47% rejection rate in our experiment. Despite differences in the job-offer procedure, the overall rejection rates are therefore similar, suggesting that our experimental findings are broadly reflective of real-world outcomes.

Can pairing mitigate these frictions? In the absence of formal enforcement, workers may adopt strategies to protect themselves against wage theft. We provide suggestive evidence for one such strategy. In a follow-up 2×2 experiment ($N=714$), we varied whether 3-day jobs were offered to a single worker or to a pair under daily-pay versus steep contracts. Pairing modestly increases

acceptance of steep contracts (by about 11% relative to when offered alone) and has little effect on acceptance of daily-pay contracts, but the difference-in-differences estimate has a p-value of 0.141 and is statistically indistinguishable from zero. Details and the full regression table are in Appendix B.2.2.

5 Firm Experiment

This section describes the results and design of the firm-side experiment to isolate how (i) worker-reneging costs and (ii) liquidity constraints and transaction costs shape firms' contractual choices.

5.1 Recruitment and Experiment design

Firm recruitment We recruited firms for the experiment using three methods. We initially approached firms that were hiring workers from the labor stand and asked them to connect us with other firms. After a few rounds of referrals, we obtained a sample of firms. We also sampled firms that were affiliated with a union of contractors in construction. Lastly, firms were recruited through direct visits to the work site. For the experiment, we focused on firms that planned to hire a worker within four weeks of the baseline survey. To confirm this, the field team contacted each firm and inquired whether they would need a worker in the next month. If the firm confirmed, we sought their consent to visit them at their office or work site for a survey and contract offer.⁴⁰ After setting the appointment, two enumerators visited the firm's site to conduct the experiment. The representative of the firm we talked to was the owner.

Design To accurately elicit the firm's willingness to hire under different contracts, we implemented a choice experiment in which firms were offered the opportunity to hire a worker under multiple contracts, and one contract was chosen at random at the end. In each contract, workers were paid a wage W for each day of work, where W represented the prevailing wage in the labor market. Since workers and firms often have differing beliefs about the going wage and negotiations over wages are common, we offered to hire workers on behalf of the firm at a daily wage of $W - 50$ INR (paid by the firm), while workers received W INR per day (with the survey firm covering the difference).

The experiment on the firm side consists of the following treatment arms (shown in Figure 7):

1. Payment structure:

- *Steep back-loaded*: the firm pays 300 INR on Day 1 and the remaining wages (i.e., $3 \times (W - 50) - 300$ for a 3-day contract or $7 \times (W - 50) - 300$ for a 7-day contract) on the last day.
- *Smooth back-loaded*: the firm pays 300 INR on each day except the last; the remaining wages are paid on the last day.

⁴⁰When calling the firm owners, we informed them about our study and obtained verbal consent. To build trust, we also provided the location of our field office, which they were welcome to visit if desired.

Figure 7: Treatment Cells in Firm Experiment

Contract	3-day	7-day	Friction addressed
Daily Pay	✓	✓	—
Guarantor Daily Pay	✓	✓	Worker reneging
Credit Daily Pay	✓	✓	Liquidity
Smooth Back-loaded	✓	✓	Worker reneging
Steep Back-loaded	✓	✓	Worker reneging + Liquidity

Notes: The figure shows the 10 treatment cells in the firm experiment (5 contracts $\times$ 2 lengths). All firms were made offers on each of the 10 contracts. After the experiment, one contract was chosen at random and the firm's choice on that contract was implemented. The total sample size is 349 firms.

- *Daily pay*: the firm pays INR $W - 50$ at the end of each day; the worker receives W per day (the experiment covers the 50 INR difference).
2. **Guarantor**: In this treatment, firms were offered 200 INR in compensation if a worker reneged on the contract at any point during the contract period. The amount was chosen such that it shut down the costs of worker reneging for 99% of firms. Additionally, the proposal offered a replacement for any absentee worker. These conditions were formalized in a written agreement and offered within the framework of the daily pay structure. Consequently, this treatment substantially reduces the costs associated with worker reneging while leaving liquidity constraints unchanged.
 3. **Credit**: In this treatment, firms were offered credit to pay worker wages. Additionally, they were given an option that enumerators could administer payments on the firm's behalf to reduce the firm's transaction costs. Firms were not given a lump sum at the beginning but instead, credit was provided on a day-to-day basis to align with the payment structure of the contract. Firms were expected to repay this credit within two weeks of the contract period ending. Regarding the payment option, if exercised, an enumerator visited the work site to pay the worker at the end of each day. These conditions were formalized in a written agreement. Consequently, the treatment alleviates both liquidity constraints and transaction costs.

Contract implementation The five contracts listed above were offered on two lengths, 7 days and 3 days. We offered the smooth contract to a subset of 202 firms to reduce respondent fatigue. All other contracts were offered to the full sample.⁴¹ We made offers to the firm on the 10 different

⁴¹In addition to the 10 cells the firm experiment elicited choices over 4 additional treatment cells, *Smooth + Credit* and *Daily Pay Credit + Guarantor*, each at both lengths. The four cells are excluded from the main analysis for expositional clarity.

contracts and asked, after each description, whether they would hire a worker under those terms. The order of contracts was randomized. After the firm responded to all contracts, one was randomly selected and the firm’s stated choice for that contract was implemented. The firms were informed about this procedure in advance. Since there was a positive probability of any stated choice being implemented, the mechanism was incentive compatible.

If the firm had chosen to hire a worker under the randomly selected contract, we informed the firm that we would recruit a worker on those terms. After the choice experiment, surveyors asked firms to provide a date when the worker would be needed; if they were unsure, firms were asked to call when demand arose. Once the experiment concluded, we also offered to recruit more than one worker for firms, if required. These additional workers were recruited under the contracts the firm had accepted in the choice experiment, including contracts that were not randomly chosen at the end of the elicitation procedure. We did this in order to match as many workers as possible. Because firms were not informed about the option to hire multiple workers before responding, their choices were not influenced by this possibility.

5.2 Results

Equation 5.1 shows our main specification. The model includes indicators for four contract types, with daily pay as the omitted category. Responses for the 7-day and 3-day contracts are pooled, and the specification includes fixed effects for contract length.

$$Y_{ik\ell} = \alpha + \beta_1 \text{Steep}_{ik\ell} + \beta_2 \text{Smooth}_{ik\ell} + \beta_3 \text{Guarantor}_{ik\ell} + \beta_4 \text{Credit}_{ik\ell} + X_i' \gamma + \delta_\ell + \epsilon_{ik\ell} \quad (5.1)$$

where $Y_{ik\ell}$ indicates whether firm i is willing to hire a worker on contract k of length ℓ . X_i denotes firm-level controls (owner’s education, number of sites, number of workers, and whether the owner is a former mason). δ_ℓ denotes contract-length fixed effects. All specifications also include fixed effects for contract presentation order.

We begin by testing [Prediction F.1](#), which posits that the probability of hiring increases with the steepness of back-loading. Figure 8a reports results from estimating specification 5.1. The acceptance rate for daily pay contracts is 43%, while acceptance is 81% for steep back-loaded contracts and 67% for smooth contracts. Next, we argue that the stark differences in preferences for these contracts are driven by a combination of concerns about worker reneging, liquidity constraints and transaction costs.

5.2.1 Effects of worker reneging on firm’s demand

The guarantor contract compensates the firm if the worker reneges while otherwise being similar to daily pay. Figure 8b compares take-up of daily-pay and guarantor contracts. The comparison is in line with [Prediction F.3](#), which implies that firms should be more willing to hire on daily-pay

contracts when the cost of worker renegeing is reduced. The effect is 26 p.p., which implies that reducing concerns about worker renegeing raises firm demand by about 60%.

The guarantor contract is also useful for interpreting the smooth contract. Like the smooth contract, it reduces the firm's exposure to worker renegeing relative to daily pay. When we compare the acceptance rates of the guarantor and smooth contracts, the two are statistically similar with a p-value of 0.494. Appendix Table B-4 shows that the two contracts also move similarly with contract length relative to daily pay. Taken together, these patterns suggest that the difference between smooth and daily-pay contracts is largely driven by reduced worker renegeing.

We can see the same pattern in firms' stated reasons for their choices. After each hiring offer, we asked firms why they accepted or rejected the contract, allowing multiple responses. These reasons are not revealed-preference measures, but they are still informative about the constraints that each contract imposes on the firm. Among firms that reject a 3-day (7-day) daily-pay contract but accept a smooth contract, 87 (91)% report liquidity constraints and 27 (22)% report concerns about worker renegeing as reasons for rejecting the daily-pay contract. At the same time, 95 (97)% of these firms report that reduced worker renegeing due to withheld payment is a reason for accepting the smooth contract. This suggests that the move from daily pay to smooth contracts is driven mainly by the perceived reduction in worker renegeing.

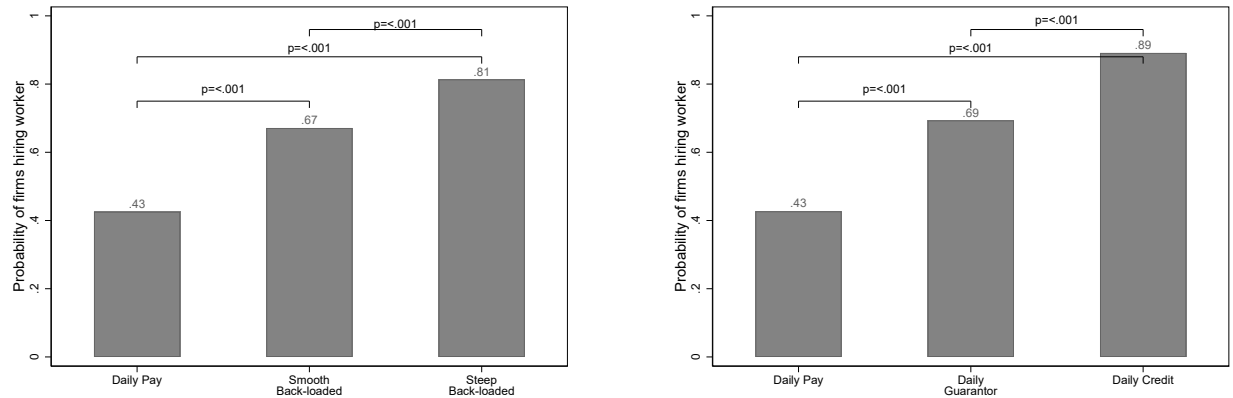
We next use the variation between the 3-day and 7-day versions of the daily-pay and smooth contracts as an additional check. Suppose the hiring gap between 3-day daily pay and smooth contracts reflects a combination of lower worker renegeing under smooth contracts and lower liquidity constraints. Moving from a 3-day to a 7-day contract tightens firms' liquidity constraints under both daily-pay and smooth contracts. Under daily pay, lengthening the contract does not increase worker lock-in through withheld pay, so the main effect should come through higher liquidity costs, which reduce acceptance. Under smooth contracts, by contrast, lengthening the contract both tightens liquidity constraints and increases the amount of wages withheld until the end, which should strengthen workers' incentives to stay and therefore reduce renegeing. Under this logic, acceptance should fall more for daily pay than for smooth, so the difference-in-differences estimate comparing 7-day to 3-day contracts across the two payment structures should be positive.

Column 1 of Appendix Table B-4 instead shows that this difference-in-differences estimate is 0.03. That small effect is hard to reconcile with a large liquidity component in the daily-smooth gap and is more consistent with the 24 p.p. gap being driven primarily by worker renegeing. Since the total gap between daily and steep contracts is 38 p.p., this implies that approximately $24/38 \approx 63\%$ of the gap between demand for daily and steep contracts can be attributed to higher costs from worker renegeing in daily-pay contracts.

5.2.2 Effects of liquidity constraints and transaction costs on firm's demand

The credit treatment arm alleviates liquidity constraints and transaction costs by saving firms from having to be present at the site at the end of the day to pay workers. Figure 8b shows that firms are 46 percentage points more likely to accept a daily-credit contract than a daily-pay contract, which

Figure 8: Probability of hiring: steepness of back-loading and role of credit/guarantor



(a) Probability of hiring and steepness of back-loading

(b) Probability of hiring on daily pay vs credit vs guarantor

Notes: Panel A and Panel B show results estimated from Eq. 5.1. Each bar is the sum of the control mean and the relevant regression coefficients. The total sample consists of 349 firms. We control for education of the owner, the number of sites the firm is operating, and the size of the firm. We use fixed effects for the order of the question, whether the respondent (the owner of the firm) is a former mason, and for contract length. Standard errors are clustered at the firm level. p -values are shown for different pairs for a two-sided test.

corresponds to an increase of about 106% relative to the acceptance rate of the daily-pay contract. Appendix Figure B-15 shows that the effect size is 40 p.p. for 3-day contracts (relative to 49% for daily pay) and 53 p.p. for 7-day contracts (relative to 36% for daily pay). Notably, credit contracts are also accepted at a higher rate than steep contracts (p -value < 0.001) which is surprising since the credit contract does not reduce worker renegeing. This may be due to two reasons. First, while the steep contract relaxes some credit constraints, it does not fully alleviate these constraints for all firms, particularly those which are paid by their clients at intervals longer than a week (40% of the sample). Second, a greater reduction in transaction costs drives acceptance of credit contracts for some firms.

We use firms' stated reasons to separate the combined effect of liquidity constraints and transaction costs. Among firms that accept the daily-credit contract but reject the daily-pay contract, 98% report the alleviation of liquidity constraints as a reason for their choice for both 3-day and 7-day contracts. Around 13% (11%) of these firms cite transaction costs as a reason for preferring the credit contract for 7-day (3-day) contracts.⁴² Thus, 85% (87%) of firms in this switching group report liquidity constraints but do not mention transaction costs. This pattern is consistent with liquidity constraints alone being sufficient to account for the switch for the majority of firms, with transaction costs playing a secondary role.

Using these shares as a reason-weighted decomposition, we interpret approximately 85% of the 53 percentage point increase in acceptance for 7-day contracts—about 45 percentage points—as being attributable to liquidity constraints. Likewise, we interpret roughly 87% of the 40 percent-

⁴²Shares may not sum to 100% because firms can report multiple reasons or none.

age point increase for 3-day contracts—about 35 percentage points—as arising from liquidity constraints. Together, these figures suggest that the bulk of the impact of the credit contract on firms’ demand operates through relaxed liquidity constraints, with transaction costs accounting for a smaller, but non-negligible, share of the overall effect. If liquidity constraints were removed then the firm’s demand for 3-day (7-day) contracts would be 84 (81)%. These patterns are consistent with [Prediction F.2](#).

To isolate the role of transaction costs, we restrict attention to weekly-paid firms and compare acceptance of three-day credit versus steep contracts. Over this horizon, liquidity differences between the two contracts are likely modest for weekly-paid firms, so the comparison largely nets out the liquidity channel. Steep contracts reduce worker renegeing through back-loaded wages but require a firm representative on site to make payments; credit contracts do not reduce renegeing but relieve firms of day-to-day payment logistics. Since steep contracts reduce renegeing but credit contracts do not, the renegeing advantage of steep works against credit acceptance. A gap in favor of credit despite this disadvantage would therefore primarily reflect the transaction-cost savings that credit provides, and the observed gap thus serves as a conservative lower bound on the transaction-cost effect. Appendix Table B-2 shows this gap is approximately 9.8 percentage points, consistent with [Prediction F.4](#). If this interpretation is correct, firms paid less frequently—who also benefit from greater liquidity relief under credit—should display a larger preference for credit over steep. The difference-in-differences estimate is around 5 percentage points in the expected direction, though not statistically significant.

What does the difference between smooth and steep contracts capture? Although in principle, the likelihood of worker renegeing could also differ between smooth and steep contracts, firms’ stated reasons point mainly to liquidity constraints and transaction costs as drivers of the shift from smooth to steep contracts. Among firms that reject a 3-day smooth contract but accept a steep contract, 79% report liquidity constraints and 9% report transaction costs as reasons for rejecting the smooth contract. This pattern suggests that, over a 3-day horizon, firms that switch from smooth to steep contracts are responding primarily to liquidity concerns rather than to worker renegeing. Among firms that reject a 7-day smooth contract but accept a steep contract, 87.5% report liquidity constraints and 21% report transaction costs as reasons for rejecting the smooth contract. Thus, even for 7-day contracts, liquidity constraints remain the dominant reported motive for switching to steep contracts, although transaction costs become more salient. Only one firm reports worker renegeing as the reason for rejecting the smooth contract, indicating that, in this comparison between smooth and steep contracts, renegeing concerns are not first-order in firms’ stated choices.

To further separate liquidity constraints from transaction costs within this comparison, we use the contrast between 3-day smooth and steep contracts, for which transaction costs should be relatively similar and, according to the stated reasons above, quantitatively small. Appendix Figure B-15 shows that acceptance of steep contracts is about 8 percentage points higher than ac-

ceptance of smooth contracts on a base of 71%. By contrast, the 21 percentage point difference between steep and smooth acceptance for 7-day contracts reflects a combination of liquidity constraints and transaction costs. Among the firms that reject the 7-day smooth contract but accept the steep contract, 6.25% report transaction costs as the *only* reason for rejecting smooth contracts. This implies that roughly 1.31 percentage points of the 21 percentage point gap can be attributed purely to transaction costs, providing a lower bound on their contribution, with the remaining difference reflecting liquidity constraints and any overlap between the two frictions.

Robustness Checks While the experiment was conducted with firms who planned to hire a worker, not all firms ended up hiring a worker on a contract. In Appendix Figure B-12, B-13 we show the results for firms which ended up hiring a worker. Naturally, the acceptance rates for all contracts for these firms is higher than the overall sample.⁴³ However, the signs and difference in magnitudes between different treatment arms remains broadly the same. Additionally, the results are robust to controlling for covariates selected by post-double selection Lasso method suggested in Belloni et al. (2014) (see Appendix D.3).

What is firms’ willingness to pay for back-loaded contracts? To quantify firms’ willingness to pay for back-loaded contracts, we fielded a structured multiple price list (MPL) module with 100 firms drawn from the main sample after the main experiment was completed. Unlike the main experiment—where we topped up wages by INR 50—the MPL was *not* incentivized and no top-up was offered. Protocol details are in Appendix B.3.4. Appendix Figure B-16 plots the share of firms willing to hire at each wage offered (INR 450–575 in INR 25 increments) for the Daily, Smooth, and Steep contracts. We convert these acceptance curves into an inverse-demand measure of willingness to pay for a given target acceptance rate p . Specifically, we use the wage that corresponds to p on the curve under each contract and compare it to the wage under Daily pay. Over the interior of the curve, the implied wage premia are modest: about INR 20 for Smooth and INR 30 for Steep—both under 6% of the daily wage used in the experiment. Our follow-up survey suggests these low premia reflect strong wage-setting norms that discourage contract-contingent pay increases (lest the “market wage” be disturbed; similar to Breza et al. (2025)), along with concerns about paying different workers different wages for the same task (Breza et al., 2018).

How do non-payment concerns, reneging, and liquidity constraints affect efficiency? A partial-equilibrium back-of-the-envelope exercise, reported in Appendix B.3.5, combines the experimentally estimated acceptance rates from the worker and firm experiments and computes implied matching rates as the minimum acceptance rate across the two sides. Among the three baseline contracts, the smooth insured contract generates the highest feasible number of matches. Once the credit and guarantor arms are included, the daily-pay credit contract yields the highest matching

⁴³Note that some firms rejected all contracts as the wage at which we offered to hire workers was not acceptable to them.

rate. These comparisons suggest that alleviating the frictions we study — particularly liquidity constraints and non-payment risk — could meaningfully increase match formation.

A natural question is whether these frictions bind in practice or whether firms can find workers outside the labor stand on favorable terms. To assess this, we surveyed a sub-sample of 120 firms.⁴⁴ Fifty-five percent reported having faced difficulty hiring workers for new projects, and 53% reported that they prefer to wait for a trusted worker — someone who has worked with them before and is more likely to accept their contractual terms — to become available before starting a new project. These patterns suggest that the frictions we study contribute to hiring delays in practice, and may also discourage firms from hiring unfamiliar workers, leading to underexploration of the labor market (Chandrasekhar et al., 2020).

Role of Relational Contracting and Reputation. Relational contracting and referrals can mitigate commitment problems when firms and workers interact repeatedly in thick networks (Heath, 2018; Macchiavello, 2022; Macchiavello and Morjaria, 2015). In our setting, however, several features sharply limit the scope for such mechanisms. Demand is highly idiosyncratic, creating frequent, unpredictable hiring needs that are difficult to satisfy through stable relationships alone. The idiosyncratic nature of demand forces firms to adopt a bifurcated labor strategy, complementing a core of long-term workers (approximately 60% of the firms hire these workers from networks and the remaining from labor stands) with a large number of short-term workers (75% of firms hire these workers from labor stands). Barker et al. (2025) document similar hiring patterns for firms in Nairobi. Our analysis focuses on the short-term contracts, where relational ties are least important in hiring practices.

Additionally, firms in our setting are typically small, lack the "thick" referral networks necessary for relational hiring, and must rely heavily on the spot market at labor stands. Waiting for trusted workers entails production delays that are costly when tasks are time-sensitive, further discouraging firms from privileging repeated matches over immediate hiring. Furthermore, the labor stands themselves are poor environments for building reputation. These markets are characterized by high worker churn, and the likelihood of repeated interaction between a specific firm and worker is low. Compounding this, the diffusion of information regarding worker quality or firm reliability is incomplete. This anonymity and high turnover prevent reputation from serving as an effective enforcement mechanism, forcing firms and workers to rely on other strategies to manage the significant commitment problems inherent in these short-term, non-relational agreements. Moreover, where longer-term ties do emerge, they need not be welfare-enhancing. Classic accounts describe "labor-tying," in which firms withhold wages or extend advances to bind workers (Bardhan, 1983). Taken together, these facts imply that relational contracting plays a second-order role for the margin we study.

⁴⁴Appendix Table D-5 shows that these firms are similar to our overall experimental sample.

6 Matching

The worker and firm experiments measure preferences over contracts but do not show how workers and firms behave once a contract is in place. In particular, do workers renege more on daily-pay contracts, and do firms extract longer hours or increase non-payment under back-loaded contracts? To answer these questions, we matched firms and workers who accepted the same contract type in the first two experiments and tracked their behavior over the duration of the contract. Matching is done at the contract level: a worker is assigned only to firms that agreed to implement the same contract and contract length as the worker accepted. This procedure ensures that every realized worker–firm pair operates under a contract that both sides had previously accepted in the experiment. The next subsection describes the matching procedure in detail.

6.1 The matching process

Matching the firm with the worker. The hiring experiment with firms and the job-offer experiment at labor stands ran in parallel. Once the hiring experiment was completed, firms reported their labor demand and desired hiring timelines. The field team then randomly contacted eligible workers from the worker-experiment pool, either by phone or at the labor stand. A worker was eligible if (i) their contract choice matched the firm’s contract offer, (ii) their experimental assignment permitted a job offer, and (iii) their location (labor stand) was sufficiently close to the firm’s construction site, since travel costs are important in this setting. When a contacted worker was already employed at another construction site and reported that they would only be available after completing their current job, we moved to the next eligible worker on the list. Because labor demand was idiosyncratic, in some cases the number of vacancies exceeded the number of available eligible workers in the pool. In these cases, we extended offers to workers who satisfied the contract-type and location criteria but whose randomization had initially resulted in a no-job outcome. In a few cases, workers who met the three eligibility criteria were offered a job on the same day as their survey.

An enumerator accompanied the worker to the work site on the first day, mirroring the typical practice of firms, which generally escort workers to the site on the first day of the contract.⁴⁵ At the site, workers received a consent form outlining the details of the contract, and firms received a corresponding consent form. Firms were not informed whether the worker’s contract included insurance. During the worker experiment, workers had been told that they would not be insured if they deliberately shirked on the job, and this condition was reiterated at the start of work. Workers were also not informed whether the firm was on a guarantor contract. After this point, our involvement in the process was minimal, and the firm and worker interacted independently, as they would have if the firm had hired the worker directly.

⁴⁵95% of the firms we surveyed reported that they take the worker to the work site on the first day of the contract.

Post-matching. An enumerator conducted a brief survey with both the firm and the worker at the end of each day of the contract, primarily over the phone. The enumerator gathered information on the number of hours the worker worked, the tasks they performed, and whether they were paid according to the contractual agreement. If the worker had not been paid, the enumerator was authorized to pay them the outstanding amount. The same set of questions was asked to the firm to ensure that there was no bias in the information provided by the worker. Throughout the analysis, we cross-check for discrepancy in reporting from both the firm and worker.

If the worker stopped coming to work before the contract ended, we offered the firm compensation if the firm was on a guarantor contract. Similarly, if the worker was removed from the job before completing the contract, we provided compensation for any unpaid work they had completed.⁴⁶ If the firm reported that a worker did not show up, we followed up by contacting the worker either by phone or at the labor stand to inquire about the reasons for their absence.

The purpose of the matching stage was to examine how workers and firms behave under different contracts. Before the experiment began, we were unsure whether there would be sufficient demand from firms for the available workers. Due to concerns about sample size, we chose to analyze firm and worker behavior after matching only for the steep and daily pay contracts. Consequently, workers and firms were not matched on smooth contracts. Since smooth contracts were excluded from the matching stage, workers who had accepted a smooth contract were upgraded to a daily pay contract, which is strictly preferred by workers. Likewise, firms were upgraded from smooth to steep contracts, as the steep contract is strictly preferred by firms.⁴⁷ Workers who were made offers in the Pair Experiment were also matched with firms (see Section B.2.2 for the description of the Pair Experiment). The matching results will include workers from the main experiment as well as the pair experiment. The matching results are robust to excluding workers from the pair experiment.

6.2 Summary of Matching

Table 6 reports descriptive statistics for the matching stage, separately for workers in the main and pair experiments. In the main experiment, 1,360 workers received job offers, of whom 882 accepted at least one. Of these, 282 were randomly selected by computer to receive a job. This share exceeds 25% of 882 because some workers accepted both offers, raising their selection probability to 43.75%. Due to idiosyncratic variation in firm demand, the pool of selected workers was sometimes insufficient, and we extended offers to additional workers who met the contract-type and location criteria. In total, 315 workers were contacted, of whom 221 were allocated work.

Contacting workers posed logistical challenges, as many did not have cell phones. To mitigate this, we provided workers with the survey firm's business card at the end of the worker experi-

⁴⁶We recorded the reasons for both the removal and any instances of non-payment. An enumerator discussed the reasons regarding non-payment with both the firm and worker. They were instructed to contact the supervisor if there was any conflicting statements about who had breached the agreement.

⁴⁷We asked both firms and workers for their consent to work on the upgraded contracts and proceeded only if they approved.

ment and asked them to contact us within two weeks. Many did so and were allocated jobs; others were located through visits to the labor stand. Among workers who were successfully contacted but did not take up the offer, 17% had left the city and 35% had found long-term employment.⁴⁸ Of the 221 workers allocated work, 16 did not reach the site, left shortly after arriving, or were turned away because the firm withdrew.⁴⁹

The pair-experiment column shows similar sample changes at each stage of the allocation process. In total, 55 workers from the pair experiment were ultimately allocated to a job. Of these, 10 were the partner of a main respondent in the pair experiment. Overall, 20 workers worked as part of a pair and the remaining 35 worked alone. In total, 260 workers in our sample completed at least one day of work. Out of the 260 workers who completed at least one day of work, 56 were allotted work on steep contracts and 204 on contracts with daily pay. The ratio of workers in the two contracts is similar to the ratio of take-up of daily pay and steep contracts in the take-up experiment.

In Appendix Table D-15 we compare the characteristics of workers who were allocated a job randomly after the experiment to the workers who were eventually matched. Workers are similar on almost all characteristics but workers who are eventually matched have higher travel costs. Appendix Table D-16 shows that this pattern holds when the comparison is done separately for steep and daily-pay contracts. Appendix Table D-14 compares observable characteristics across steep and daily-pay workers within the matched sample. Out of 12 variables, only the fraction of days worked prior to the experiment differs at the 5% level and all other characteristics are statistically indistinguishable.

Panels B and D show the details of matching for firms. A total 349 firms were surveyed, out of which 335 agreed to hire a worker on at least one contract. All firms, which had accepted at least one contract, were contacted after the experiment to provide them workers on the contract chosen by the random draw. Firms provided us with demands as far as one month in the future. Some firms told us that they had already hired a worker, and would contact us as per their needs. Some firms were too far away from the labor stand, and workers refused to work at those sites. Idiosyncrasies in demand meant that hiring plans changed for several firms and they were unable to hire workers. Due to these reasons, we were able to match a worker to 83 firms. This is 25% of the firms which had agreed to hire at least one worker. More firms were matched on daily pay contracts than guarantor contracts. This is because the random draw of contract had higher drawing chances for daily contract.

Several firms ultimately hired more than one worker in the matching stage. This decision was taken only after the firm experiment had concluded and therefore does not affect firms' experimental choices or the estimation of firm-side treatment effects. The decision reflected two considerations. First, a number of firms requested permission to hire more than one worker. Second, to increase statistical power in the matching stage, we sought to allocate as many workers to

⁴⁸Appendix Table B-6 summarizes the reasons for workers forgoing job offers.

⁴⁹In cases where the firm reneged on the contract, workers received compensation equal to 60% of the daily wage.

firms as possible. Additional workers were assigned under any of the contracts that the firm had accepted in the firm experiment, regardless of whether that contract had been randomly selected for implementation in the main firm experiment. Because multiple workers can be assigned to the same firm, we cluster standard errors at the firm level to account for within-firm correlation in the matching-stage regressions.

Table 6: Summary statistics of Matching

<i>Panel A: Workers</i>		
	Main Experiment	Pair Experiment
Number of workers eligible for a job offer	1,378	506
No of workers offered a 3 day job	1,360	506
No of workers offered a 7 day job	1060	0
No of workers who accepted any job	882	295
No of workers chosen randomly for jobs	282	75
No of workers contacted for jobs	315	122
No of workers who were allotted a job	221	55
No of workers who completed at least one day of work	205	55
<i>Panel B: Firms</i>		
	Number of Firms	
Number of Firms surveyed	349	
Firms which agree to hire on at least one contract	335	
Firms with a successful random draw (hired a worker)	258	
Firms contacted for hiring	335	
Number of firms which hired at least one worker	83	
<i>Panel C: Summary of contracts for matched workers</i>		
	Steep	Daily Pay
Total workers matched	56	204
Total workers on insurance contracts	24	84
<i>Panel D: Summary of contracts for matched firms</i>		
	Steep	Daily Pay
Workers matched on Credit contracts	0	80
Workers matched on Guarantor contracts	0	40
Workers matched on control contracts	56	84

Notes: The Table shows the number of workers and firms at each stage of the experiment. Panel A shows the number of workers at each stage of the experiment separately for the main experiment and the pair experiment. Panel B shows the number of firms at each stage of the experiment. *Accepted any job* refers to accepting at least one offered contract in the worker-side experiment. *Chosen randomly for jobs* refers to accepted offers selected for implementation by the random draw. *Contacted* denotes workers successfully reached for job offer. *Allotted a job* denotes workers assigned to a firm for matching. Panel C shows the number of workers matched under different contracts. Panel D shows the number of workers matched to firms under different contracts.

6.3 Results

The matching sample consists of firms and workers who were ultimately matched on either steep or daily pay contracts. Because firms and workers chose which contracts to accept before matching, they may differ systematically in ways that influence both their initial contract choice and their subsequent behavior. A randomized experiment that assigns workers to contracts would identify the causal effect of contract type. Examining the matching sample that we have, in contrast, reflects realized-match outcomes with endogenous selection into contracts.

To separate the effects of worker selection into contracts from the effects of contract themselves, we show results for two samples: (i) the complete matching sample, which reflects realized outcomes under endogenous contract selection, and (ii) a restricted sub-sample in which final contract allocation is determined by the experimental randomization and is plausibly exogenous to workers' unobserved characteristics. The latter sub-sample includes workers whose final job contract either differed from the contract they initially accepted in the worker experiment due to deterministic upgrade rules, or who would be willing to accept both daily pay and a back-loaded contract. For these workers, the random assignment of the initial offer in the worker experiment determines whether they enter the matching stage under a steep or a daily pay contract. Conditional on being matched, variation in final contract type within this group is therefore quasi-random and should not reflect differences in workers' preferences over contract types.

This sub-sample consists of three groups of workers. First, workers who were allotted a job on a steep contract. Since daily pay strictly dominates steep contracts from the worker's perspective, and the initial offer of a steep contract was random, these workers form a group that would also have accepted daily pay had it been offered on the same terms. Second, workers who were simultaneously offered a daily pay contract and a steep or smooth contract, accepted both, but were ultimately allotted the daily pay contract according to the experimental protocol. Third, workers who had accepted a smooth contract in the worker experiment but were allocated a daily pay job because all smooth contracts were upgraded to daily pay at the matching stage.⁵⁰ Focusing on these workers allows us to compare outcomes across contract types for individuals with revealed or implied willingness to accept both daily pay and back-loaded contracts. In this sense, our design uses experimental variation in contract terms, in the spirit of [Karlan and Zinman \(2009\)](#), to separate the effect of selection into contracts and the effects of contract features on worker and firm behavior.

The specification for the analysis is shown in equation 6.1. Here, D_{ij} is an indicator that takes value 1 if the contract is a steep contract. X_i includes controls for the worker i , such as the time it takes the worker to reach the labor stand from their home, and μ_{ij} are fixed effects for insured

⁵⁰Categories 2 and 3 include workers who accepted a smooth contract but were ultimately matched to a daily-pay job; we treat their willingness to accept smooth as proxy evidence of willingness to accept steep. This proxy is strongest for 3-day contracts, which account for roughly 75% of matches, because take-up of smooth and steep is nearly identical at that horizon (Appendix Figures B-4a and B-4b). Appendix Table D-11 confirms that restricting the quasi-experimental sample to 3-day contracts yields quantitatively similar estimates.

contract, length of the contract, the daily wage and the month of work.⁵¹

$$Y_{ij} = \alpha + \beta_m D_{ij} + X_i + \mu_{ij} + \epsilon_{ij} \quad (6.1)$$

The coefficient of interest is β_m which captures the effect of a steep contract on the outcomes of the worker/firm. The outcome variables we show in Table 7 were pre-specified in the AEA RCT registry. Panel A of Table 7 shows the results for the entire sample, and Panel B shows the results for the quasi-experimental sample.

Worker Separation Columns 1 and 2 of Table 7 test [Prediction WE.1](#), which predicts higher contract completion under back-loaded (steep) contracts than under daily pay. In Column 1 (Panel A), we code a contract as completed if the worker completes each day of work in the contract. When the firm ends the contract early (e.g., because the work ends or the firm is dissatisfied), we treat this as if the worker would have completed the contract. The estimates show that the contract completion rate is 34% under daily pay and is higher by 23.5 percentage points under steep contracts. This corresponds to roughly a 69% increase in completion relative to daily pay. In Column 2, we drop observations in which the firm breaks the contract, and the estimated steep-daily difference is very similar. Across both columns, the coefficients for the quasi-experimental sample are statistically and quantitatively similar to those for the full matching sample, suggesting that differences in completion are unlikely to be driven solely by selection into contracts and are consistent with an important role for contract terms.

Appendix Table B-7 lists self-reported reasons for renegeing by workers. In approximately 30% of cases, workers did not go to the work site because they preferred to stay at home or reported a family emergency or illness. In about 25% of cases, workers renegeed because they felt the contractor was assigning too much work or requiring them to work very long hours. To probe this mechanism more directly, we plot in Appendix Figure B-18 the relationship between excess hours worked and the probability of showing up for work the next day. Workers who work longer than 8 hours are less likely to return the following day, consistent with the idea that being asked to work longer hours increases renegeing. A further 10% of separations occur because workers find work at another site, consistent with workers responding to outside opportunities. Taken together, these patterns suggest that renegeing is driven largely by a combination of excess work hours, negative shocks at home, and arrival of better outside opportunities, consistent with the large contractual lock-in costs estimated in the worker experiment.

Working hours The contractual agreement between firms and workers, typically made orally, is for an 8-hour workday, but firms often ask workers to stay longer, a concern that workers also raise in the preliminary survey. Next, we test [Prediction FE.1](#)—that workers work longer hours under

⁵¹Our experiment was conducted during peak periods and during this period the going daily wage at the labor stand increased from 500 INR to 550 INR. The going wage remained at INR 550 for the remainder of the experiment. Therefore, we changed the wage in our job offer to INR 550 after this shift.

steep back-loaded contracts than under daily pay—by comparing the number of hours workers work under different contracts. We measure daily hours by asking workers how many hours they worked on each surveyed day and averaging over the days observed in the contract. In Panel A, workers on daily pay work an average of 8.19 hours per day, and those on steep contracts work 0.274 hours more (p-value = 0.011). Using hours reported by firms yields a very similar estimate.⁵² In the quasi-experimental sub-sample (Panel B), the estimated effect is 0.182 hours (p-value = 0.149), which is smaller and imprecisely estimated. In this sub-sample, daily-pay workers work somewhat longer hours on average. This pattern is consistent with some workers who are unwilling to work longer days selecting into daily pay contracts. Even in the quasi-experimental sub-sample, however, the point estimate remains positive, suggesting that steep contracts still allow firms to obtain additional hours of work from workers. Workers work more than 8 hours on average even under daily pay, and long working hours are one of the main reasons they report for breaking contracts before completion.

The joint finding that workers are more likely to complete steep contracts than daily wage contracts and that they work longer hours under steep contracts suggests that the payment structure in steep contracts allows firms both to reduce worker reneging and to extract more work. By holding back part of the wage until the end of the contract, steep contracts make premature exit costly for workers, which discourages reneging. In the quasi-experimental sub-sample (Panel B), the estimated effect on daily hours worked is 0.182 hours. Although this may appear small relative to an eight-hour day, hours worked vary little in our data, and all but two workers work at least seven hours. It is therefore more informative to interpret additional work relative to the contractual eight-hour day, treating eight hours as the baseline against which extra work under steep contracts is measured.

As complementary evidence, we now examine self-reported excess work, defined as additional hours worked beyond what workers initially expected over the course of the contract. We asked workers whether the firm had asked them to work more hours than initially agreed. On each day of the contract, workers who answered yes were then asked to report the number of minutes they worked beyond what they had expected, and we sum these reports over the duration of the contract.⁵³ Under daily pay contracts, workers report working on average 0.09 hours more than expected in total over the contract. Column 4 of Table 7 reports estimates of equation 6.1 using this excess-hours measure as the outcome. The coefficient for steep contracts is 0.146, so workers on steep contracts report about 0.24 excess hours in total over the contract, compared to 0.09 hours under daily pay. The estimates for the quasi-experimental sample are quantitatively similar to those for the full sample, suggesting that the higher excess work under steep contracts is not primarily driven by worker selection into contracts but by the contract structure itself. On the extensive margin, about 11.2% of workers on daily pay contracts state that they were asked to work more hours than initially agreed, compared with 23.3% under steep contracts.

⁵²Not shown in the table; the coefficient is 0.239 with a p-value of 0.013.

⁵³This includes workers who report no excess work, for whom we code excess hours as zero.

Table 7: Outcomes for workers under steep vs daily contracts

Panel A: Steep vs Daily					
	Contract completed	Contract completed (excluding firm rejections)	Hours worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.235*** (0.085)	0.266*** (0.090)	0.274** (0.105)	0.146** (0.064)	-0.509 (0.365)
Observations	260	233	259	260	250
Control group mean	0.34	0.26	8.19	0.09	6.97
Fixed Effect	Yes	Yes	Yes	Yes	Yes
Panel B: Steep vs Daily (Workers in the quasi-experiment sub sample)					
	Contract completed	Contract completed (excluding firm rejections)	Hours worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.225** (0.100)	0.230** (0.102)	0.182 (0.125)	0.193** (0.079)	-0.747 (0.468)
Observations	128	117	127	128	124
Control group mean	0.33	0.28	8.30	0.07	7.05
Fixed Effect	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is measured by surveying workers and firms which were matched with each other. The independent variable is a dummy for the *Steep contract*. *Contract completed* captures whether the worker turned up on each day of the contract and was present till the end of each day. *Hours Worked* is winsorized at the 1st and 99th percentile, and captures the average number of hours over the contract period as reported by the worker. *Excess work* captures the sum of excess hours worked over the contract period as reported by the worker. *Worker rating* captures the average of rating given on a scale of 1-10 (higher is better) by the firm to the worker over the contract period. We use fixed effects for whether the contract was insured, length of contract, daily wage offered, month and control for the time it takes the worker to get to the labor stand. Panel A shows results for the entire sample of workers who were part of the matching stage. Panel B restricts the sample to workers whose revealed choices show willingness to accept both daily pay and back-loaded contracts. This group includes: (i) workers allotted a steep contract who would also have accepted daily pay, (ii) workers who accepted a smooth contract but were upgraded to daily pay, and (iii) workers simultaneously offered daily pay and a steep or smooth contract, who accepted both but were ultimately allocated daily pay. Standard errors are clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A natural question is whether the structure of contracts also affects work quality. We examine this using quality ratings reported by firms. Results in Table 7 indicate that, on a 10-point quality scale, the point estimate for workers on steep contracts is 0.509 points lower than for workers on daily pay contracts (p-value = 0.168), with a quantitatively similar coefficient in the quasi-experimental sub-sample. Thus, firms' subjective ratings are, if anything, slightly lower for steep-contract workers, but the difference is modest and imprecisely estimated. Moreover, when we control for these ratings in the regressions for completion and hours (Appendix Table D-9), the main coefficients of interest change very little, suggesting that the observed differences in ratings do not drive the retention and hours effects.

To assess whether the rating pattern reflects genuine differences in on-the-job performance, we turn to a direct measure of worker productivity. We use detailed task-level data collected by enumerators who observed workers for one day at the construction sites, recording activities during two separate hours (once in the first half of the day and once in the second half). Enumerators documented the nature, intensity, and difficulty of each task, and these measures were combined into a productivity index. Appendix Figure B-19 plots the distribution of this index by contract type. We find no statistically significant differences in productivity between steep and daily-pay contracts (the standardized mean difference is small and the p-value is 0.45), indicating that steep-contract workers are not less productive on the job.⁵⁴ Taken together, the small, noisy differences in firm ratings, the stability of our main coefficients when controlling for ratings, and the absence of productivity differences suggest that ratings are subject to measurement noise or idiosyncratic evaluation, rather than reflecting systematic differences in worker quality across contract types.

Non-payment on wages Across the 260 worker–firm matches and approximately 800 workdays we observe 24 instances in which workers report not receiving their full wage for the day. In roughly two-thirds of these cases, the firm’s representative was either absent from the site at the end of the day or reported not having enough cash to pay the worker. In about one-quarter of cases, firms paid less because they believed the worker had not completed the full amount of work expected. Given the small number of events, we do not detect any statistically significant difference in the frequency of non-payment between daily pay and back-loaded contracts (results not shown in Table).

While realized non-payment during matched jobs is infrequent, non-payment concerns can still be first-order for contract choice because workers decide *ex ante* based on expected losses from non-payment. In addition, non-payment may be attenuated in our matched sample because firms knew the match was organized through the survey firm, and wage payment is a particularly salient and easily verifiable outcome relative to hours worked, reducing firms’ willingness to underpay even if other margins of behavior continue to respond to contract terms.

Robustness Checks We check the robustness of the results for different specifications. Results in Table 7 are robust to controlling for education, age and a dummy for backward caste (see Appendix D.5), variables selected by the post-double lasso method of Belloni et al. (2014) (see Appendix D.4), correcting p-values for multiple testing (see Appendix D-8) and only including the sample from the main worker experiment (see Appendix D-13). Providing insurance to workers might change their productivity or actions during a job. We formally test for this in Appendix Table D-12 and find that insured and uninsured workers do not differ in their actions and outcomes. Another possibility is that firms who were offered guarantor contracts might misuse those

⁵⁴The productivity results here contrast with Kaur et al. (2025), who find that financial concerns (i.e., liquidity constraints) reduce worker productivity. One plausible explanation is that, unlike Kaur et al. (2025), where workers make disposable plates by stitching together leaves from trees, the work in our setting is paid by time rather than by output and thus offers fewer margins for effort variation.

contracts to extract excess hours of work from workers and that might change the results. In Appendix Table D-10 we control for whether the contract is a guarantor contract and find that our results do not change substantially.

The matching results, together with the high demand for low lock-in contracts documented earlier, help explain the coexistence of high unemployment, unmet labor demand, and worker absenteeism in low-wage labor markets. Appendix Figure B-10 shows that on a typical day at the labor stand, many workers do not find work. Yet this does not mean that firms can hire freely on their preferred terms: in a survey of a sub-sample of firms, 56% report having been unable to find workers on suitable contracts or wages, and 54% report waiting for a trusted worker to become available before starting new projects. Contract terms shape both whether a match forms and whether it survives. Many unemployed workers reject back-loaded contracts because they value the option to exit freely, generating unmet demand for firms that hire only on back-loaded terms. When matches do form on daily pay, they frequently dissolve: the completion rate under daily pay is just 34%, compared with 57.5% under steep contracts. Workers who renege return to the labor stand to search again, while firms must restart the hiring process. The result is a market in which unemployment, unmet labor demand, and high turnover persist simultaneously.

7 Conclusion

This paper examines how wage theft, worker renegeing, and liquidity constraints jointly shape labor supply and demand in informal labor markets. Using data from informal labor markets in India, we establish that non-payment by firms and high rates of renegeing by workers generate mutual distrust. Workers are poor and require frequent payments for consumption, while firms are liquidity constrained and averse to paying frequently. We incorporate these facts into a theoretical framework that generates predictions about preferences of firms and workers for contracts with different payment structures and on-the-job behavior on these contracts.

We test these predictions using an experiment with firms and workers in construction in Patna, India, varying the payment structure of contracts. Firms are 38 p.p. more likely to hire on steep back-loaded contracts than on daily-pay contracts: about 24 p.p. of this gap reflects reduced costs from worker renegeing (daily-versus-smooth contrast) and the remaining 14 p.p. reflects liquidity constraints and transaction costs (smooth-versus-steep contrast). Fully alleviating firm liquidity constraints and transaction costs increases the probability of hiring by 46 percentage points, from a baseline of 43%.

In the experiment, workers are less likely to accept the back-loaded contracts preferred by firms. Insurance against non-payment increases take-up, with larger effects for back-loaded contracts, confirming that wage-theft concerns reduce labor supply. Consumption liquidity constraints further suppress acceptance of back-loaded contracts, but the largest factor is contractual lock-in costs: workers who exit a back-loaded contract forfeit accumulated wages, and three circumstances make exit a salient risk — excess work extraction by firms, health or family shocks,

and better outside options. Among surveyed workers who rejected offers, 72% earned less over the following seven days than the offer would have paid, suggesting these frictions impose income losses. The conflicting preferences of firms and workers over payment structure reduce match rates in equilibrium, helping explain the coexistence of high unemployment, unmet labor demand, and high worker renegeing in our setting and potentially in similar informal labor markets in LMICs.

After the first two experiments, workers and firms that accept the contracts are matched with each other, and we compare the actions of firms and workers under different contracts. Workers are more likely to renege on daily-pay contracts before completion than on back-loaded contracts. Under back-loaded contracts, workers work longer hours on average than under daily-pay contracts, as reported by both firms and workers. These patterns are consistent with the model prediction that back-loading both reduces worker renegeing and strengthens firms' incentives to extract longer hours.

Our sample consists of extremely poor urban job-seekers in construction in LMICs. External validity is likely to be strongest for workers at the bottom of the informal labor market. For better resourced workers, short run consumption liquidity constraints may be less binding, but non-payment concerns and contractual lock-in costs should remain important determinants of labor supply in the informal economy. Evidence from more formal settings shows that absenteeism persists even where wage theft and firm-side liquidity constraints are less salient ([Adhvaryu et al., 2024](#); [Goraya et al., 2025b](#)). Our findings are consistent with this: surveys with workers who quit during the matching stage indicate that separation is driven by health shocks, family emergencies, and outside options—factors that operate across labor market settings regardless of payment structure.

Policy Implications Our results point to several policy levers. First, our findings suggest that institutions capable of enforcing contractual commitments — labor courts, intermediaries, or unions — can reshape beliefs about non-payment and renegeing on both sides ([Fernando and Singh, 2025](#); [Casaburi and Macchiavello, 2019](#)). Written contract templates alone are unlikely to be effective unless paired with low-cost verification, mediation, or payment-guarantee mechanisms administered by a party with enforcement capacity. In our setting, labor unions are absent, yet workers express near-universal support for unionization, primarily because they believe unions can address non-payment. Notably, 90% of firms in a random sample of 96 also report that they would support a worker union, citing improved accountability, lower renegeing and absenteeism, reduced search costs, and support in recovering unpaid sums from their own clients ([Lagos, 2024](#)). Digital platforms that build reputation-based trust could complement these institutions, though their reach in our setting is limited: 90% of workers lack smartphones and only 56% have any phone.

Second, improving firms' access to working capital could yield significant efficiency gains. However, the frictions we document are embedded in a hierarchy of payment timing: large clients pay small firms only at project completion, and firms in turn back-load payments to workers.

Liquidity and enforcement risk thus flow down this chain and are ultimately borne by the poorest agents. When upstream clients delay or withhold payment, firms respond by delaying wages or reducing hiring, amplifying shocks that originate higher in the chain. Policies targeting only worker-firm contracts may therefore have limited impact if payment practices at the top of the chain remain unchanged.

References

- Abebe, Girm, A. Stefano Caria, and Esteban Ortiz-Ospina**, “The Selection of Talent: Experimental and Structural Evidence from Ethiopia,” *American Economic Review*, June 2021, 111 (6), 1757–1806.
- , **A Stefano Caria, Marcel Fafchamps, Paolo Falco, Simon Franklin, and Simon Quinn**, “Anonymity or Distance? Job Search and Labour Market Exclusion in a Growing African City,” *The Review of Economic Studies*, May 2021, 88 (3), 1279–1310.
- Adhvaryu, Achyuta, Jean-François Gauthier, Anant Nyshadham, and Jorge Tamayo**, “Absenteeism, Productivity, and Relational Contracts Inside the Firm,” *Journal of the European Economic Association*, August 2024, 22 (4), 1628–1677.
- Aghion, Philippe and Richard Holden**, “Incomplete Contracts and the Theory of the Firm: What Have We Learned over the Past 25 Years?,” *Journal of Economic Perspectives*, June 2011, 25 (2), 181–197.
- Baijal, Shishir and Ashwani Awasthi**, “Skilled Employment in Construction Sector in India,” *Frank Knight, RICS*, 2023.
- Banerjee, Abhijit and Sandra Sequeira**, “Learning by searching: Spatial mismatches and imperfect information in Southern labor markets,” *Journal of Development Economics*, September 2023, 164, 103111.
- Bardhan, Pranab K.**, “Labor-Tying in a Poor Agrarian Economy: A Theoretical and Empirical Analysis,” *The Quarterly Journal of Economics*, 1983, 98 (3), 501–514. Publisher: Oxford University Press.
- Barker, Nathan, Inbar Amit, Alison Andrew, Robert Garlick, Carol Nekesa, and Kate Orkin**, “Flexibility versus Performance: The Determinants of Labor Contracts in Nairobi, Kenya,” February 2025. Job market paper.
- Becker, Gary S. and George J. Stigler**, “Law Enforcement, Malfeasance, and Compensation of Enforcers,” *The Journal of Legal Studies*, 1974, 3 (1), 1–18. Publisher: [University of Chicago Press, University of Chicago Law School].

- Belloni, Alexandre, Victor Chernozhukov, and Christian Hansen**, “Inference on Treatment Effects after Selection among High-Dimensional Controls,” *The Review of Economic Studies*, April 2014, 81 (2), 608–650.
- Bhorat, Haroon, Ravi Kanbur, and Benjamin Stanwix**, “Compliance with Labor Laws in Developing Countries,” *IZA World of Labor*, 2019, (80).
- Boudreau, Laura**, “Multinational Enforcement of Labor Law: Experimental Evidence on Strengthening Occupational Safety and Health Committees,” *Econometrica*, 2024, 92 (4), 1269–1308.
- , **Rachel Heath**, and **Tyler H. McCormick**, “Migrants, experience, and working conditions in Bangladeshi garment factories,” *Journal of Economic Behavior & Organization*, March 2024, 219, 196–213.
- Breza, Emily and Supreet Kaur**, “Labor Markets in Developing Countries,” *Annual Review of Economics*, 2025.
- , —, and **Nandita Krishnaswamy**, “Social Norms as a Determinant of Aggregate Labor Supply: Evidence from Decentralized Spot Markets,” *Revise and Resubmit, Journal of Political Economy*, August 2025.
- , —, and **Yogita Shamdasani**, “The Morale Effects of Pay Inequality*,” *The Quarterly Journal of Economics*, May 2018, 133 (2), 611–663.
- , —, and —, “Labor Rationing,” *American Economic Review*, October 2021, 111 (10), 3184–3224.
- Brune, Lasse and Jason T. Kerwin**, “Income Timing and Liquidity Constraints: Evidence from a Randomized Field Experiment,” *Journal of Development Economics*, 2019, 138, 294–308.
- , **Eric Chyn**, and **Jason Kerwin**, “Pay Me Later: Savings Constraints and the Demand for Deferred Payments,” *American Economic Review*, July 2021, 111 (7), 2179–2212.
- Caria, Stefano A. and Paolo Falco**, “Skeptical Employers: Experimental Evidence on Biased Beliefs Constraining Firm Growth,” *The Review of Economics and Statistics*, September 2024, 106 (5), 1352–1368.
- Caria, Stefano, Kate Orkin, Alison Andrew, Robert Garlick, Rachel Heath, and Niharika Singh**, “Barriers to Search and Hiring in Urban Labour Markets,” *VoxDevLit*, 10(1), 2024.
- Carranza, Eliana, Robert Garlick, Kate Orkin, and Neil Rankin**, “Job Search and Hiring with Limited Information about Workseekers’ Skills,” *American Economic Review*, November 2022, 112 (11), 3547–3583.
- Carroll, Christopher D., Martin B. Holm, and Miles S. Kimball**, “Liquidity constraints and precautionary saving,” *Journal of Economic Theory*, July 2021, 195, 105276.

- Casaburi, Lorenzo and Jack Willis**, “Time versus State in Insurance: Experimental Evidence from Contract Farming in Kenya,” *American Economic Review*, December 2018, 108 (12), 3778–3813.
- **and Rocco Macchiavello**, “Demand and Supply of Infrequent Payments as a Commitment Device: Evidence from Kenya,” *American Economic Review*, 2019, 109 (2), 523–555.
- Cefala, Luisa, Supreet Kaur, Heather Schofield, and Yogita Shamdasani**, “Habit Formation in Labor Supply,” 2025.
- Chandrasekhar, Arun, Melanie Morten, and Alessandra Peter**, “Network-Based Hiring: Local Benefits; Global Costs,” February 2020, (w26806), w26806.
- Danzer, Alexander M.**, “Can Secondary Jobs Smooth Consumption? Evidence from Unanticipated Wage Arrears,” *Economic Development and Cultural Change*, 2019, 67 (3), 571–594.
- Donald, Aletheia and Florian Grosset**, “Complementarities in Labor Supply,” 2024.
- Earle, John S and Klara Sabirianova Peter**, “Complementarity and Custom in Wage Contract Violation,” *The Review of Economics and Statistics*, November 2009, 91 (4), 832–849.
- Farazi, Subika**, “Informal Firms and Financial Inclusion: Status and Determinants,” 2014.
- Fernando, A Nilesh and Niharika Singh**, “Regulation by Reputation? Intermediaries, Labor Abuses, and International Migration,” *The Review of Economics and Statistics*, 2025.
- Fernando, A. Nilesh, Niharika Singh, and Gabriel Tourek**, “Hiring Frictions and the Promise of Online Job Portals: Evidence from India,” *American Economic Review: Insights*, December 2023, 5 (4), 546–562.
- Goldar, Bishwanath, Prateek Kukreja, and Havishaye Puri**, “Labour Reforms and Growth in Manufacturing Employment in India,” *The Indian Journal of Labour Economics*, 2025, 68, 45–73.
- Goraya, Sampreet Singh, Suanna Oh, and Yogita Shamdasani**, “Absent but Not Idle: the Social Roots of Demand for Flexible Work,” 2025.
- , — , **Nandita Krishnaswamy, and Yogita Shamdasani**, “Worker Absenteeism and Firm Outcomes: Evidence from the Indian Manufacturing Sector,” 2025.
- Government of Bihar, Finance Department**, “Bihar Economic Survey 2024–25,” Technical Report, Government of Bihar, Patna 2025.
- Heath, Rachel**, “Why Do Firms Hire Using Referrals? Evidence from Bangladeshi Garment Factories,” *Journal of Political Economy*, August 2018, 126 (4), 1691–1746. Publisher: The University of Chicago Press.
- Holm, Sture**, “A Simple Sequentially Rejective Multiple Test Procedure,” *Scandinavian Journal of Statistics*, January 1979, 6, 65–70.

- Kanbur, Ravi and Lucas Ronconi**, "Enforcement matters: The effective regulation of labour," *International Labour Review*, 2018, 157 (3), 331–356.
- Karlan, Dean and Jonathan Zinman**, "Observing Unobservables: Identifying Information Asymmetries With a Consumer Credit Field Experiment," *Econometrica*, 2009, 77 (6), 1993–2008. : <https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA5781>.
- Kaur, Supreet**, "Nominal Wage Rigidity in Village Labor Markets," *American Economic Review*, October 2019, 109 (10), 3585–3616.
- , **Sendhil Mullainathan, Suanna Oh, and Frank Schilbach**, "Do Financial Concerns Make Workers Less Productive?," *The Quarterly Journal of Economics*, January 2025, 140 (1), 635–689.
- Krishnaswamy, Nandita**, "Missing and Fired: Worker Absence, Labor Regulation, and Firm Outcomes," 2019.
- Kumar, Sunil and Melissa Fernandez**, "The Urbanisation Construction Migration Nexus in 5 Cities in South Asia," 2015.
- Lagos, Lorenzo**, "Union Bargaining Power and the Amenity-Wage Tradeoff," *SSRN Electronic Journal*, 2024.
- Lazear, Edward P.**, "Agency, Earnings Profiles, Productivity, and Hours Restrictions," *The American Economic Review*, 1981, 71 (4), 606–620. Publisher: American Economic Association.
- Macchiavello, Rocco**, "Relational Contracts and Development," *Annual Review of Economics*, 2022, 14 (1), 337–362. : <https://doi.org/10.1146/annurev-economics-051420-110722>.
- and **Ameet Morjaria**, "The Value of Relationships: Evidence from a Supply Shock to Kenyan Rose Exports," *American Economic Review*, 2015, 105 (9), 2911–2945.
- Murthy, S V Ramana**, "Measuring Informal Economy in India, Indian Experience," 2019.
- Musa, Dauda and Daniel J. Sonnenstuhl**, "Working Without Wages: The Consequences of Widespread Pay Delays," November 2025. Job market paper (JMP), University of Chicago. Version: November 30, 2025.
- Naidu, Suresh and Noam Yuchtman**, "Coercive Contract Enforcement: Law and the Labor Market in Nineteenth Century Industrial Britain," *American Economic Review*, February 2013, 103 (1), 107–144.
- Praveen, M. P.**, "Migrant workers lodge complaints of 'wage theft' amounting to INR 2.41 crore in 18 months," *The Hindu*, May 2024.
- Romano, Joseph P. and Michael Wolf**, "Stepwise multiple testing as formalized data snooping," *Econometrica*, 2005, 73 (4), 1237–1282.

- **and** —, “Efficient computation of adjusted p-values for resampling-based stepdown multiple testing,” *Econometrica*, 2016, 84 (2), 645–683.
- Shapiro, Carl and Joseph E. Stiglitz**, “Equilibrium Unemployment as a Worker Discipline Device,” *The American Economic Review*, 1984, 74 (3), 433–444. Publisher: American Economic Association.
- Sharma, Anisha, Manisha Shah, and Beata Łuczywek**, “Understanding the Impact of Low-Cost Loans on Forced Labor,” 2024.
- Surendra, Vaishnavi**, “The Moneylender as Middleman: Formal Credit Supply and Informal Loans in Rural India,” 2025.
- Thomas, Jonathan and Tim Worrall**, “Self-Enforcing Wage Contracts,” *The Review of Economic Studies*, 1988, 55 (4), 541–553. Publisher: [Oxford University Press, Review of Economic Studies, Ltd.].
- Varun, K**, “Negotiations and undercutting in low wage labor markets,” *Working Paper*, 2025.
- Wells, Jill**, “Labour contracting, migration and wage theft in the construction industry in Qatar, China, India, US and the EU,” in “Routledge Handbook on Labour in Construction and Human Settlements,” 1 ed., London: Routledge, November 2023, pp. 114–136.
- World Bank**, “World Development Indicators: GDP per capita (current US\$) – India,” <https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?locations=IN> 2025.

Part I

Appendix

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A Model Appendix

A.1 Firm non-payment

There are two firm types: a fraction $\pi_F \in (0, 1)$ are defaulting (F) and the remainder $\pi_H = 1 - \pi_F$ are compliant (H). Compliant firms pay as promised. A defaulting firm pays through a self-chosen default day $\tau_j^* \in \{1, \dots, L\}$ under schedule $j \in \{\text{daily, smooth, steep}\}$ and then withholds what is owed on that day. Let $\omega_{j\tau}$ be the payment scheduled at the end of day τ under j , let r_j

denote the worker's reneging probability on day τ (conditional on payment through $\tau - 1$), and define the arrival probability to the start of day τ by $S_j(\tau) = \prod_{s=1}^{\tau-1} (1 - r_{j,s})$. Let $\tilde{\omega}$ be the per-day wage implied by the contract total. The unpaid accrual (accrued-but-unpaid balance) just before the day- τ payment is

$$H_j(\tau) \equiv \tau \tilde{\omega} - \sum_{s=1}^{\tau-1} \omega_{j,s}.$$

A defaulting firm that defaults at τ can appropriate $H_j(\tau)$ only if the worker arrives, so it chooses

$$\tau_j^* \in \arg \max_{\tau=1, \dots, L} S_j(\tau) H_j(\tau).$$

Optimal default timing: Because $H_{\text{daily}}(\tau)$ is constant and $S_{\text{daily}}(\tau)$ is weakly decreasing, a defaulting firm defaults on day 1 under daily: $\tau_{\text{daily}}^* = 1$. Under steep and smooth, $H_j(\tau)$ is increasing in τ (slope W for steep from $\tau \geq 2$; slope $W - 350 > 0$ for smooth), so the choice is between day 1 and day L :

$$\tau_{\text{steep}}^* = \begin{cases} 1, & \text{if } S_{\text{steep}}(1) W \geq S_{\text{steep}}(L) (LW - 350), \\ L, & \text{otherwise,} \end{cases}$$

$$\tau_{\text{smooth}}^* = \begin{cases} 1, & \text{if } S_{\text{smooth}}(1) W \geq S_{\text{smooth}}(L) (LW - 350(L - 1)), \\ L, & \text{otherwise.} \end{cases}$$

A.2 What would an optimal contract look like?

Given that firms and workers have opposing preferences, it is important to understand what contract a social planner could implement to maximize aggregate welfare. The planner's instrument is the set of all possible payment schedules $\omega = \{\omega_t\}_{t=1}^L$ with a fixed average daily wage $\tilde{\omega}$, such that $\sum_{t=1}^L \omega_t = L \tilde{\omega}$ and $\omega_t \geq 0$, holding all other primitives fixed. Let $j(\omega)$ denote the contract corresponding to a given schedule ω .

Objective and decomposition. Social welfare aggregates worker and firm values (defined in 3.8 and 3.10):

$$\text{Welfare}(\omega) = \int_i W_i^{j(\omega)} dA(i) + \int_k T_k^{j(\omega)} dB(k). \quad (\text{A.1})$$

For a given ω , welfare moves through two margins emphasized by the model: (i) the *matching margin*, captured by the acceptance probability $P(j(\omega))$ on the worker side and by $P_f(j(\omega))$ on the firm side; and (ii) the *continuation margin*, via the equilibrium reneging probability $r_{j(\omega)}$ that determines replacement/search losses (the λ_s term) and foregone continuation surplus. The payment schedule ω also directly affects other frictions: per-payment administrative costs (λ_{pf}), firm working-capital costs ($\lambda_f[B]^-$), and worker borrowing penalties ($\lambda_B \mathbb{1}\{A < 0\}$). The outside option is the spot market where the wage is ω_s .

Benchmark (no frictions). If there is no limited commitment (workers never renege; firms always pay, i.e. $q = 1$), no liquidity frictions ($\lambda_{pf} = 0$, $\lambda_f = 0$, $\lambda_B = 0$), and no discounting ($\rho \approx 1$), then all schedules ω with the same total pay are welfare-equivalent.

Turning frictions on, one at a time. (i) *Worker exit only* ($q = 1$, no liquidity frictions). Unpaid accrual (the amount still owed) acts as a bond, raising the worker's cost of exiting. A larger bond reduces reneging $r_{j(\omega)}$ and the implied replacement losses (λ_s) for firms. Hence welfare rises when payment mass is shifted *later*: the fully back-loaded schedule is optimal.

(ii) *Adding firm liquidity frictions* ($\lambda_{pf} > 0$, $\lambda_f > 0$). Shifting payment mass later also reduces the number and size of interim payouts, lowering the firm's per-payment administrative costs and liquidity exposure. This reinforces (i): concentrating mass at the end remains optimal.

(iii) *Add worker liquidity needs* ($\lambda_B > 0$). Now, a trade-off emerges. Shifting some mass *earlier* raises worker acceptance $P(j(\omega))$ and lowers expected borrowing penalties. However, this adjustment can increase worker reneging $r_{j(\omega)}$ and raise firm-side costs. The welfare-improving adjustment is to introduce a *small early payment* while preserving a *large terminal payment* that sustains continuation incentives.

A schedule that nests these forces and is optimal within a broad class is:

$$\omega_t^* = \begin{cases} s & \text{for } t = 1, \dots, \tau^*, \\ 0 & \text{for } t = \tau^* + 1, \dots, L - 1, \\ X & \text{for } t = L, \end{cases} \quad s \cdot \tau^* + X = L\tilde{\omega}, \quad (\text{A.2})$$

with (s, τ^*) chosen as the *minimal* early mass that (in expectation) keeps most workers from hitting $A_{it} < 0$ while maintaining a large terminal payment X to deter exit. This “stipend-and-bonus” schedule uses few payout dates (lowering λ_{pf} costs), eases worker liquidity (addressing λ_B), and preserves strong continuation incentives (addressing λ_s).

Firm reneging If $q < 1$, earlier mass further raises acceptance by lowering exposure to non-payment and the associated excess-hours risk perceived by workers. The optimal ω^* therefore shifts *weakly* more mass to the stipend component relative to (A.2).

In summary, with fixed total pay $L\tilde{\omega}$, the welfare-maximizing schedule must balance the worker's need for early payments (to address liquidity λ_B and non-payment $q < 1$ frictions) against the firm's need for later payments (to ensure worker continuation λ_s and manage firm-side λ_f, λ_{pf} costs).

A.3 Value functions

This subsection formally defines the value functions referenced in the main text. All notation follows Section 3.

Worker value functions. The value of continuing with contract $j \in \{1, \dots, J-1\}$ to worker i at time t , conditional on having been paid in the previous period, is:

$$V_{ij}^t(A_{it}, e_j) = U_{ijt} + \epsilon_{ijt} + \rho \{ \mathbb{E}[q * \max_{d_{it+1}} [d_{it+1} * V_{ij}^{t+1}(A_{it+1}, e_j) + (1 - d_{it+1}) V_{ir}^{t+1}(A_{it+1})] + (1 - q) * V_{ir}^{t+1}(A_{it+1})] \} \quad (\text{A.3})$$

where ϵ_{ijt} is an idiosyncratic shock with a Type-1 extreme value distribution, i.i.d. across i, t . $V_{ir}^{t+1}(A_{it+1})$ is the value of renegeing at the beginning of period $t+1$, and $V_{ij}^{t+1}(A_{it+1}, e_j)$ is the continuation value under contract j .

The value of renegeing on any contract $j \in \{1, \dots, J-1\}$ at time t is:

$$V_{ir}^t(A_{it}) = \max\{V_{i0}^t(A_{it}), V_{iJ}^t(A_{it})\} \quad (\text{A.4})$$

where the values of searching at the labor stand and staying at home are:

$$V_{i0}^t(A_{it}) = U_{i0t} + \epsilon_{i0t} + \rho \mathbb{E} V_{ir}^{t+1}(A_{it+1}) \quad (\text{A.5})$$

$$V_{iJ}^t(A_{it}) = U_{iJt} + \epsilon_{iJt} + \rho \mathbb{E} V_{ir}^{t+1}(A_{it+1}) \quad (\text{A.6})$$

The value of rejecting contract j at $t = 0$ is:

$$V_{i0}(A_{i0}, a_i = 0) = U_{i00} + \epsilon_{i00} + \rho \mathbb{E} V_{ir}^1(A_{i1}) \quad (\text{A.7})$$

After rejecting, the worker searches at the stand on day 0 and chooses between search and home in subsequent periods.

The value of accepting contract j at $t = 0$ is:

$$V_{ij}(A_{i0}, a_i = 1, e_j) = U_{ij0} + \epsilon_{ij0} + \rho \{ \mathbb{E}[q * \max_{d_{it+1}} [d_{it+1} * V_{ij}^{t+1}(A_{it+1}, e_j) + (1 - d_{it+1}) V_{ir}^{t+1}(A_{it+1})] + (1 - q) V_{ir}^{t+1}(A_{it+1})] \} \quad (\text{A.8})$$

Firm states and transitions. Let B_{kt} denote the firm's cash-on-hand and A_{kt}^f outstanding accounts receivable at the beginning of day t . The firm's employer settles receivables stochastically: let $\xi_t \in \{0, 1\}$ be a settlement indicator at the end of day t with hazard $h_t = P(\xi_t = 1)$, and $\phi_t \in (0, 1]$ the settlement fraction. States evolve as:

$$A_{k,t+1}^f = (1 - \xi_t \phi_t) A_{kt}^f + y_t, \quad (\text{A.9})$$

$$B_{k,t+1} = B_{kt} - \omega_{jt} - \lambda_f [B_{kt}]^- + \xi_t \phi_t A_{kt}^f, \quad (\text{A.10})$$

where $\lambda_f \geq 0$ is the borrowing/overdraft rate and $[x]^- = \max\{-x, 0\}$. At $t = L$, residual receivables are written off with probability δ_e (employer default):

$$\Pi_j^{L+1}(B_{k,L+1}, A_{k,L+1}^f) = \Pi_{0k}^{L+1}(B_{k,L+1}, A_{k,L+1}^f) \equiv -\delta_e A_{k,L+1}^f. \quad (\text{A.11})$$

Firm value functions. Total gross output over the contract is:

$$M = \sum_{t=1}^L \nu \hat{\theta}(H + e_{kj}) \cdot \mathbb{1}\{\text{worker present at } t\}. \quad (\text{A.12})$$

The value of hiring a worker on contract j at day t is:

$$\Pi_j^t(B_{kt}, A_{kt}^f; r_j) = f_{kjt} + \eta_{kjt} + \rho \mathbb{E}[(1 - r_j) \Pi_j^{t+1}(B_{k,t+1}, A_{k,t+1}^f) + r_j (\Pi_{0k}^{t+1} - \lambda_s)], \quad (\text{A.13})$$

where η_{kjt} is an idiosyncratic shock (Type-1 extreme value, i.i.d. across k, j, t), and Π_{0k}^{t+1} is the firm's outside option. Note that the firm does not bear search costs when hiring through the experiment, as the field team brings the worker to the site.

The firm's outside option is:

$$\Pi_{0k}^t = F_k + \eta_{k0t} + \rho \mathbb{E}[\Pi_{0k}^{t+1}], \quad (\text{A.14})$$

where F_k is a firm-specific component that captures resources and hiring networks, treated as a monotone function of initial liquidity B_{k0} .

At $t = L$, residual receivables are written off with probability δ_e (employer default):

$$\Pi_j^{L+1}(B_{k,L+1}, A_{k,L+1}^f) = \Pi_{0k}^{L+1}(B_{k,L+1}, A_{k,L+1}^f) \equiv -\delta_e A_{k,L+1}^f. \quad (\text{A.15})$$

A.4 Proofs

We prove the predictions stated in the main text. Throughout, all contracts have the same length L and the same average daily wage $\tilde{\omega}$, so that $\sum_{t=1}^L \omega_{jt} = L\tilde{\omega}$ for each schedule j . We use the shorthand

$$\text{"daily"} \equiv \mathcal{C}^D, \quad \text{"smooth"} \equiv \mathcal{C}^{Sm}, \quad \text{"steep"} \equiv \mathcal{C}^S.$$

By construction, the daily schedule pays (weakly) earlier than the smooth schedule, which in turn pays (weakly) earlier than the steep schedule: for each $\tau \leq L$,

$$\sum_{t=1}^{\tau} \omega_{\mathcal{C}^D,t} \geq \sum_{t=1}^{\tau} \omega_{\mathcal{C}^{Sm},t} \geq \sum_{t=1}^{\tau} \omega_{\mathcal{C}^S,t}, \quad \text{with equality at } \tau = L.$$

Common assumptions. (i) Worker per-day utility under contract j is

$$U_{ijt} = c_i - \lambda_B \mathbb{1}\{A_{it} < 0\} - \lambda_e e_{ijt}^2,$$

with c_i the minimum daily consumption level.

(ii) Payment risk: conditional on working, $\chi_t \in \{0, 1\}$ indicates whether the promised wage ω_{jt} is paid at the end of day t . The sequence $\{\chi_t\}_{t=1}^L$ is i.i.d. across t , independent of worker assets and types, with $\mathbb{E}[\chi_t] = q$. If $\chi_t = 0$ at any date, the worker sets the continuation value with that firm

to zero and does not return (Assumption 3 in the worker section).

(iii) Outside option: the value of searching at the labor stand, $V_{i0}^t(A_{it})$, is increasing in type θ_i via the job-finding probability $p(\theta_i)$, as assumed in the worker section.

(iv) Let

$$U_{j,t} \equiv \sum_{s=t+1}^L \mathbb{E}[\chi_s \omega_{js}]$$

denote the expected unpaid accrual at the start of day t under contract j . Because $\mathbb{E}[\chi_s] = q$ for all s , this can be written as

$$U_{j,t} = q \sum_{s=t+1}^L \omega_{js}.$$

There exists a continuation-surplus function $\Delta_t(A, U)$ such that the worker continues at t under schedule j if and only if

$$\Delta_t(A_{it}, U_{j,t}) \geq 0,$$

and

$$\frac{\partial \Delta_t}{\partial U}(A, U) \geq 0 \quad \text{for all relevant } (A, U).$$

(v) Discounting $\rho \in (0, 1)$ is fixed and, given the short horizon in the experiment ($L \leq 7$), pure time discounting effects are negligible relative to the liquidity and wage-theft channels.

A.4.1 Worker side predictions

Prediction (W.1). *Holding $(L, \tilde{\omega})$ fixed, the acceptance probabilities satisfy*

$$P(\mathcal{C}^D) \geq P(\mathcal{C}^{Sm}) \geq P(\mathcal{C}^S).$$

Proof. Fix worker i and a contract $j \in \{\mathcal{C}^D, \mathcal{C}^{Sm}, \mathcal{C}^S\}$. Let $\{\omega_{jt}\}_{t=1}^L$ denote the per-day payments, with identical totals $\sum_{t=1}^L \omega_{jt} = L\tilde{\omega}$. Define the sums

$$S_{j,\tau} \equiv \sum_{t=1}^{\tau} \omega_{jt} \quad (\tau = 1, \dots, L).$$

By construction,

$$S_{\mathcal{C}^D,\tau} \geq S_{\mathcal{C}^{Sm},\tau} \geq S_{\mathcal{C}^S,\tau} \quad \text{for all } \tau \leq L, \quad \text{with equality at } \tau = L.$$

Define the future sums

$$F_{j,t} \equiv \sum_{s=t+1}^L \omega_{js} = L\tilde{\omega} - S_{j,t}.$$

The ordering of $S_{j,t}$ implies

$$F_{\mathcal{C}^D,t} \leq F_{\mathcal{C}^{Sm},t} \leq F_{\mathcal{C}^S,t} \quad \forall t < L,$$

and, using the definition of $U_{j,t}$,

$$U_{\mathcal{C}^D,t} \leq U_{\mathcal{C}^{Sm},t} \leq U_{\mathcal{C}^S,t} \quad \forall t < L. \quad (\text{A.16})$$

Asset paths. Let $\{\chi_t\}_{t=1}^L$ be the payment-success process and let $C_i \equiv c_i + m_i$ denote per-day cash needs, where m_i is the commuting cost (identical across contracts that involve working). The asset recursion under contract j is

$$A_{i\tau}^{(j)} = A_{i0} + \sum_{t=1}^{\tau} \chi_t \omega_{jt} - \tau C_i \quad (\tau = 1, \dots, L).$$

Define

$$\mu_{j,t} \equiv \mathbb{E}[\chi_t \omega_{jt} \mid \mathcal{F}_{t-1}].$$

Because the contracts differ only in $\{\omega_{jt}\}$ and the process $\{\chi_t\}$ is common, and if χ_t are i.i.d. with mean q , then $\mu_{j,t} = q \omega_{jt}$ and the sums inherit the same ordering:

$$\sum_{t=1}^{\tau} \mu_{\mathcal{C}^D,t} \geq \sum_{t=1}^{\tau} \mu_{\mathcal{C}^{Sm},t} \geq \sum_{t=1}^{\tau} \mu_{\mathcal{C}^S,t} \quad \forall \tau \leq L. \quad (\text{A.17})$$

Let

$$N_i^{(j)} \equiv \sum_{\tau=1}^L \mathbb{1}\{A_{i\tau}^{(j)} < 0\}$$

be the number of days in which assets are negative. Since expected assets under daily are at least as high as under smooth, and under smooth at least as high as under steep, at every date, the probability that assets are negative at any date is lowest under daily and highest under steep (given FOSD of Daily over smooth, and of smooth over steep). Summing over dates,

$$\mathbb{E}[N_i^{(\mathcal{C}^D)}] \leq \mathbb{E}[N_i^{(\mathcal{C}^{Sm})}] \leq \mathbb{E}[N_i^{(\mathcal{C}^S)}]. \quad (\text{A.18})$$

Exit and forfeiture. At the start of day t , the expected unpaid accrual under contract j is $U_{j,t} = qF_{j,t}$. If the worker exits at some day $\tau < L$, the expected loss of promised payments is $U_{j,\tau}$. The outside option value at τ does not depend on j , so by (A.16) the loss from exiting is smallest under $\mathcal{C}^D$ and largest under $\mathcal{C}^S$ at every possible exit date.

The expected value from accepting j for worker i , V_{ij}^{accept} , is lower when the worker expects more days with $A_{it} < 0$ and higher when exiting is less costly. By (A.18) and (A.16),

$$V_{i,\mathcal{C}^D}^{\text{accept}} \geq V_{i,\mathcal{C}^{Sm}}^{\text{accept}} \geq V_{i,\mathcal{C}^S}^{\text{accept}}.$$

Since the outside option is the same across schedules, acceptance is monotone in V_{ij}^{accept} and therefore

$$P(\mathcal{C}^D) \geq P(\mathcal{C}^{Sm}) \geq P(\mathcal{C}^S).$$

□

Prediction (W.2). *If q increases, the acceptance probability $P(j)$ (for any fixed j) weakly increases.*

Proof. The asset law of motion under contract j is

$$A_{i,t+1} = A_{it} - c_i - m_{ij} + \chi_t \omega_{jt},$$

with $m_{ij} = m_i$ for all contracts that involve working and $\mathbb{E}[\chi_t] = q$. A higher q raises expected wages on every day and shifts the distribution of $A_{i,t+1}$ upward at every date. This reduces the frequency and duration of episodes with $A_{it} < 0$ and increases the worker's continuation value under j . The outside option does not depend on q , so the set of initial states (A_{i0}, θ_i) for which

$$V_{ij}^0(A_{i0}; e_j^*) \geq V_{i0}^0(A_{i0})$$

holds is (weakly) larger when q is higher. Hence $P(j)$ is weakly increasing in q . □

Prediction (W.2.1). *For a smooth contract, providing insurance (setting $q = 1$) increases acceptance.*

Proof. This follows directly from W.2: moving from $q < 1$ to $q = 1$ increases the continuation value under $\mathcal{C}^{Sm}$ while leaving the outside option unchanged, so $P(\mathcal{C}^{Sm})$ weakly increases. The effect is larger when a larger share of total pay is scheduled for later dates. □

Prediction (W.3). *If $q = 1$, acceptance for smooth exceeds steep:*

$$P(\mathcal{C}^{Sm}) \geq P(\mathcal{C}^S).$$

Proof. When $q = 1$, we have $\chi_t \equiv 1$ so there is no payment risk. The asset recursion under contract j becomes

$$A_{i,t+1} = A_{it} - c_i - m_{ij} + \omega_{jt},$$

so only the timing of payments matters. Relative to $\mathcal{C}^S$, the smooth contract delivers (weakly) more pay earlier given the same total. As in the argument above, this implies that assets under $\mathcal{C}^{Sm}$ are at least as high as under $\mathcal{C}^S$ at every date, so expected borrowing losses are lower under $\mathcal{C}^{Sm}$.

For any potential exit date $t < L$, the expected unpaid accrual satisfies $U_{\mathcal{C}^{Sm},t} \leq U_{\mathcal{C}^S,t}$ (because $F_{\mathcal{C}^{Sm},t} \leq F_{\mathcal{C}^S,t}$), so the expected loss from exiting at t is smaller under $\mathcal{C}^{Sm}$. Both effects increase $V_{i,\mathcal{C}^{Sm}}^{\text{accept}}$ relative to $V_{i,\mathcal{C}^S}^{\text{accept}}$, holding the outside option fixed. Therefore $P(\mathcal{C}^{Sm}) \geq P(\mathcal{C}^S)$. □

Prediction (W.4). *With $q = 1$, acceptance for daily exceeds smooth:*

$$P(\mathcal{C}^D) \geq P(\mathcal{C}^{Sm}).$$

Proof. When $q = 1$, the comparison between $\mathcal{C}^D$ and $\mathcal{C}^{Sm}$ is parallel to the comparison between smooth and steep. The daily schedule delivers (weakly) more pay earlier than smooth for the

same total, which implies that assets under $\mathcal{C}^D$ are at least as high as under $\mathcal{C}^{Sm}$ at every date. Expected borrowing losses are therefore lower under $\mathcal{C}^D$.

For any $t < L$, the expected unpaid accrual satisfies $U_{\mathcal{C}^D,t} \leq U_{\mathcal{C}^{Sm},t}$, so the expected loss from exiting at t is smaller under $\mathcal{C}^D$. The value of the option to exit is therefore higher under daily. Combining the lower expected borrowing losses and the lower expected loss from exiting yields

$$V_{i,\mathcal{C}^D}^{\text{accept}} \geq V_{i,\mathcal{C}^{Sm}}^{\text{accept}},$$

and hence $P(\mathcal{C}^D) \geq P(\mathcal{C}^{Sm})$. □

Prediction (W.5). *Higher-type workers accept less: if $\theta_1 > \theta_2$, then*

$$P(j \mid \theta_1) \leq P(j \mid \theta_2) \quad \text{for any fixed contract } j.$$

Proof. By Assumption (iii), the value of the outside option, $V_{i0}^t(A_{it})$, is increasing in type θ_i through the job-finding probability $p(\theta_i)$ and potential wages in the spot market. For a fixed contract j and fixed primitives, the value of accepting j , V_{ij}^{accept} , does not increase with θ because the contract terms and e_j^* are type-agnostic.

Acceptance occurs when

$$V_{ij}^{\text{accept}} \geq V_{i0}^0(A_{i0}).$$

The right-hand side is increasing in θ_i , while the left-hand side is not. Thus, as θ rises, the set of initial asset realizations A_{i0} for which the inequality holds shrinks. Consequently,

$$P(j \mid \theta_1) \leq P(j \mid \theta_2) \quad \text{whenever } \theta_1 > \theta_2.$$

□

LEMMA A.1 (Unpaid accrual ordering). *For any $t < L$ and any schedules $j \in \{\mathcal{C}^D, \mathcal{C}^{Sm}, \mathcal{C}^S\}$ with the same total $\sum_{s=1}^L \omega_{js}$, the expected unpaid accrual satisfies*

$$U_{\mathcal{C}^S,t} \geq U_{\mathcal{C}^{Sm},t} \geq U_{\mathcal{C}^D,t}.$$

Proof. By definition,

$$U_{j,t} = \sum_{s=t+1}^L \mathbb{E}[\chi_s \omega_{js}] = q \sum_{s=t+1}^L \omega_{js}.$$

The future sums satisfy

$$\sum_{s=t+1}^L \omega_{\mathcal{C}^D,s} \leq \sum_{s=t+1}^L \omega_{\mathcal{C}^{Sm},s} \leq \sum_{s=t+1}^L \omega_{\mathcal{C}^S,s}$$

because daily pays more earlier and steep pays more later, conditional on the same total. Multiplying by the common scalar q preserves the ordering of $U_{j,t}$. □

Prediction (WE.1). *In equilibrium, workers renege at a higher rate under daily than under steep, holding $(L, \tilde{\omega})$ fixed:*

$$r_{\mathcal{C}^D} \geq r_{\mathcal{C}^S}.$$

Proof. By the Lemma, for any $t < L$ and any realization of assets A_{it} ,

$$U_{\mathcal{C}^S, t} \geq U_{\mathcal{C}^{Sm}, t} \geq U_{\mathcal{C}^D, t}.$$

Assumption (iv) states that $\Delta_t(A, U)$ is weakly increasing in U , and that the worker continues at t under schedule j if and only if $\Delta_t(A_{it}, U_{j, t}) \geq 0$. Thus, for any fixed A_{it} and t ,

$$\Delta_t(A_{it}, U_{\mathcal{C}^S, t}) \geq \Delta_t(A_{it}, U_{\mathcal{C}^{Sm}, t}) \geq \Delta_t(A_{it}, U_{\mathcal{C}^D, t}).$$

It follows that the set of states in which the worker continues is largest under $\mathcal{C}^S$ and smallest under $\mathcal{C}^D$. Aggregating over types and states, the implied reneging hazards satisfy

$$r_{\mathcal{C}^D} \geq r_{\mathcal{C}^{Sm}} \geq r_{\mathcal{C}^S}.$$

□

A.4.2 Firm side predictions

Prediction (F.1). *Holding $(L, \tilde{\omega})$ fixed, the firm hiring probabilities satisfy*

$$P_f(\mathcal{C}^D) \leq P_f(\mathcal{C}^{Sm}) \leq P_f(\mathcal{C}^S).$$

Proof. Let Π^j denote firm value at $t = 0$ under schedule $j \in \{\mathcal{C}^D, \mathcal{C}^{Sm}, \mathcal{C}^S\}$, as defined by the firm's Bellman equation.

(i) *Per-payment administrative costs.* The expected number of payout events obeys

$$\sum_{t=1}^L \mathbb{1}\{\omega_{\mathcal{C}^D, t} > 0\} \geq \sum_{t=1}^L \mathbb{1}\{\omega_{\mathcal{C}^{Sm}, t} > 0\} \geq \sum_{t=1}^L \mathbb{1}\{\omega_{\mathcal{C}^S, t} > 0\}.$$

The per-payment cost is incurred only when $\kappa_{kjt} = 1$, so the total administrative cost under contract j is

$$\sum_{t=1}^L \kappa_{kjt} \lambda_{pf} \mathbb{1}\{\omega_{jt} > 0\}.$$

Assuming that κ_{kjt} is identically distributed across contracts and days and independent of the payment schedule conditional on $\omega_{jt} > 0$, the expected administrative cost under contract j is proportional to the number of payment dates:

$$\mathbb{E}\left[\sum_{t=1}^L \kappa_{kjt} \lambda_{pf} \mathbb{1}\{\omega_{jt} > 0\}\right] = \lambda_{pf} \mathbb{E}[\kappa_{kjt} \mid \omega_{jt} > 0] \sum_{t=1}^L \mathbb{1}\{\omega_{jt} > 0\}.$$

Hence expected exposure to per-payment administrative costs is still largest under daily and smallest under steep.

(ii) *Working-capital costs.* Earlier and more frequent payouts reduce B_{kt} in expectation, increasing the likelihood and magnitude of overdrafts $[B_{kt}]^-$. Given the cash recursion

$$B_{k,t+1} = B_{kt} - \omega_{jt} - \lambda_f [B_{kt}]^- + \xi_t \phi_t A_{kt}^f,$$

the expected overdraft costs $\lambda_f [B_{kt}]^-$ are (weakly) largest under daily and smallest under steep.

(iii) *Continuation and reneging.* By the Lemma and WE.1, expected unpaid accrual and continuation probabilities are (weakly) highest under steep and lowest under daily. Thus, worker reneging and the associated search/replacement losses (captured by λ_s and foregone continuation surplus in the Bellman) are (weakly) smallest under steep and largest under daily.

All three channels increase firm value as we move from daily to smooth to steep. Because the hiring decision is increasing in firm value, the firm hiring probabilities satisfy

$$P_f(\mathcal{C}^D) \leq P_f(\mathcal{C}^{Sm}) \leq P_f(\mathcal{C}^S).$$

□

Prediction (F.2). *Providing credit to firms (reducing λ_f) increases their probability of hiring workers.*

Proof. From the cash recursion,

$$B_{k,t+1} = B_{kt} - \omega_{jt} - \lambda_f [B_{kt}]^- + \xi_t \phi_t A_{kt}^f,$$

a lower λ_f reduces the size of the overdraft penalty for any given $B_{kt} < 0$, and therefore raises $B_{k,t+1}$ along every sample path. Higher cash levels relax liquidity constraints and increase the continuation value Π_j^t under every contract j , while the outside option Π_{0k}^t is unchanged.

Consequently, the set of initial states (B_{k0}, A_{k0}^f) for which

$$\Pi_j^0(B_{k0}, A_{k0}^f; e_j^k, r_j) \geq \Pi_{0k}^0$$

holds becomes larger. Aggregating over firms, $P_f(j)$ increases for each contract j . □

Prediction (F.3). *Providing compensation to firms when workers renege (for example, via a guarantor) increases their probability of hiring workers.*

Proof. In the firm's Bellman, the expected loss associated with worker reneging appears in the branch where a separation occurs:

$$\rho \mathbb{E}[r_j(\Pi_{0k}^{t+1} - \lambda_s)].$$

Providing compensation upon reneging either reduces the effective search/replacement cost λ_s or offsets part of the foregone continuation surplus. In both cases, the continuation value Π_j^t increases

for all contracts j , while the outside option Π_{0k}^t is unchanged. Therefore the region of (B_{k0}, A_{k0}^f) in which hiring is optimal expands, and $P_f(j)$ increases for each j . $\square$

Prediction. F.4: *Alleviating transaction costs of firms increases their probability of hiring workers.*

Proof. Consider a change that alleviates transaction costs by reducing the per-payment administrative cost from λ_{pf} to $\lambda'_{pf} < \lambda_{pf}$, holding all other parameters fixed. For any firm k , contract j , and day t , the one-period flow profit under the baseline and under the alleviated-transaction-cost regime are

$$f_{kjt} = y_t - \omega_{jt} - \kappa_{kjt} \lambda_{pf} \mathbb{1}\{\omega_{jt} > 0\},$$

and

$$f'_{kjt} = y_t - \omega_{jt} - \kappa_{kjt} \lambda'_{pf} \mathbb{1}\{\omega_{jt} > 0\},$$

respectively. Taking the difference,

$$f'_{kjt} - f_{kjt} = \kappa_{kjt} (\lambda_{pf} - \lambda'_{pf}) \mathbb{1}\{\omega_{jt} > 0\} \geq 0,$$

with strict inequality whenever there is a payment on day t and the transaction cost is realized (i.e. $\omega_{jt} > 0$ and $\kappa_{kjt} = 1$). Thus per-period flow profit is (weakly) higher under the alleviated-transaction-cost regime for every contract j and every realization of κ_{kjt} .

Let Π_j^t and $\Pi_j'^t$ denote the firm's continuation values at time t under the baseline and under the alleviated-transaction-cost regime, respectively, as given by the Bellman equation. At the terminal date, by construction of the problem, $\Pi_j^{L+1} = \Pi_j'^{L+1}$ for all contracts j . Working backwards in t , the Bellman recursion implies

$$\Pi_j'^t = f'_{kjt} + \rho \mathbb{E}[\text{continuation value under } \lambda'_{pf}] \geq f_{kjt} + \rho \mathbb{E}[\text{continuation value under } \lambda_{pf}] = \Pi_j^t,$$

where the inequality follows because $f'_{kjt} \geq f_{kjt}$ in every period and the same argument applies recursively to all future periods. Hence $\Pi_j'^t \geq \Pi_j^t$ for all t and all contracts j .

Assuming that the firm's outside option Π_{0k}^t is unaffected by the transaction-cost alleviation, the set of initial states (B_{k0}, A_{k0}^f) for which

$$\Pi_j'^0(B_{k0}, A_{k0}^f) \geq \Pi_{0k}^0(B_{k0}, A_{k0}^f)$$

holds is (weakly) larger than the corresponding set under the baseline with λ_{pf} . Aggregating over firms, this implies that the hiring probability $P_f(j)$ weakly increases for each contract j . Therefore, alleviating transaction costs of firms increases their probability of hiring workers. $\square$

Prediction (FE.1). *In equilibrium, optimal excess hours are (weakly) higher under steep than under daily:*

$$e_{CS}^* \geq e_{CD}^*.$$

Proof. For a given schedule j and day t , let

$$U_{j,t} \equiv \sum_{s>t} \mathbb{E}[\chi_s \omega_{js}]$$

denote the expected unpaid accrual faced by the worker. By the Lemma, $U_{CS,t} \geq U_{CD,t}$ for all t .

For the firm, consider the expected one-period objective as a function of the requested excess hours e and the worker's expected unpaid accrual U :

$$V(e; U) \equiv R(e) - C(e; U),$$

where $R(e) = \nu\hat{\theta}e$ captures gross revenue from extra hours and $C(e; U)$ summarizes the expected continuation-related loss induced by requesting e hours when the worker faces unpaid accrual U .

Assume:

- (i) For each U , $V(\cdot; U)$ is quasi-concave in e on $[0, \bar{H} - H]$ and has a nonempty compact argmax.
- (ii) V has increasing differences in (e, U) :

$$\frac{\partial^2 V}{\partial e \partial U}(e, U) \geq 0 \iff \frac{\partial^2 C}{\partial e \partial U}(e, U) \leq 0.$$

This means that when more pay is held for future dates (larger U), raising e today leads to a smaller marginal loss of expected continuation value.

Under (i)–(ii), Topkis's Monotonicity Theorem (Topkis, 1978)⁵⁵ implies that the set of maximizers

$$e^*(U) \in \arg \max_{e \in [0, \bar{H} - H]} V(e; U)$$

is nondecreasing in U . Since $U_{CS,t} \geq U_{CD,t}$ for all t , it follows that the optimal requested excess hours satisfy

$$e_{CS}^* \geq e_{CD}^*.$$

If the optimum is interior, the first-order condition is

$$\nu\hat{\theta} = \frac{\partial C}{\partial e}(e^*, U).$$

Because $\partial C / \partial e$ is weakly decreasing in U by (ii), a higher U shifts the right-hand side down; to restore equality, the optimal e^* must weakly increase. Corner solutions preserve the weak inequality. $\square$

⁵⁵Topkis's Monotonicity Theorem states that if an objective function $V(e; U)$ is supermodular (has increasing differences) in the decision variable e and parameter U , then the set of maximizers $e^*(U) \in \arg \max_e V(e; U)$ is nondecreasing in U . In this context, this implies that a higher expected unpaid accrual U leads the firm to choose weakly higher optimal excess hours.

B Additional Results

B.1 Contract Design

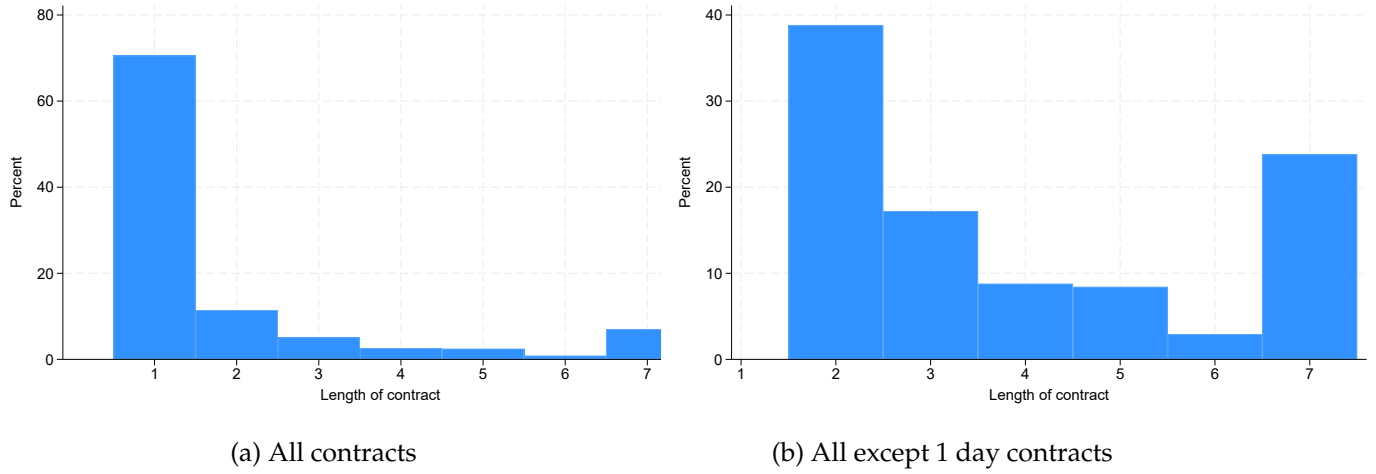
Contract length at labor stands. To understand the distribution of contract lengths in the spot labor market, we surveyed all workers arriving at two labor stands over a ten-day period in July–September 2023. Each day, we asked every worker at the stand whether they had found a job on the previous day, and if so, the wages earned and the total length of the job in days. This yielded 928 realized job spells.

Figure B-1a plots the distribution of contract lengths. Panel (a) shows that approximately 70% of jobs last a single day. However, many of these one-day jobs involve inherently short-term tasks—such as unloading bricks, breaking rocks, digging pits, or cleaning houses—that are completed within a few hours, or single-day tasks such as casting that must be finished in one session. For such jobs, multi-day contracts would not be appropriate, and back-loading of wages is not meaningful. The relevant question for our experiment is the distribution of contract lengths *among multi-day jobs*. Panel (b) shows this distribution. Among multi-day contracts, 3-day contracts account for approximately 18% and contracts of 7 or more days account for roughly 25%. These are the two contract lengths used in our experiment.

Payment structure used by firms. To document the payment structures that firms use for multi-day contracts, we surveyed a separate sample of construction firms in May 2023, before the main experiment. Each firm was asked: *If you were hiring a worker for a 7-day job that pays a total of 5,000 INR, what would you pay the worker on each day of the contract?* Firms first stated a daily wage—95% chose 400, 450, or 500 INR—and then described how they would distribute the total payment across the seven days.

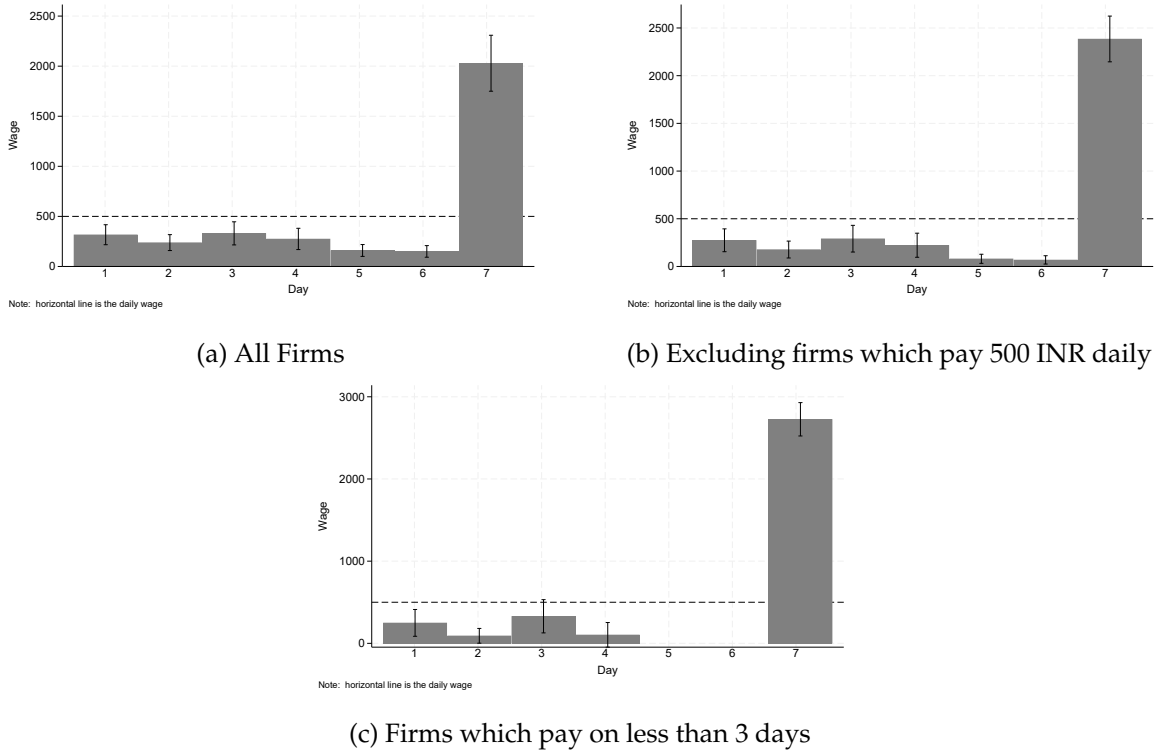
Figure B-2a plots the average daily payment across all surveyed firms, rescaled so that total payment equals 3,500 INR (i.e., 500 INR per day). Panel (a) averages over all firms, including the approximately 18% that pay a flat daily wage of 500 INR on each day. Because these flat-daily firms pull the average toward the daily wage line, Panel (a) understates the degree of back-loading practiced by the majority of firms. Panel (b) excludes flat-daily firms and shows a clearer pattern: firms pay substantially less than the daily wage in the early days of the contract and concentrate payments toward the end. Panel (c) further restricts to firms that make payments on fewer than three distinct days, representing the most steeply back-loaded structures. In this group, nearly all wages are paid on the final day. Overall, approximately 14% of firms reported paying the entire wage on the last day. These patterns confirm that back-loaded payment structures are common among firms for multi-day contracts, and that the steep and smooth contracts used in our experiment reflect payment schedules that firms actually employ.

Figure B-1: Length of contracts of workers at the labor stand



Note: The figure is based on data collected in July-September 2023. We surveyed all workers that arrived at 2 labor stands for 10 days. The survey question included whether they found a job on the day before, wages earned, length of the job. The figure shows the distribution of the number of days of the job accepted by workers. These jobs were offered by firms at the labor stand. The total number of jobs in part a) is 928. We pool all contracts of 7 or greater days into a 7 day contract.

Figure B-2: Payment structure of contracts offered by firms



Note: Firms were asked to tell us the total wages and the payment structure at which they usually hire workers for a job of 7 days. Firms first told us the total wages, their reasons for choosing that figure and then the structure. The reasons indicated that almost all firms first decided a daily wage and then multiplied it with 7 to give us the total wages. The figure shows the payment structure averaged over all firm choices for a 7 day contract. We rescale the wages offered by firms so that the total payment is INR 3500. The data was collected in May 2023.

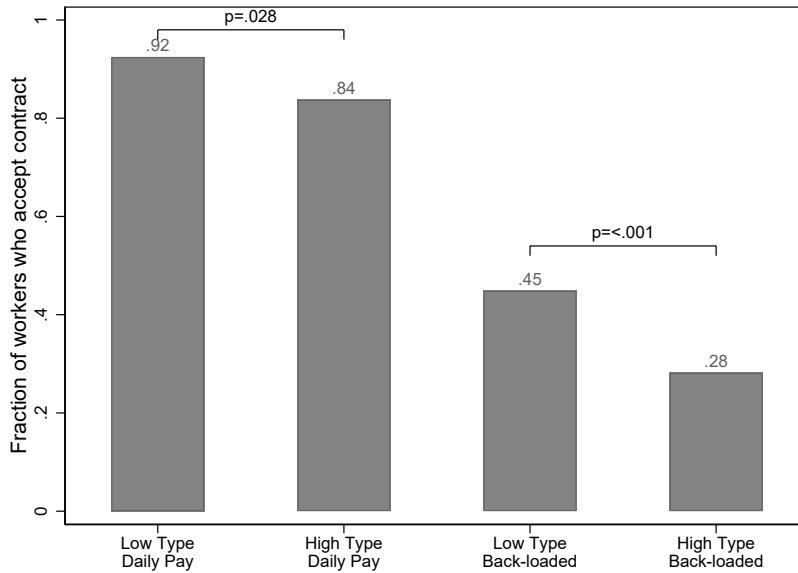
B.2 Worker side

B.2.1 Acceptance by worker type

We classify workers in the top decile of earnings over the four days prior to the survey as high type. This income-based measure was pre-registered.⁵⁶ High-type workers are 8 percentage points less likely to accept a daily-pay job and 17 percentage points less likely to accept a back-loaded job than workers below the 90th percentile, even though all of them are actively seeking work at the labor stand. The decline in take-up by type is more pronounced for back-loaded contracts than for daily-pay contracts (p-value = 0.09). One interpretation is that higher-income (and likely higher-productivity) workers face larger losses from potential non-payment or from foregone alternative opportunities under back-loaded contracts than do low-type workers.

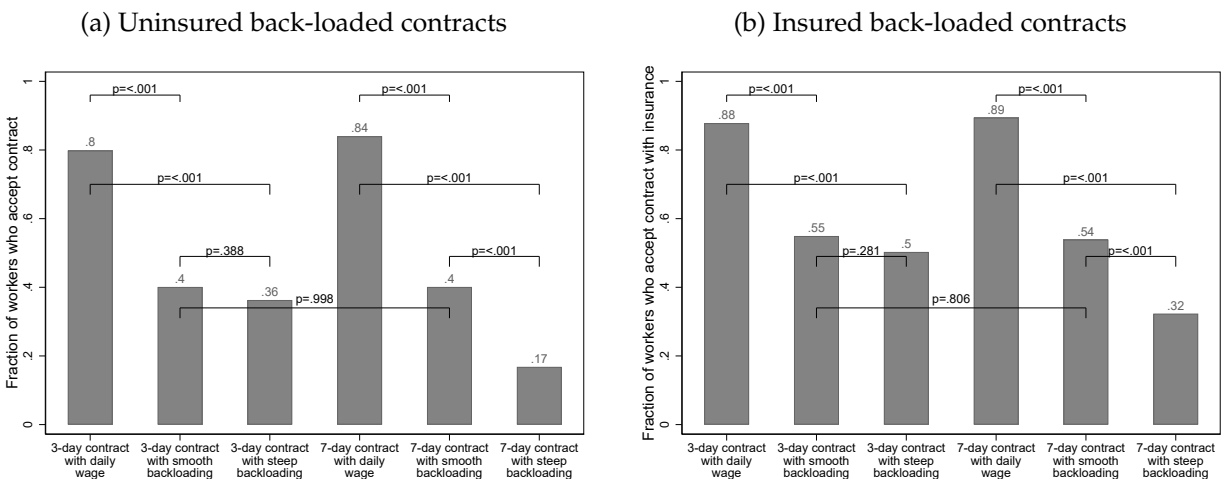
⁵⁶Alternate measures for type, such as including those in the top quartile, give similar patterns.

Figure B-3: Take-up of daily pay and back-loaded contracts by worker type



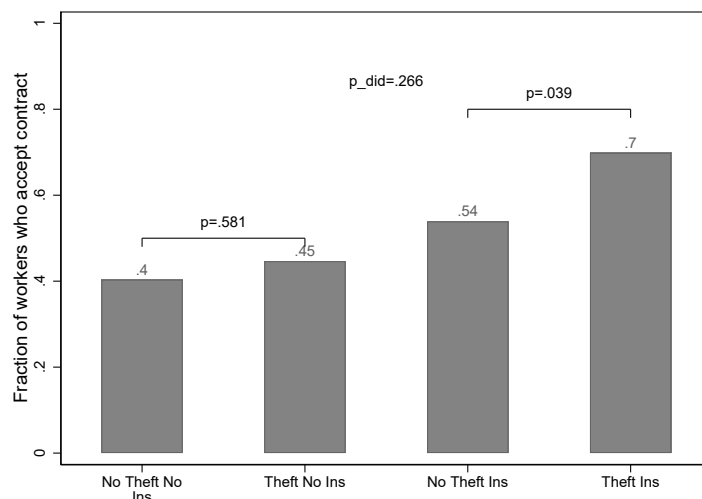
Notes: The figure shows results for take-up of daily pay and back-loaded contracts by worker type. We pool steep and smooth back-loaded contracts for ease of representation. Each bar is the sum of the control mean and the relevant regression coefficients. We use fixed effects for the labor stand, length of contract, whether the contract was insured and control for respondent age, education and half hour of survey. The p-value of difference in difference test between take-up of daily pay and back-loaded contracts for low and high type workers is 0.09. Each respondent (except the first 300) accounts for 2 observations (one for three day contract and one for seven day contract). Standard errors are clustered at the respondent level. p-values are shown for different pair of treatments for a two-sided test.

Figure B-4: Take-up of back-loaded contracts: uninsured vs. insured



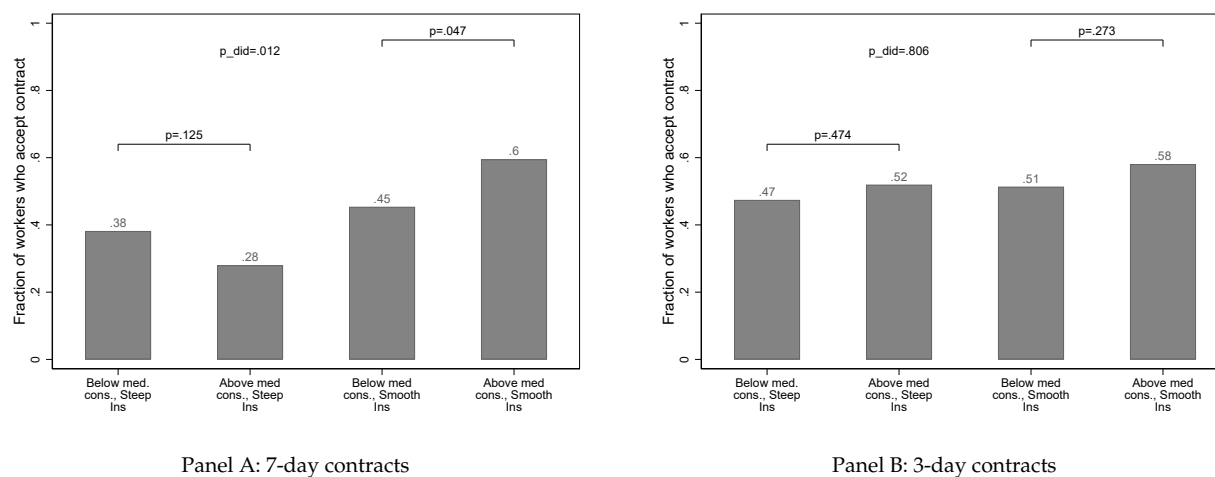
Notes: Panel (a) shows results for a regression of take up on treatment for uninsured contracts. Panel (b) shows results for a regression of take up on treatment for insured contracts. In both panels, each bar is the sum of the control mean and the relevant regression coefficients. We use fixed effects for the labor stand, and control for respondent age, education and half hour of survey. Each respondent (except the first 300) accounts for 2 observations (one for three day contract and one for seven day contract). Standard errors are clustered at the respondent level. p-values are shown for different pair of treatments for a two-sided test.

Figure B-5: Previous experience with Wage Theft and effect of insurance on smooth contracts



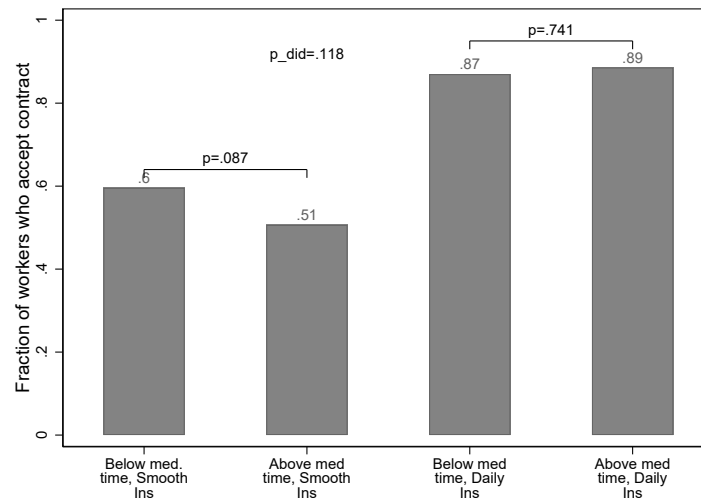
Notes: The figure shows results for take-up of smooth uninsured and insured contracts for workers who experienced (or did not) non-payment in the month before the survey. Each bar is the sum of the control mean and the relevant regression coefficients. We use fixed effects for the labor stand, length of the contract and control for respondent age, education and half hour of survey. Each respondent (except the first 300) accounts for 2 observations (one for three day contract and one for seven day contract). *No theft* captures the response of workers who did not experience non-payment in the month prior to the survey, and *Theft* captures response of workers who did. Standard errors are clustered at the respondent level. p-values are shown for different pair of treatments for a two-sided test.

Figure B-6: Daily consumption, liquidity constraints and effect of smooth vs steep



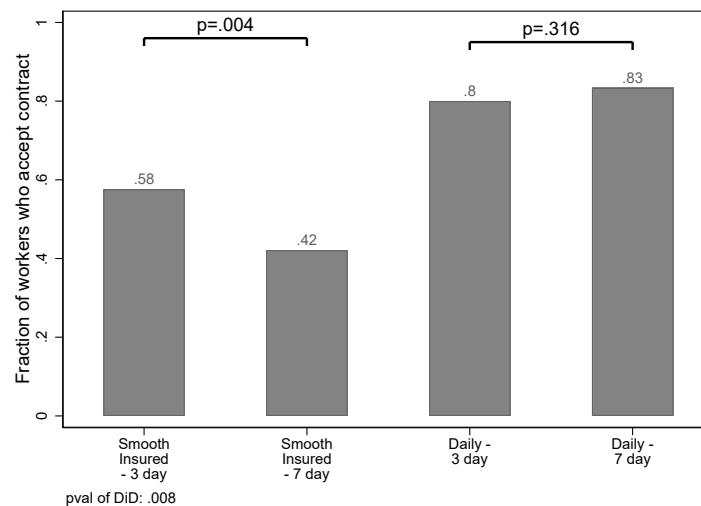
Notes: Panel A shows take-up of smooth insured and steep insured 7-day contracts for workers who are below or above the median of the daily consumption distribution. Panel B shows the corresponding results for 3-day contracts. Each bar is the sum of the control mean and the relevant regression coefficients. We use fixed effects for the labor stand, and control for respondent age, education, and half hour of survey time. Each respondent (except the first 300) contributes two observations (one for the 3-day contract and one for the 7-day contract). *Below med. cons.* captures the response of workers who are below the median in the daily consumption distribution, and *Above med. cons.* captures the response of workers who are above the median. Standard errors are clustered at the respondent level. p-values are shown for different pairs of treatments for a two-sided test.

Figure B-7: Lock-in costs and time to market



Notes: The figure shows results for take-up of insured smooth and insured daily-pay contracts for workers whose one-way travel time to the labor stand is above or below the sample median. Each bar is the sum of the control mean and the relevant regression coefficients. We use fixed effects for the labor stand and the length of the contract, and control for respondent age, education, and half hour of survey. Each respondent (except the first 300) accounts for 2 observations (one for the three-day contract and one for the seven-day contract). Standard errors are clustered at the respondent level. p -values are shown for different pairs of treatments for a two-sided test.

Figure B-8: Lock-in costs and length of contract



Notes: We restrict sample to workers who: i) were offered two jobs ii) both jobs were on smooth insured contract or both jobs were on a daily pay contract. The figure shows results for take-up of daily pay and smooth insured contracts. Each bar is the sum of the control mean and the relevant regression coefficients. We use fixed effects for the labor stand, and control for respondent age, education and half hour of survey. Standard errors are clustered at the respondent level. p -values are shown for different pair of treatments for a two-sided test.

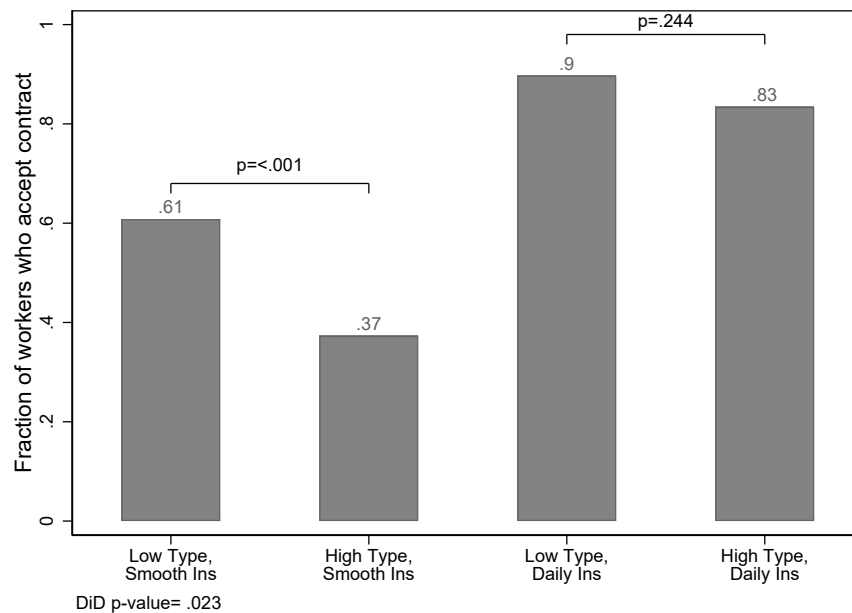
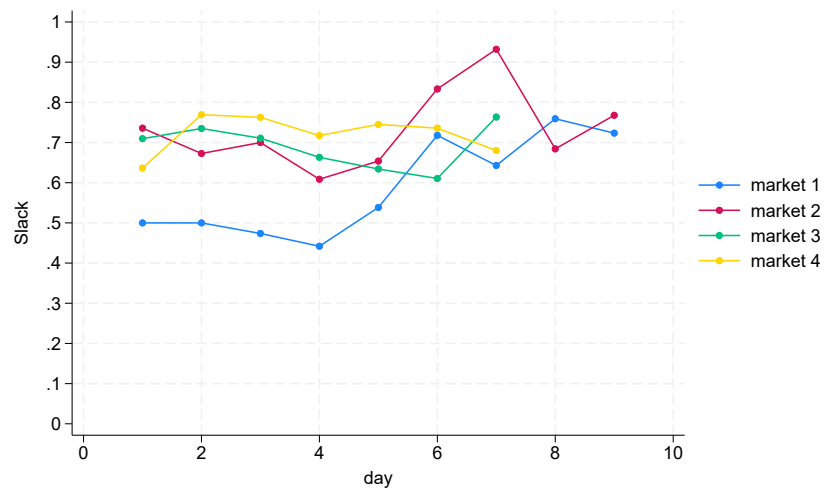


Figure B-9: Lock-in costs by worker type

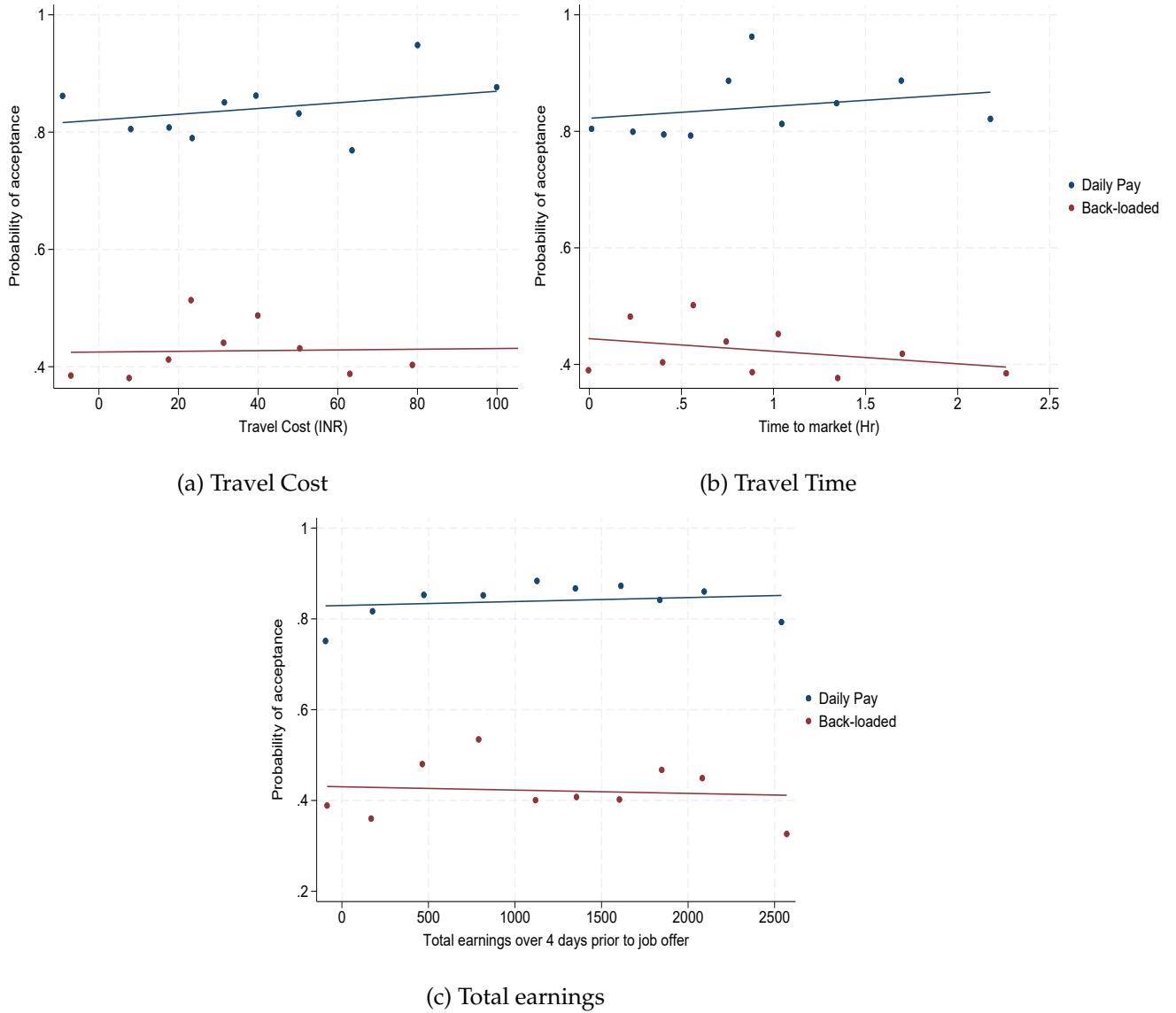
Notes: The figure shows results for take-up of smooth-insured and daily-insured contracts by worker type. High-type workers are defined as those in the top decile of earnings over the four days prior to the survey, following the pre-registered measure used in the main text. This measure captures workers with stronger outside options. Each bar is the sum of the control mean and the relevant regression coefficients. We use fixed effects for the labor stand and length of contract, and control for respondent age, education, and half hour of survey time. Each respondent (except the first 300) accounts for 2 observations (one for three-day contracts and one for seven-day contracts). Standard errors are clustered at the respondent level. p-values are shown for different pairs of treatments for a two-sided test. The p-value of the difference-in-differences test between take-up of daily-insured and smooth-insured contracts for high- and low-type workers is 0.023.

Figure B-10: Unemployment at the market level



Notes: The figure shows the unemployment at four spot markets in Patna. To plot the figure, we measure the total labor supply by collecting data for each worker who arrived at the stand for the period of our survey. We contacted each worker who visited the stand—either by phone or in-person on subsequent days—to track whether they found work. We collected data for markets 3 and 4 in the last week of July 2023, and for markets 1 and 2 in the last week of August and first week of September 2023.

Figure B-11: Correlations between accepting offers and worker characteristics



Notes: The figure shows the correlation between accepting daily pay and back-loaded contract with worker characteristics. We control for worker age, education and hour of survey and use fixed effects for whether contract was an insurance contract and labor stand.

Can Hyperbolic discounting explain the results Using the potential-earnings data for workers who rejected the offers, we compute the average outside option of rejectors. Fixing the long-run daily discount factor at $\delta = 0.9917$ (25% per-month benchmark offered by moneylenders) and assuming workers earn this amount each day, the implied present-bias parameters that would rationalize rejection are: (i) Smooth (3-day): $\beta = 0.038$; (ii) Smooth (7-day): $\beta = 0.017$; (iii) Steep (3-day): $\beta = 0.038$; (iv) Steep (7-day): $\beta = 0.018$. These β values are implausibly small relative to standard calibrations (e.g., $\beta \approx 0.7$), so present-bias/time preference alone is unlikely to explain rejection. Therefore, frictions arising from the payment structure are more likely to explain these preferences.

Using the same potential-earnings data, we instead fix a standard present-bias parameter at $\beta = 0.7$ and solve for the implied daily discount factors. The values that would rationalize rejection are: (i) Smooth (3-day): $\delta = 0.246$; (ii) Smooth (7-day): $\delta = 0.590$; (iii) Steep (3-day): $\delta = 0.468$; (iv) Steep (7-day): $\delta \approx 0.853$. Compared with the moneylender benchmark $\delta \approx 0.9917$, these implied δ values are far too low (i.e., require extreme impatience), again indicating that payment-structure frictions, rather than time preference, are the more plausible drivers.

B.2.2 Pairing experiment

The results in Section 4 demonstrate that non-payment, liquidity constraints, and contractual lock-in costs are significant frictions for workers. In response, workers may adopt strategies to mitigate these risks when they are hired by firms. We study one such strategy: a preference for working in pairs rather than alone. Previous work by [Donald and Grosset \(2024\)](#) has shown that workers may jointly want to supply labor to reduce non-pecuniary costs from commuting and social isolation. In our setting, working in a pair has several potential advantages, including the ability to put joint pressure on firms if they do not pay on time (or at all), as well as non-pecuniary benefits. In the words of one worker at the stand:

We prefer to work with others because having someone to talk to helps keep the mind at ease. Additionally, we can support each other if a problem arises. A lone worker can be easily pressured by the firm, but if the firm refuses to pay, he might be more hesitant to do so when dealing with two people rather than just one.

We implemented a 2×2 job offer experiment that varied (i) the payment schedule—either the 3-day steep back-loaded contract used in the main experiment or a 3-day daily-pay contract—and (ii) the hiring mode—pair (two workers hired together) versus alone. In the pair arm, workers were told they would be hired with one additional worker from the stand on the same contract, and that this second worker did not have to be someone they knew as we did not want to impose the cost of finding a friend on the worker. The remaining procedures mirrored the main experiment, and each offer had a 25% chance of being realized. Workers for whom the randomization yielded a realized offer were then matched with firms based on the contract they accepted.⁵⁷ The

⁵⁷This add-on experiment was conducted at the stands after the main experiment. Because it ran in July–August 2024, whereas the main experiment ran in April–June 2024, the control mean may be higher here.

key idea is that pairing can lower the expected costs of accepting a back-loaded (steep) offer—by improving informal enforcement, creating joint bargaining leverage if payment is delayed, and providing mutual monitoring and insurance—while offering comparatively fewer gains for daily-pay offers where payment risk is already low. Accordingly, the primary prediction is that pairing should raise take-up more for steep contracts than for daily-pay contracts and we should expect a positive interaction effect on acceptance.

Table B-1: Take-up of steep contracts in pair vs alone

	Take-up	
<i>Steep contract</i>	-0.508*** (0.041)	-0.510*** (0.045)
<i>Paired contract</i>	0.002 (0.017)	-0.000 (0.017)
<i>Steep × Paired</i>	0.042 (0.035)	0.050 (0.034)
Observations	506	506
Mean: Daily Single	0.952	0.952
Fixed effects	Stand	Stand + Day

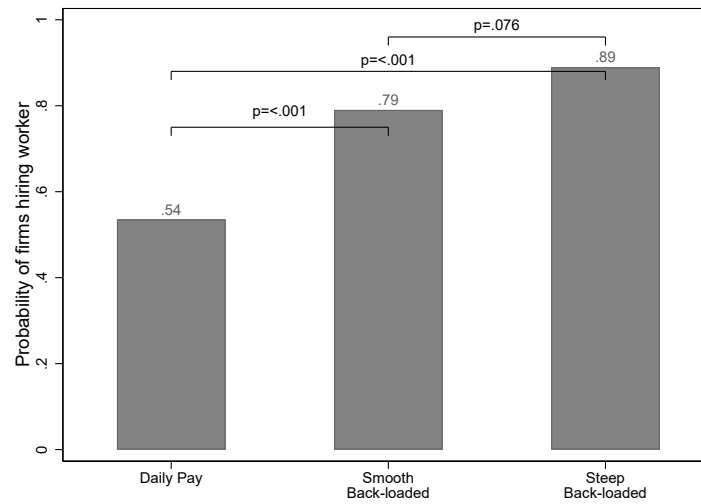
Notes: The dependent variable equals 1 if the worker accepts the offered contract. Sample is restricted to 3-day offers in the 2×2 pairing experiment. *Paired contract* indicates the worker is told they will be hired together with another worker from the stand on the same contract. Column 1 includes labor-stand fixed effects and column 2 includes labor-stand and survey-day fixed effects. *Mean: DailySingle* is the acceptance rate in the daily-pay, single-worker arm. We control for age, education and half hour of survey time. Standard errors are clustered at worker level and are shown in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Consistent with earlier results, Table B-1 shows that steep contracts substantially reduce take-up relative to daily-pay contracts. Offering the job in a pair has essentially no independent effect on take-up of daily-pay offers. The Steep × Pair interaction is positive and non-trivial in magnitude—about 4–5 p.p., or roughly 11% of the non-paired steep-contract take-up, but it is statistically indistinguishable from zero (p-value = 0.141), and the estimates are stable across specifications with and without labor-stand fixed effects.

B.3 Firm side

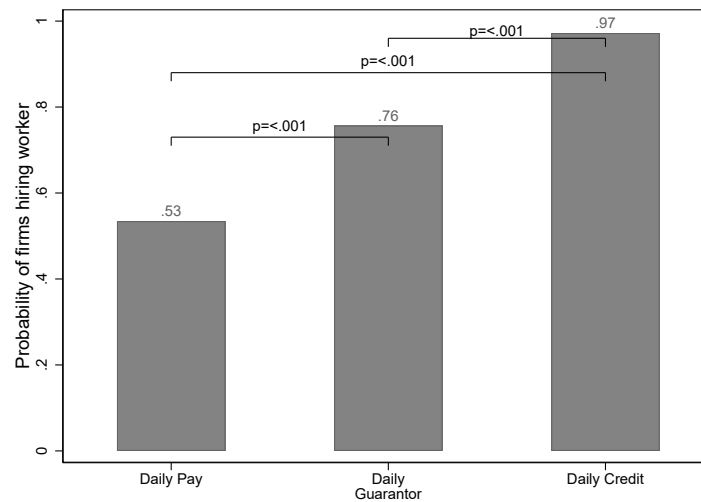
B.3.1 Results for firms which hired a worker in the matching stage

Figure B-12: Labor demand of firms which hired a worker in the matching stage



Notes: The figure shows results estimated from Eq: 5.1. Each bar is the sum of the control mean and the relevant regression coefficients. The total sample consists of 83 firms which ended up hiring at least one worker. We control for education of the owner, number of sites the firm is operating and size of the firm. We use fixed effects for order of the question, whether the respondent (the owner of the firm) is a former mason and for contract length. Standard errors are clustered at the firm level. p-values are shown for different pairs for a two-sided test.

Figure B-13: Labor demand of credit and guarantor contracts of firms which hired a worker in the matching stage



Notes: The figure shows results estimated from Eq: 5.1. Each bar is the sum of the control mean and the relevant regression coefficients. The total sample consists of 83 firms. We control for education of the owner, number of sites the firm is operating and size of the firm. We use fixed effects for order of the question, whether the respondent (the owner of the firm) is a former mason and for contract length. Standard errors are clustered at the firm level. p-values are shown for different pairs for a two-sided test.

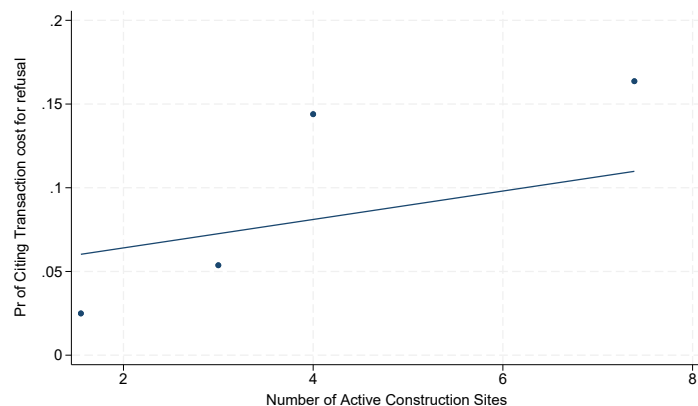
B.3.2 Transaction costs of firms

Table B-2: Take-up of steep vs. credit contracts to bound transaction costs

	Take up for 3 day contracts	Take up for 3 day contracts
<i>Steep contract</i>	-0.154*** (0.051)	-0.098*** (0.036)
Frequency > 1 week		-0.065 (0.050)
Steep × Frequency > 1 week		-0.056 (0.065)
Observations	104	317
Mean for Credit contract	0.942	
Mean for Credit × Frequency ≤ 1 week		0.941

The dependent variable is an indicator for whether the firm accepted the offer. First column reports results from a regression of take-up on a steep-contract dummy, restricted to daily credit and steep offers for firms which are paid weekly by their employers. Second column reports results from a regression of take-up on a steep-contract dummy, interacted with indicator for frequency of payment by employer and restricted to daily credit and steep offers for firms. Standard errors clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure B-14: Pr. of rejecting contracts due to transaction costs vs active number of construction sites.



Notes: This figure displays a binned scatterplot analyzing the relationship between firm's size and incidence of transaction costs. The x-axis represents the number of active construction sites currently managed by the firm and the y-axis represents the proportion of firms that explicitly cited the difficulty of "going from site to site" as a reason for their decision to reject specific contract offers. The sample is restricted to the sample of rejected offers. The slope of the line is 0.008 and the p-value is 0.001.

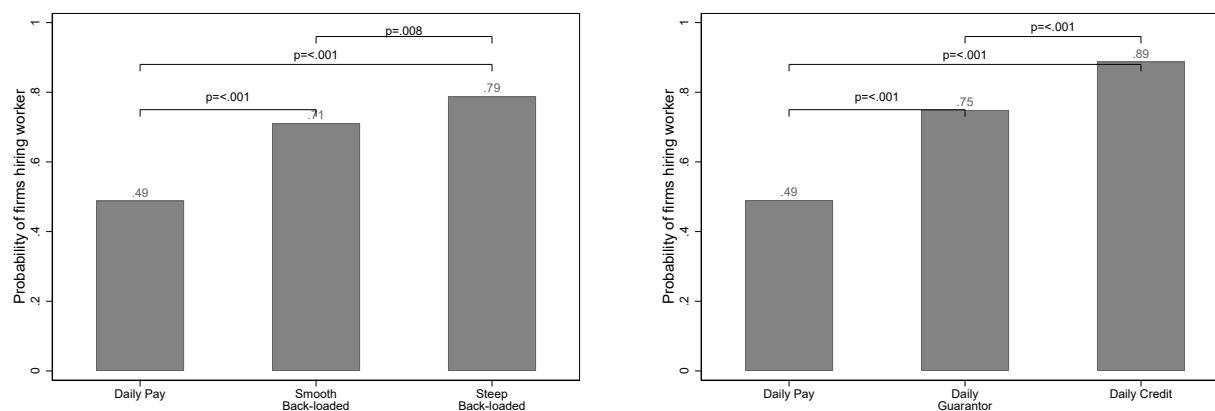
B.3.3 Contract length and Firm Demand

Table B-3: Firm labor demand and length of contracts

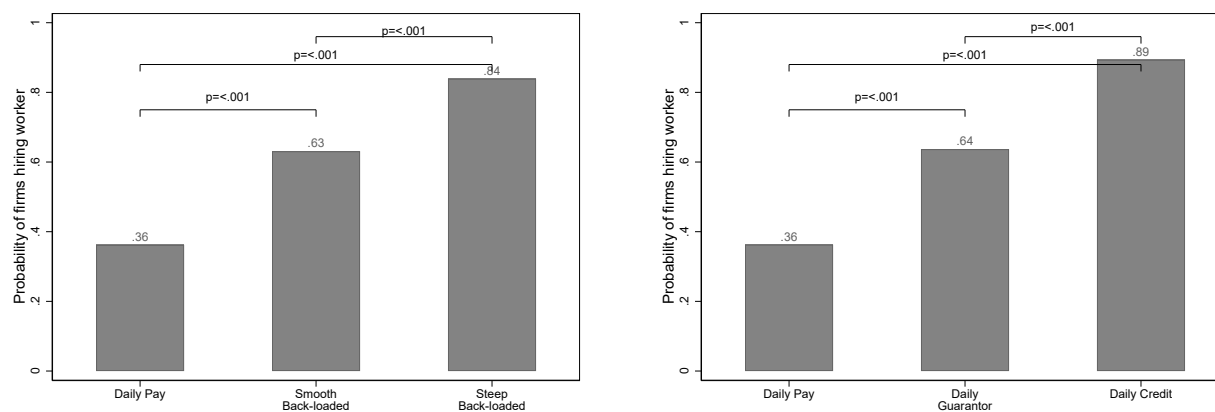
	Daily	Credit	Guarantor	Steep	Smooth
<i>Long contract</i>	-0.128*** (0.020)	0.005 (0.012)	-0.112*** (0.019)	0.050** (0.022)	-0.078*** (0.024)
Observations	690	679	678	673	402
Mean - 3 day	0.49	0.89	0.75	0.79	0.71
Fixed Effect	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is an indicator for whether the firm accepted the offer. Each column reports results from a regression of take-up on whether the contract is a long (7 day) contract while restricting the sample to the contract in the column heading. We control for education of the owner, number of sites the firm is operating and size of the firm. We use fixed effects for order of the question, and whether the respondent (the owner of the firm) is a former mason. Standard errors are clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure B-15: Labor demand of firms by length of contract



Take-up for 3-day contracts



Take-up for 7-day contracts

Notes: The dependent variable is an indicator for whether the firm accepted the offer. The figure shows results estimated from Eq: 5.1. Each bar is the sum of the control mean and the relevant regression coefficients. The total sample consists of 349 firms. We control for education of the owner, number of sites the firm is operating and size of the firm. We use fixed effects for order of the question, whether the respondent (the owner of the firm) is a former mason. Standard errors are clustered at the firm level. p-values are shown for different pairs for a two-sided test.

Table B-4: Difference in difference on Contract type and length

	Take Up of contract		
<i>Long contract</i>	-0.106*** (0.022)	-0.128*** (0.020)	-0.075*** (0.023)
<i>Smooth</i>	0.226*** (0.037)		
<i>Guarantor</i>		0.253*** (0.025)	
<i>Long × Smooth</i>	0.030 (0.029)		
<i>Long × Guarantor</i>		0.015 (0.021)	
<i>Steep</i>			0.083*** (0.027)
<i>Long × Steep</i>			0.123*** (0.030)
Observations	799	1,368	1,073
Mean- 3 day Daily	0.48	0.49	
Mean- 3 day Smooth			0.71
sample_s	Smooth + Daily	Guarantor + Daily	Smooth + Steep

Notes: The dependent variable is an indicator for whether the firm accepted the offer. Each column reports results from a regression of take-up on the interaction of the contract length on contract type. We use fixed effects for the firm. Standard errors are clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

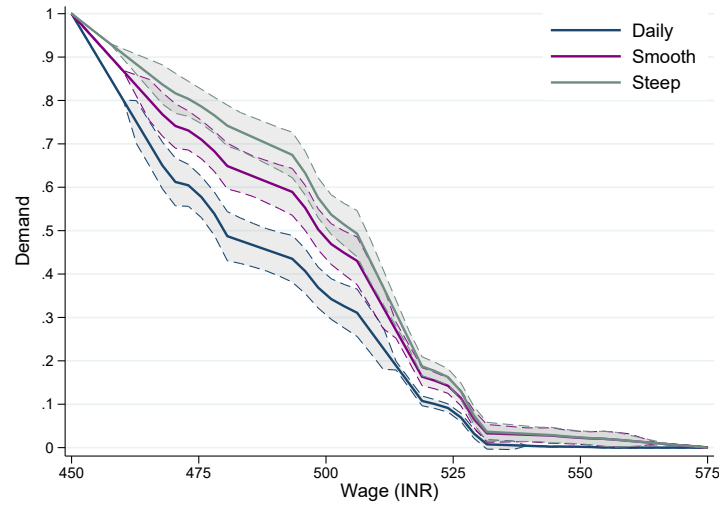
B.3.4 Firm Demand for contracts using Multiple price list

After completing the main experiment, we administered a structured multiple price list (MPL) module to 100 firms (from our main sample) to map hiring decisions over a posted wage schedule. The module, which was implemented by enumerators and was not incentivized, presented wages from INR 450 to INR 650 in INR 25 increments (450, 475, ... , 650). Starting at INR 450, firms were asked at each successive wage whether they would hire at that wage. Enumerators enforced monotonicity in real time: the sequence proceeded upward until the first “not hire,” at which point the module stopped and no higher wages were asked for that contract. If a firm indicated “hire” at all listed wages, the questions continued until the top of the grid. The resulting data provide a simple, stated measure of how hiring decisions vary with posted wages, which we use descriptively to complement the revealed preference evidence from the main experiment.

We use the MPL curves to convert stated acceptance into an *inverse demand* for each contract and then read off the wage premium that holds acceptance fixed. Let $w_s(p)$ denote the wage at which the share of firms willing to hire equals p under contract $s \in \{\text{Daily, Smooth, Steep}\}$. The willingness-to-pay (WTP) premium of contract s over Daily at acceptance level p is

$$\Delta_s(p) \equiv w_s(p) - w_{\text{Daily}}(p).$$

Figure B-16: Labor Demand of a sub-sample of firms estimated using Multiple Price List



Notes: The Figure plots the labor demand curve for a sub-sample of 100 firms. The underlying data was collected using a multiple price list method at intervals of 25 INR from 450 to 650. The experiment was not incentivized. For each contract arm, we first residualize demand on firm covariates: education of firm owner, size of the firm, number of sites that the firm is operating, and use fixed effects for former mason and contract length. We then smooth adjusted demand over wages using local-linear kernel regression (Epanechnikov; a bandwidth of 30) and plot conditional means with 95% pointwise confidence intervals.

Figure B-16 shows that demand for contract varies over INR 450 to INR 525. The Steep curve lies to the right of Daily and Smooth, and Smooth lies to the right of Daily. At any point p on the curve, the maximum wage premium for smooth contract is around INR 20, and for Steep contract is INR 30.

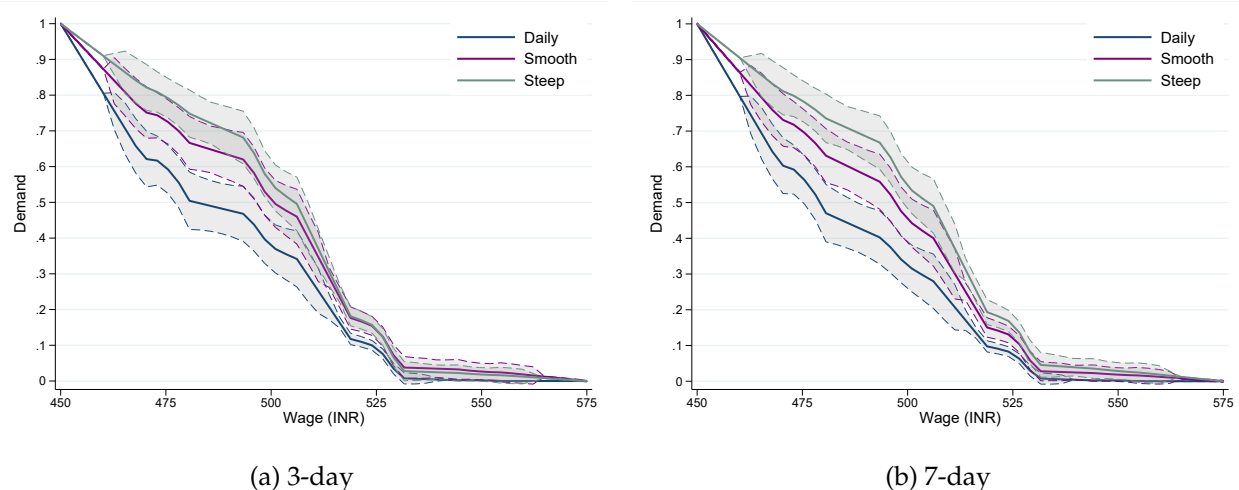


Figure B-17: Demand curves by contract length.

Notes: The Figure plots the labor demand curve for a sub-sample of 100 firms. The underlying data was collected using a multiple price list method at intervals of 25 INR from 450 to 650. The experiment was not incentivized. For each contract arm, we first residualize demand on firm covariates: education of firm owner, size of the firm, number of sites that the firm is operating, and use fixed effects for former mason. We then smooth adjusted demand over wages using local-linear kernel regression (Epanechnikov; a bandwidth of 30) and plot conditional means with 95% pointwise confidence intervals.

B.3.5 Hypothetical matching rates under different contracts

In this part of the paper, we conduct a straightforward exercise. We use the results from our worker and firm-side experiments to determine the matches each contract can generate in equilibrium. One caveat to keep in mind before we proceed is that the firms were made offers at a daily wage which was 50 INR below the worker's offer. Therefore, their demand at the wage offered to workers is going to be lower.

To calculate the matching rates, we take the minimum over the acceptance rate of a contract on both sides of the market. Table B-5 shows the results from this exercise. Within our three base contracts, the smooth insurance contract has the highest matching rate. If we include *credit* and *guarantor* contracts then daily pay credit contract has the highest matching rate.

Table B-5: Hypothetical matching rates under different contracts, by contract length

Contract	Worker acceptance (%)		Firm acceptance (%)		Max. possible matches (%)	
	3-day	7-day	3-day	7-day	3-day	7-day
<i>No Insurance</i>						
Daily Pay	80	84	49	36	49	36
Smooth	40	40	71	63	40	40
Steep	36	17	79	84	36	17
<i>Insurance</i>						
Daily Pay	88	89	49	36	49	36
Smooth (insured)	55	54	71	63	55	54
Steep (insured)	50	32	79	84	50	32
Credit to Firms (daily)	80	84	89	89	80	84
Guarantor for Firms (daily)	80	84	75	64	75	64

Notes: For each contract-length cell, the “Max. possible matches” entry equals the minimum of worker and firm acceptance rates. As in the main text, firm offers were made at a daily wage 50 INR below the worker offers, so firm demand at the worker wage would be weakly lower.

B.4 Matching

Table B-6: Reasons for non-allotment of jobs

	(1)	
	Not allotted work b	pct
Went out of state	8	8.25
Went home, will come after a long time	9	9.28
Found work for a long-term	34	35.05
Refused contract: Wages are low	1	1.03
Do not want to work on steep contract	5	5.15
Do not want to work with us	17	17.53
Do not want to work as daily wage labor	11	11.34
Did not like work	1	1.03
Not working as a labor anymore	7	7.22
Worker unwell	4	4.12
Total	97	100.00

Notes: The table shows the break down of reasons for which workers who were contacted for job were not allotted work. These reasons were reported by workers either in-person or on phone.

Table B-7: Reasons for Non-fulfillment of contracts

	(1)	
	Contract not fulfilled	
	b	pct
Do not know	1	0.60
other reason	8	4.76
Stayed at home	14	8.33
Did not want to go	15	8.93
allotted a lot of work	25	14.88
Got work at other place	16	9.52
Fired from work by the Contractor	3	1.79
Site is far away	11	6.55
Family emergency/Unwell	19	11.31
Labor could not be contacted	18	10.71
Contractor was making the labor do illegal work	2	1.19
Contractor asked to come early to work	1	0.60
Contractor asked to stay late for work	17	10.12
The labor ran away in between the work	8	4.76
Contractor did not pay	2	1.19
Contractor did not have work	8	4.76
Total	168	100.00

Notes: The table shows the break down of reasons for which workers who started working did not fulfill the contract. Reasons were collected by surveying workers in-person or on phone.

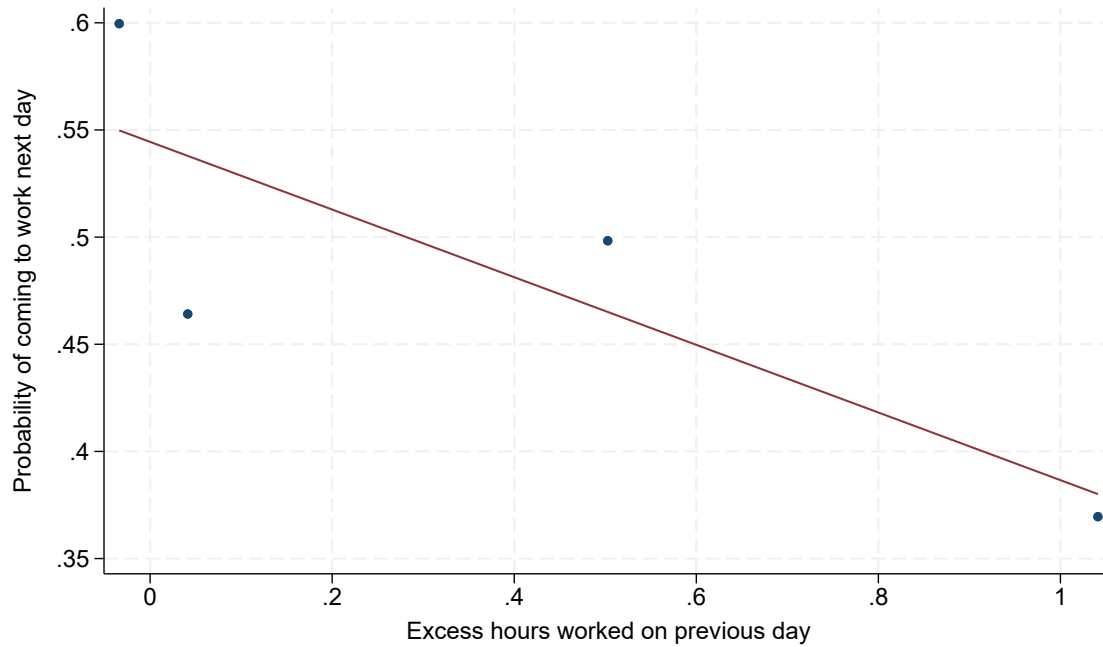


Figure B-18: Excess hours worked and worker probability of renegeing

Notes: The figure shows the correlation between number of excess hours (above 8) worked on day 1 of a *Daily Pay* contract and turning up for work on the next day. We control for travel cost and use fixed effects for whether contract was an insurance contract. The p-value for the corresponding regression of excess hours on turning up on the next day is 0.054.

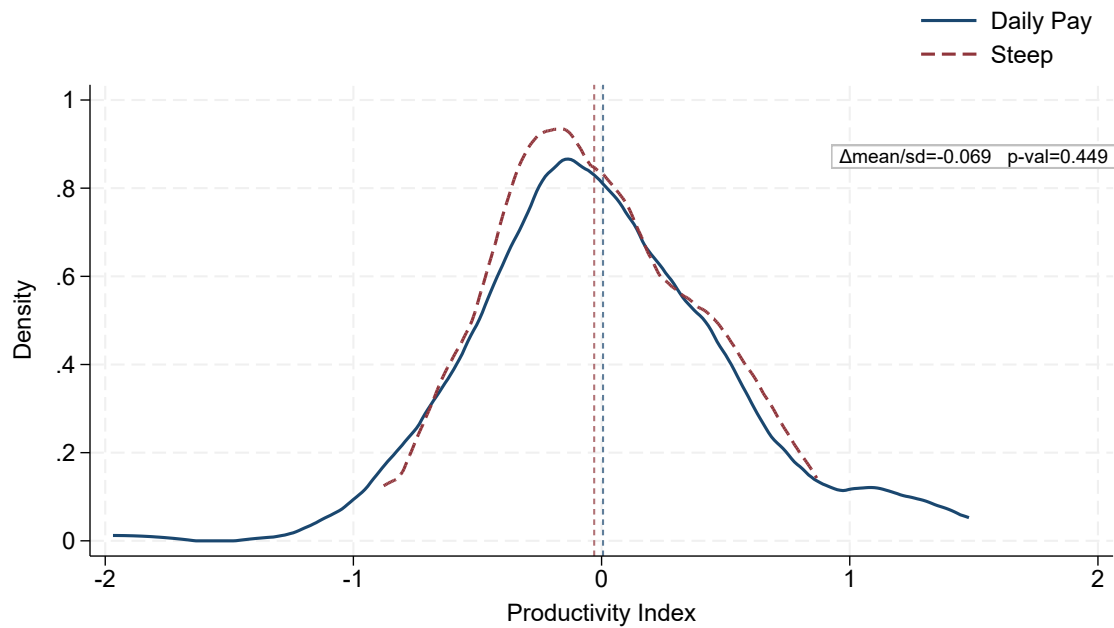


Figure B-19: Productivity Index of workers across contracts

Notes: The productivity index is based on detailed task-level data collected by enumerators who observed workers for one day, recording all activities and their intensity separately for two hours at construction sites. Task measures (e.g., bricks passed, bowls of plaster carried) were adjusted for distance traveled and bowl size, with negative productivity indicators (e.g., idle time, broken bricks) inverted. Standardized measures were then combined using principal components analysis (PCA), with the first component used as the index.

C Randomization Balance

Table C-1: Balance

	Means				Pairwise differences P-values			
	Insured	Daily payment	Low type	Smooth contract	Insured - Uninsured	Daily - back-loaded	Smooth vs Steep	High vs Low
Age	32.26	32.26	32.22	32.29	0.99	0.99	0.84	0.69
Years of education	4.09	4.17	4.28	4.26	0.27	0.91	0.34	0.04
Backward caste	0.70	0.68	0.69	0.69	0.55	0.32	0.32	0.05
Scheduled caste/tribe	0.25	0.26	0.26	0.26	0.85	0.47	0.30	0.02
Not paid at least once (in last month)	0.12	0.10	0.11	0.11	0.90	0.23	0.20	0.23
Daily consumption cost	191.13	191.08	187.64	187.63	0.26	0.52	0.50	0.02
Total days present	2.45	2.43	2.39	2.34	0.20	0.67	0.80	0.02
Total earnings (INR)	1172.27	1166.66	1168.19	1143.07	0.81	0.74	0.54	0
Loan due	1.66	1.66	1.66	1.65	0.76	0.92	0.78	0.06
Already have a job	0.11	0.11	0.12	0.12	0.09	0.56	0.31	0.32
Time of Survey	4.64	4.63	4.60	4.62	0.97	0.77	0.19	0.13
Time to market (hrs)	0.91	0.98	0.90	0.93	0.99	0.01	0.49	0

Notes: This table assesses balance in baseline worker characteristics across randomly assigned contract attributes. Columns under *Means* report average values for workers assigned to: (i) insured contracts, (ii) daily-payment contracts, (iii) low-type workers, and (iv) smooth back-loaded contracts. Columns under *Pairwise differences P-values* report two-sided p-values from tests of equality of means between: insured vs. uninsured, daily-payment vs. back-loaded (smooth or steep), smooth vs. steep (within back-loaded contracts), and high-type vs. low-type workers. “Not paid at least once (in last month)” is a self-reported indicator of prior non-payment. Monetary variables are in INR and time-to-market is measured in hours. All variables are measured in the survey conducted at the time of job offers.

Table C-2: Balance along main treatment cells of worker side experiment

	Pairwise differences p-values									
	UD - U SM	UD - U ST	U SM - U ST	ISM - IST	ISM - UD	IST - UD	IST - U SM	ISM - U SM	ISM - U ST	IST - U ST
Age	0.59	0.48	0.84	0.88	0.84	0.73	0.82	0.71	0.58	0.68
Years of education	0.37	0.55	0.82	0.27	0.14	0.63	0.65	0.50	0.41	0.86
Backward caste	0.89	0.32	0.23	0.76	0.99	0.78	0.65	0.88	0.29	0.44
Scheduled caste/tribe	0.90	0.29	0.21	0.81	0.69	0.54	0.42	0.58	0.46	0.61
Not paid at least once (in last month)	0.92	0.34	0.26	0.48	0.89	0.42	0.33	0.80	0.38	0.82
Daily consumption cost	0.74	0.62	0.40	0.80	0.35	0.48	0.70	0.52	0.16	0.23
Total days present	0.82	0.30	0.38	0.53	0.16	0.41	0.52	0.21	0.82	0.75
Total earnings (INR)	0.51	0.76	0.74	0.65	0.84	0.83	0.35	0.62	0.90	0.59
Loan due	0.89	0.67	0.55	0.37	0.79	0.57	0.64	0.66	0.85	0.31
Already have a job	0.38	0.52	0.84	0.33	0.77	0.23	0.03	0.20	0.32	0.06
Time of Survey	0.78	0.53	0.37	0.35	0.74	0.57	0.38	0.96	0.33	0.91
Time to market (hrs)	0.24	0.40	0.78	0.25	0.42	0.05	0.43	0.70	0.93	0.31

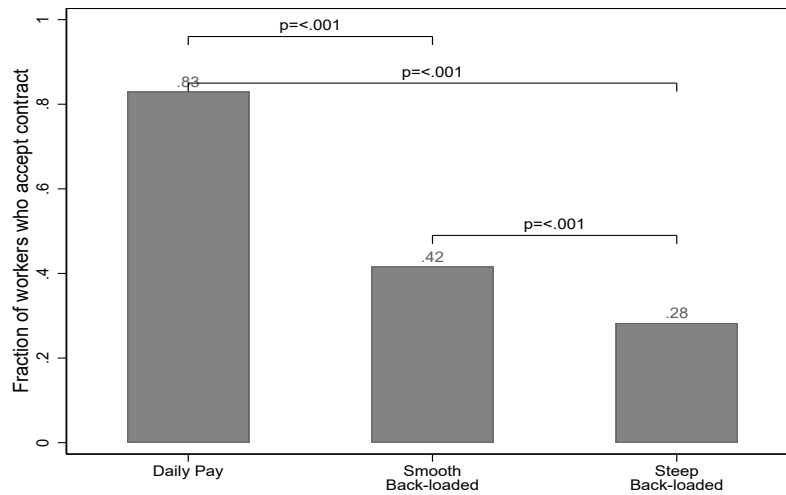
	ID - UD	ID - U SM	ID - U ST	ID - ISM	ID - IST
Age	0.21	0.44	0.95	0.31	0.44
Years of education	0.21	0.48	0.59	0.85	0.26
Backward caste	0.50	0.71	0.22	0.50	0.42
Scheduled caste/tribe	0.98	0.65	0.61	0.88	0.87
Not paid at least once (in last month)	0.43	0.53	0.10	0.34	0.24
Daily consumption cost	0.95	0.52	0.74	0.39	0.48
Total days present	0.20	0.40	0.76	0.88	0.97
Total earnings (INR)	0.31	0.98	0.57	0.61	0.29
Loan due	0.93	0.95	0.38	0.56	0.89
Already have a job	0.26	0.99	0.96	0.20	0.07
Time of Survey	0.98	0.69	0.51	0.77	0.68
Time to market (hrs)	0.23	0.07	0.09	0.13	0.02

Notes: This table shows p-values for worker characteristics in different treatment cells. We combine 3 and 7 days into a single treatment and show the p-values for a pairwise comparison of these cells. Overall we have 15 pairs of treatment cells. Notation is as follows: *I* - Insurance contracts, *U* - Uninsured contracts, *SM* - Smooth contracts, *ST* - Steep contracts, *D* - Daily contracts.

D Robustness

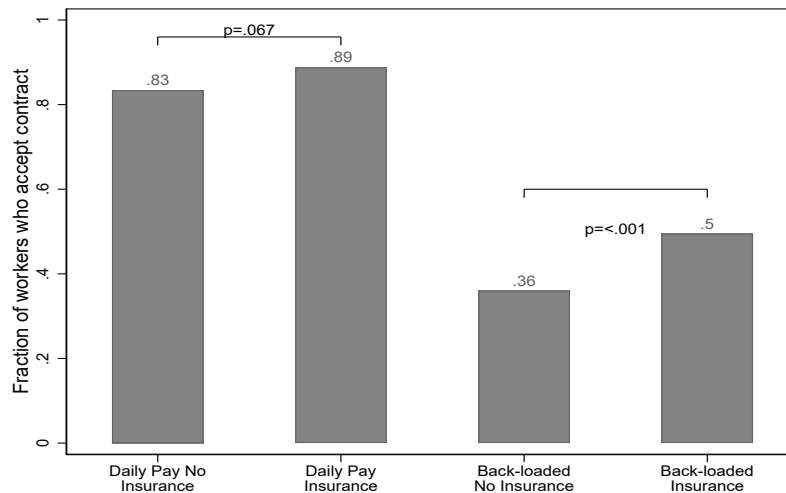
D.1 Worker Side - Controlling for covariate selected using post-double selection Lasso

Figure D-1: Take up of uninsured daily pay vs back-loaded contracts - Lasso controls



Notes: The figure shows results from regression 4.1 for uninsured contracts. Each bar is the sum of the control mean and the relevant regression coefficients. We use fixed effects for the labor stand and length of contract, and control for variable selected by post-double selection Lasso method (Belloni et al. (2014)). Each respondent (except the first 300) accounts for 2 observations (one for three day contract and one for seven day contract). Standard errors are clustered at the respondent level. p-values are shown for different pair of treatments for a two-sided test.

Figure D-2: Effect of insurance on take-up of daily pay and back-loaded contracts - Lasso controls



Notes: The figure shows results from regression 4.1 for comparison between insured and uninsured contracts. Each bar is the sum of the control mean and the relevant regression coefficients. We pool steep and smooth back-loaded contracts for ease of representation. We use fixed effects for the labor stand and length of contract, and control for variable selected by post-double selection Lasso method (Belloni et al. (2014)). The p-value of difference in difference test between take-up of daily pay and back-loaded contracts for uninsured vs insured contracts is 0.033. Each respondent (except the first 300) accounts for 2 observations (one for three day contract and one for seven day contract). Standard errors are clustered at the respondent level. p-values are shown for different pair of treatments for a two-sided test.

Table D-1: Wage Theft concerns, Liquidity constraints and Lock-in costs

	Wage Theft Concerns			Liquidity constraints			Lock-in costs		
<i>Smooth insured vs uninsured</i>	0.144*** (0.033)	0.163*** (0.044)	0.121** (0.049)						
<i>Smooth vs Steep (Insured)</i>				0.115*** (0.033)	0.039 (0.043)	0.216*** (0.047)			
<i>Daily insured vs Smooth insured</i>							0.336*** (0.034)	0.323*** (0.045)	0.349*** (0.046)
Observations	928	522	401	923	518	404	692	386	306
Control group mean	0.40	0.39	0.41	0.42	0.50	0.32	0.55	0.55	0.54
Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Complete 3 days 7 days			Complete 3 days 7 days			Complete 3 days 7 days		

The dependent variable is an indicator for whether the worker accepted the contract offered. The first row reports coefficients from regressions of the dependent variable on the insurance indicator for the sample of smooth contracts, estimated separately by day and for the pooled sample. The second row reports coefficients from regressions of the dependent variable on the smooth-contract indicator for the sample of steep and smooth insured contracts, again estimated separately by day and for the pooled sample. The third row reports coefficients from regressions of the dependent variable on the daily-pay contract indicator for the sample of daily-pay and smooth insured contracts, estimated separately by day and for the pooled sample. We use fixed effects for labor stand, and control for variable selected by post-double selection Lasso method (Belloni et al. (2014)). Each respondent accounts for 2 observations, except for the first 300 observations. Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.2 Worker Side - Controlling for additional covariates

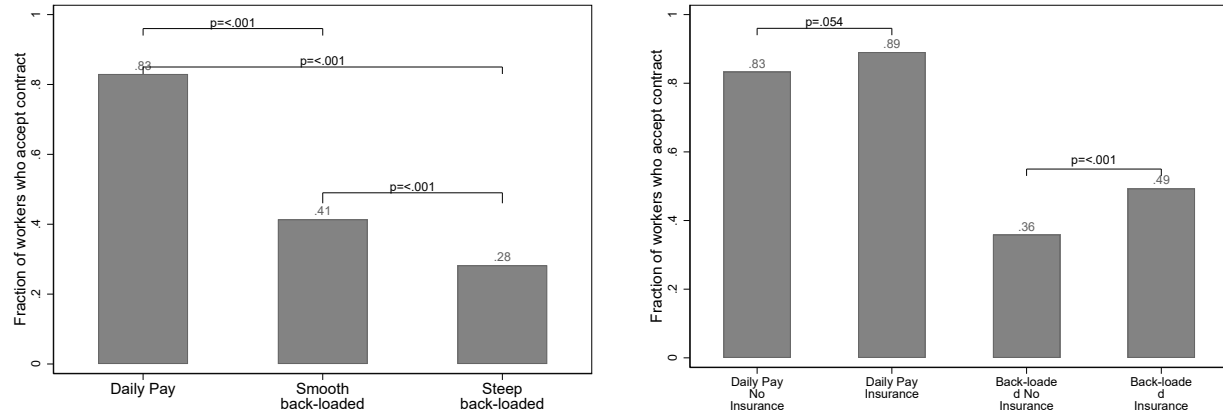
Table D-2: Wage Theft concerns, Liquidity constraints and Lock-in costs

	Wage Theft Concerns			Liquidity constraints			Lock-in costs		
<i>Smooth insured vs uninsured</i>	0.144*** (0.033)	0.159*** (0.044)	0.119** (0.050)						
<i>Smooth vs Steep (Insured)</i>				0.118*** (0.033)	0.048 (0.044)	0.218*** (0.047)			
<i>Daily insured vs Smooth insured</i>							0.341*** (0.034)	0.327*** (0.045)	0.359*** (0.049)
Observations	924	519	400	919	516	402	689	384	305
Control group mean	0.40	0.39	0.41	0.42	0.50	0.32	0.55	0.55	0.54
Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Complete 3 days 7 days			Complete 3 days 7 days			Complete 3 days 7 days		

The dependent variable is an indicator for whether the worker accepted the contract offered. The first row reports coefficients from regressions of the dependent variable on the insurance indicator for the sample of smooth contracts, estimated separately by day and for the pooled sample. The second row reports coefficients from regressions of the dependent variable on the smooth-contract indicator for the sample of steep and smooth insured contracts, again estimated separately by day and for the pooled sample. The third row reports coefficients from regressions of the dependent variable on the daily-pay contract indicator for the sample of daily-pay and smooth insured contracts, estimated separately by day and for the pooled sample. We use fixed effects for labor stand and length of the contract, and control for offered wage; age, marital status, caste and education of the respondent; hour of survey, whether worker has a loan due, travel cost to market, no of years worked as a labor and whether they are from outside the district. Each respondent accounts for 2 observations, except for the first 300 observations. Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure D-3: Worker take-up: uninsured contracts and the effect of insurance

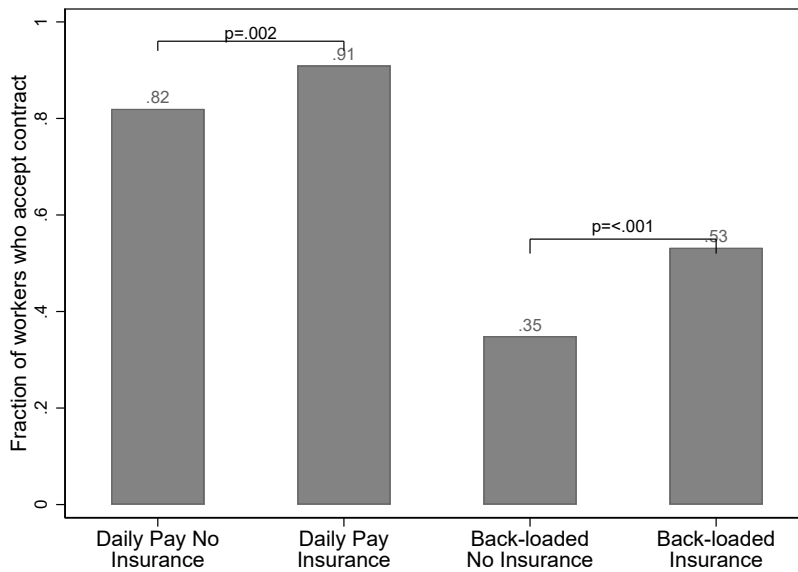
- (a) Take up of uninsured daily pay vs back-loaded contracts (b) Effect of insurance on take-up of daily pay and back-loaded contracts



Notes: Both panels show results from regression 4.1. In each panel, each bar is the sum of the control mean and the relevant regression coefficients. Specifications include fixed effects for labor stand and contract length, and control for offered wage; respondent age, marital status, caste, and education; hour of survey; whether the worker has a loan due; travel cost to market; years worked as a laborer; and whether the worker is from outside the district. Each respondent (except the first 300) accounts for 2 observations (one 3-day and one 7-day contract). Standard errors are clustered at the respondent level. P-values shown correspond to two-sided tests for the indicated pairwise treatment comparisons. Panel (a) reports results for uninsured contracts only. Panel (b) compares insured and uninsured contracts. For ease of representation, steep and smooth back-loaded contracts are pooled. The p-value for the difference-in-differences test comparing the daily-pay vs back-loaded take-up gap between uninsured and insured contracts is 0.043.

D.2.1 Estimates had the workers trusted the insurance arm completely

Figure D-4: Effect of insurance on labor supply - Accounting for mistrust in survey firm



Notes: The figure shows results from regression 4.1 for comparison between insured and uninsured contracts. Each bar is the sum of the control mean and the relevant regression coefficients. We pool steep and smooth back-loaded contracts for ease of representation. We use fixed effects for the labor stand and length of contract, and control for respondent age, education and half hour of survey. We assume that workers who deny taking up an insurance contract because they do not trust us would have taken up the job if they had trusted us and assign them a take-up of 1. The p-value of difference in difference test between take-up of daily pay and back-loaded contracts for uninsured vs insured contracts is 0.01. Each respondent (except the first 300) accounts for 2 observations (one for three day contract and one for seven day contract). Standard errors are clustered at the respondent level. p-values are shown for different pair of treatments for a two-sided test.

Table D-3: Wage theft concerns, liquidity constraints and Lock-in costs - Accounting for mistrust in survey firm

	Wage Theft Concerns			Liquidity constraints			Lock-in costs		
<i>Smooth insured vs uninsured</i>	0.193*** (0.032)	0.215*** (0.043)	0.173*** (0.048)						
<i>Smooth vs Steep (Insured)</i>				0.109*** (0.033)	0.028 (0.043)	0.217*** (0.047)			
<i>Daily insured vs Smooth insured</i>							0.325*** (0.032)	0.305*** (0.043)	0.350*** (0.046)
Observations	973	549	420	964	540	424	717	400	316
Control group mean	0.40	0.39	0.41	0.48	0.57	0.36	0.59	0.60	0.58
Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Complete 3 days 7 days			Complete3 days 7 days			Complete 3 days 7 days		

The dependent variable is an indicator for whether the worker accepted the contract offered. The first row reports coefficients from regressions of the dependent variable on the insurance indicator for the sample of smooth contracts, estimated separately by day and for the pooled sample. The second row reports coefficients from regressions of the dependent variable on the smooth-contract indicator for the sample of steep and smooth insured contracts, again estimated separately by day and for the pooled sample. The third row reports coefficients from regressions of the dependent variable on the daily-pay contract indicator for the sample of daily-pay and smooth insured contracts, estimated separately by day and for the pooled sample. We use fixed effects for labor stand, and control for respondent age, education, and half hour of survey time. We assume that workers who deny taking up an insurance contract because they do not trust us would have taken up the job if they had trusted us and assign them a take-up of 1. Each respondent accounts for 2 observations, except for the first 300 observations. Standard errors are clustered at the individual level. * p < 0.10, ** p < 0.05, *** p < 0.01.

D.2.2 Robustness to including surveyor fixed effects

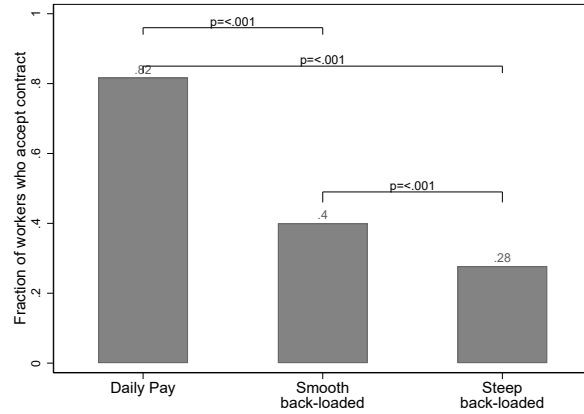
Table D-4: Wage theft concerns, liquidity constraints and Lock-in costs - Surveyor fixed effects

	Wage Theft Concerns			Liquidity constraints			Lock-in costs		
<i>Smooth insured vs uninsured</i>	0.156*** (0.033)	0.169*** (0.043)	0.139*** (0.049)						
<i>Smooth vs Steep (Insured)</i>				0.125*** (0.033)	0.050 (0.043)	0.225*** (0.046)			
<i>Daily insured vs Smooth insured</i>							0.336*** (0.036)	0.329*** (0.047)	0.341*** (0.051)
Observations	973	549	420	964	540	424	645	328	316
Control group mean	0.40	0.39	0.41	0.42	0.50	0.32	0.55	0.55	0.54
Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Complete 3 days 7 days			Complete3 days 7 days			Complete 3 days 7 days		

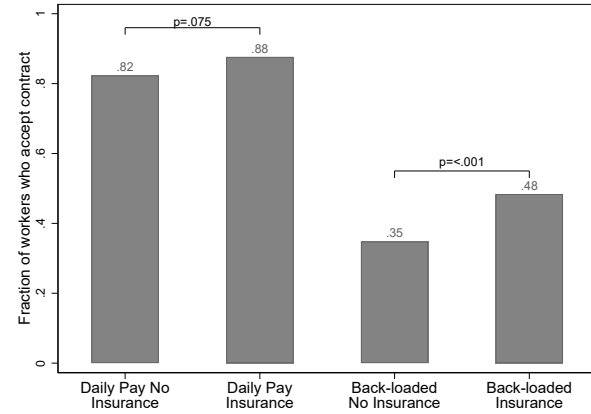
The dependent variable *take-up* is a binary variable which captures whether the worker was willing to accept the contract offered. We use fixed effects for labor stand, surveyor, and control for respondent age, education, and half hour of survey time. Each respondent accounts for 2 observations, except for the first 300 observations. Standard errors are clustered at the individual level. * p < 0.10, ** p < 0.05, *** p < 0.01.

Figure D-5: Worker take-up with surveyor fixed effects: uninsured contracts and the effect of insurance

(a) Take up of uninsured daily pay vs back-loaded contracts



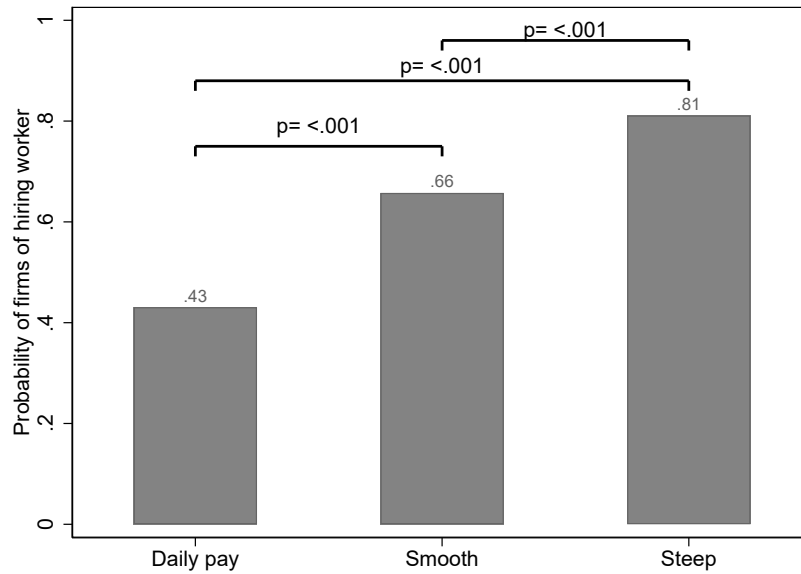
(b) Effect of insurance on take-up of daily pay and back-loaded contracts



Notes: Both panels report results from specifications that include fixed effects for labor stand, surveyor, and contract length, and control for respondent age, education, and half-hour of survey time. Each respondent (except the first 300) contributes two observations (one 3-day and one 7-day contract). Standard errors are clustered at the respondent level. P-values shown correspond to two-sided tests for the indicated pairwise treatment comparisons. Panel (a) compares take-up of uninsured daily-payment and back-loaded contracts. Panel (b) compares insured and uninsured contracts; for ease of representation, steep and smooth back-loaded contracts are pooled. The p-value for the difference-in-differences test comparing the daily-pay vs back-loaded take-up gap between uninsured and insured contracts is 0.017.

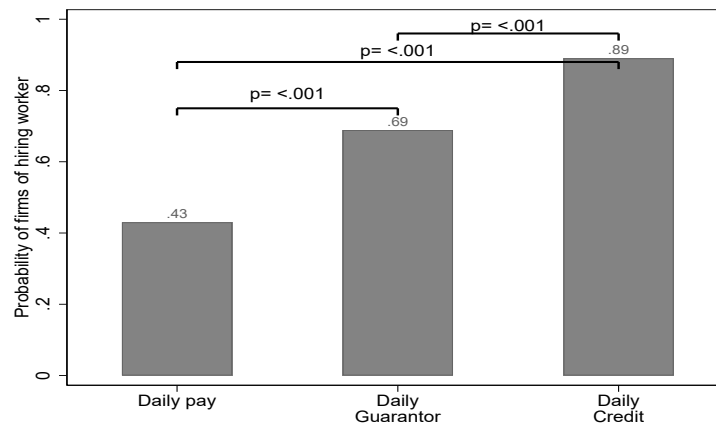
D.3 Firm Side - covariate selected using post-double selection Lasso

Figure D-6: Probability of hiring and steepness of back-loading



Notes: The figure shows results estimated from Eq: 5.1. Each bar is the sum of the control mean and the relevant regression coefficients. The total sample consists of 349 firms. We control for variables selected by post-double selection Lasso method (Belloni et al. (2014)). We use fixed effects for order of the question, whether the respondent (the owner of the firm) is a former mason and for contract length. Standard errors are clustered at the firm level. p-values are shown for different pairs for a two-sided test.

Figure D-7: Probability of hiring on daily pay vs credit vs guarantor



Notes: The figure shows results estimated from Eq: 5.1. Each bar is the sum of the control mean and the relevant regression coefficients. The total sample consists of 349 firms. We control for variables selected by post-double selection Lasso method (Belloni et al. (2014)). We use fixed effects for order of the question, whether the respondent (the owner of the firm) is a former mason and for contract length. Standard errors are clustered at the firm level. p-values are shown for different pairs for a two-sided test.

Table D-5: Summary stats of firms surveyed for hiring delays

	Mean: Surveyed	Mean: Not Surveyed	Difference	p-value
Age	41.463	40.892	0.571	0.603
Years of education	7.326	6.856	0.470	0.336
Backward caste	0.734	0.692	0.042	0.409
Scheduled caste/tribe	0.162	0.233	-0.072	0.102
Years working as a contractor	14.087	15.333	-1.246	0.226
Worked as a mason before	0.825	0.883	-0.058	0.156
Firm is registered	0.127	0.102	0.025	0.501
Hire most laborers from stand	0.721	0.807	-0.086	0.135
No of workers in the firm	11.048	9.850	1.198	0.291
No of operational construction sites	3.157	3.367	-0.209	0.428
Firm has a manager	0.132	0.050	0.082	0.032
Worker did not turn up (in last two months)	0.807	0.756	0.050	0.517
Instances of worker not turning up (in last two months)	5.036	5.417	-0.380	0.549
Instances of Employer reneging on contract	1.042	1.210	-0.168	0.400
Amount not paid by employer	47123.244	45183.618	1939.626	0.752
Costs borne if worker does not turn up (INR)	91.667	103.056	-11.389	0.278
Income over last month	116724.891	145025.000	-28300.109	0.304
Pct. Employers do not pay on time	27.895	31.571	-3.677	0.403

Notes: The table shows summary statistics for firms who were asked about delays in hiring and compares it to summary statistics of firms which were part of the experiment but were not asked these questions. Column 3 captures the difference in means and Column 4 shows the p-value of that difference.

D.4 Matching - covariate selected using post-double selection Lasso

Table D-6: Outcomes for workers under steep vs daily contracts

Panel A: Steep vs Daily

	Contract completed	Contract completed (excluding firm rejections)	Hr worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.206** (0.092)	0.242** (0.093)	0.263** (0.104)	0.184*** (0.069)	-0.607 (0.424)
Observations	240	215	239	240	231
Control group mean	0.34	0.26	8.19	0.09	6.97
Fixed Effect	Yes	Yes	Yes	Yes	Yes

Panel B: Steep vs Daily (Workers in the quasi-experiment sub-sample)

	Contract completed	Contract completed (excluding firm rejections)	Hr worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.218** (0.102)	0.225** (0.104)	0.193 (0.122)	0.200** (0.081)	-0.717 (0.479)
Observations	121	112	120	121	117
Control group mean	0.33	0.28	8.30	0.07	7.05
Fixed Effect	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is measured by surveying workers and firms which were matched with each other. *Hours Worked* is winsorized at the 1st and 99th percentile. We use fixed effects for whether the contract was insured, length of contract, daily wage offered and control for variables selected by post-double selection Lasso method (Belloni et al. (2014)). Standard errors are clustered at the firm level. * p < 0.10, ** p < 0.05, *** p < 0.01.

D.5 Matching - controlling for additional covariates

Table D-7: Outcomes for workers under steep vs daily contracts - controlling for additional covariates

Panel A: Steep vs Daily					
	Contract completed	Contract completed (excluding firm rejections)	Hr worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.198** (0.093)	0.235** (0.095)	0.260** (0.112)	0.176*** (0.066)	-0.555 (0.408)
Observations	238	213	237	238	230
Control group mean	0.34	0.26	8.19	0.09	6.97
Fixed Effect	Yes	Yes	Yes	Yes	Yes
Panel B: Steep vs Daily (Workers in the quasi-experiment sub-sample)					
	Contract completed	Contract completed (excluding firm rejections)	Hr worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.210** (0.104)	0.218** (0.106)	0.186 (0.128)	0.194** (0.075)	-0.715 (0.458)
Observations	121	112	120	121	117
Control group mean	0.33	0.28	8.30	0.07	7.05
Fixed Effect	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is measured by surveying workers and firms which were matched with each other. *Hours Worked* is winsorized at the 1st and 99th percentile. We use fixed effects for whether the contract was insured, length of contract, daily wage offered and control for religion, age and caste of worker, time taken to come to the labor stand, and whether worker belongs to district other than Patna. Standard errors are clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D-8: Outcomes for workers under steep vs daily contracts with p-values adjusted for multiple testing

	Contract completed	Contract completed (excluding firm rejections)	Hr worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.235*** (0.085)	0.266*** (0.090)	0.274** (0.105)	0.146** (0.064)	-0.509 (0.365)
Observations	260	233	259	260	250
Control group mean	0.34	0.26	8.19	0.09	6.97
Fixed Effect	Yes	Yes	Yes	Yes	Yes
p-val Bonferroni-Holm	.028		.033	.052	.17
p-val Romano-Wolf	.004		.004	.01	.074

Notes: The dependent variable is measured by surveying workers and firms which were matched with each other. *Hours Worked* is winsorized at the 1st and 99th percentile. We use fixed effects for whether the contract was insured, length of contract, daily wage offered and control for the time it takes the worker to get to the labor stand. Adjusted p-values are computed using the Bonferroni-Holm method (Holm (1979)) and Romano-Wolf (step-down) method (Romano and Wolf (2005), Romano and Wolf (2016)) to control the Family Wise Error Rate. We apply these adjustments to the four outcome variables for which we jointly test differences. Standard errors are clustered at the firm level. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table D-9: Outcomes for workers under steep vs daily contracts after controlling for worker rating

Panel A: Steep vs Daily

	Contract completed	Contract completed (excluding firm rejections)	Hr worked	Excess Work (beyond 8 hrs)
<i>Steep contract</i>	0.222** (0.084)	0.287*** (0.081)	0.189** (0.086)	0.144** (0.068)
Work quality rating	0.031* (0.016)	0.067*** (0.015)	-0.002 (0.022)	0.012 (0.013)
Observations	249	225	249	249
Control group mean	0.34	0.26	8.19	0.09
Fixed Effect	Yes	Yes	Yes	Yes

Panel B: Steep vs Daily (Workers in the quasi-experiment sub-sample)

	Contract completed	Contract completed (excluding firm rejections)	Hr worked	Excess Work (beyond 8 hrs)
<i>Steep contract</i>	0.235** (0.097)	0.277*** (0.096)	0.153 (0.117)	0.196** (0.086)
Work quality rating	0.056** (0.022)	0.084*** (0.021)	-0.003 (0.023)	0.003 (0.015)
Observations	124	114	124	124
Control group mean	0.33	0.28	8.30	0.07
Fixed Effect	Yes	Yes	Yes	Yes

Notes: The dependent variable is measured by surveying workers and firms which were matched with each other. *Hours Worked* is winsorized at the 1st and 99th percentile. We use fixed effects for whether the contract was insured, length of contract, daily wage offered and month. We control for average rating (on a scale of 1 (bad) - 10 (good)) over the contract period. This rating is given by the firm for the worker's work quality at the end of each day. Standard errors are clustered at the firm level. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table D-10: Outcomes for workers under steep vs daily contracts after controlling for Guarantor contract

Panel A: Steep vs Daily

	Contract completed	Contract completed (excluding firm rejections)	Hr worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.217** (0.089)	0.257*** (0.092)	0.277*** (0.093)	0.168** (0.064)	-0.506 (0.377)
Guarantor contract	-0.123 (0.088)	-0.049 (0.083)	-0.043 (0.223)	0.118 (0.074)	0.032 (0.358)
Observations	260	233	259	260	250
Control group mean	0.34	0.26	8.19	0.09	6.97
Fixed Effect	Yes	Yes	Yes	Yes	Yes

Panel B: Steep vs Daily (Workers in the quasi-experiment sub-sample)

	Contract completed	Contract completed (excluding firm rejections)	Hr worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.218** (0.102)	0.226** (0.104)	0.181 (0.132)	0.189** (0.081)	-0.682 (0.472)
Guarantor contract	-0.106 (0.134)	-0.055 (0.128)	-0.023 (0.196)	-0.062 (0.063)	0.931* (0.544)
Observations	128	117	127	128	124
Control group mean	0.33	0.28	8.30	0.07	7.05
Fixed Effect	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is measured by surveying workers and firms which were matched with each other. *Hours Worked* is winsorized at the 1st and 99th percentile. We use fixed effects for whether the contract was insured, length of contract, daily wage offered and month. We control for whether the contract offered to the firm is a guarantor contract or not. Standard errors are clustered at the firm level. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table D-11: Outcomes for workers under steep vs daily contracts for 3-day contracts

Steep vs Daily (Workers in the quasi-experiment sub-sample)

	Contract completed	Contract completed (excluding firm rejections)	Hr worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.251** (0.112)	0.257** (0.117)	0.173 (0.155)	0.208** (0.089)	-0.756 (0.559)
Observations	98	90	97	98	96
Control group mean	0.35	0.31	8.25	0.09	7.00
Fixed Effect	Yes	Yes	Yes	Yes	Yes

Notes: The sample is restricted to 3-day contracts, for which take-up of smooth and steep back-loaded contracts is nearly identical, strengthening the proxy used to construct the quasi-experimental sample. The dependent variable is measured by surveying workers and firms which were matched with each other. *Hours Worked* is winsorized at the 1st and 99th percentile. We use fixed effects for whether the contract was insured, daily wage offered and month. We control for whether the contract offered to the firm is a guarantor contract or not. Standard errors are clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D-12: Insurance contracts and worker outcomes

	Contract completed	Hr worked	Worker Rating	Excess Work (beyond 8 hrs)
<i>Insurance contract</i>	-0.092 (0.055)	-0.012 (0.095)	-0.024 (0.187)	-0.017 (0.038)
Observations	260	259	250	260
Control group mean	0.44	8.23	6.84	0.13
Fixed Effect	Yes	Yes	Yes	Yes

Notes: The dependent variable is measured by surveying workers and firms which were matched with each other. *Hours Worked* is winsorized at the 1st and 99th percentile. We use fixed effects for the length of contract, daily wage offered and month. Standard errors are clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D-13: Outcomes under steep vs daily contracts after excluding workers matched in pair-worker experiment

	Contract completed	Contract completed (excluding firm rejections)	Hours worked	Excess Work (beyond 8 hrs)	Worker Rating
<i>Steep contract</i>	0.211** (0.097)	0.232** (0.099)	0.238** (0.105)	0.209*** (0.074)	-0.748* (0.446)
Observations	205	187	204	205	197
Control group mean	0.31	0.25	8.24	0.08	6.95
Fixed Effect	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is measured by surveying workers and firms which were matched with each other. We use fixed effects for whether the contract was insured, length of contract, daily wage offered and control for the time it takes the worker to get to the labor stand. We include workers who were matched in the worker experiment and exclude workers who were matched in the paired-worker experiment. Standard errors are clustered at the firm level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D-14: Balance for the matching sample

	Mean of Daily Pay	Coef	p-val
Age	31.96	-0.07	0.96
Years of education	5.17	0.11	0.90
Backward caste	0.65	0.05	0.52
Scheduled caste/tribe	0.30	0.02	0.81
Travel Cost (INR)	47.49	-3.70	0.51
Time to market (hrs)	1.03	-0.09	0.44
Not paid at least once (in last month)	0.12	0.06	0.32
Worked with contractor who held payment	0.43	-0.01	0.93
Daily consumption cost	192.08	-11.19	0.36
Others have reneged on work	0.58	-0.07	0.42
Fraction days worked	0.48	0.11	0.05
Total earnings (INR)	1380.85	170.36	0.12

Notes: Balance tests for baseline worker characteristics in the realized matching sample. Column 1 shows the Daily Pay mean. Columns 2–3 report the coefficient and p-value from regressing each characteristic on an indicator for being matched to a steep contract. The coefficient is the difference in means (Steep minus Daily Pay) with labor stand fixed effects.

Table D-15: Summary stats of workers who were allotted jobs randomly and who were matched

	Job Sample	Matched Sample	p-value
Age	32.25887	31.42273	.3479045
Years of education	4.480427	5.154545	.2999241
Backward caste	.6950355	.7090909	.9699702
Scheduled caste/tribe	.2446809	.2681818	.428366
two sided travel cost (INR)	37.525	45.75342	.4078227
Time to market (hrs)	.848227	.9999239	.4349086
Baseline Wage Theft	.0996441	.1181818	.2645706
Have worked with firm who back-loaded payment	.4341637	.4454545	.7410002
Daily consumption	188.5536	194.2727	.7044717
Know workers who have reneged on work	.5464286	.5550459	.6791339
Fraction days worked	.5718085	.5375	.5644897
Total earnings (previous 4 days) (INR)	1175.929	1171.964	.5520587

Notes: The table shows summary statistics for workers who were allotted job after the experiment and random selection (Job Sample) and the workers who were eventually matched (Matched Sample). Column 3 captures the p-value of that difference with labor stand fixed effects.

Table D-16: Summary stats of workers who were allotted jobs randomly and who were matched by contracts

	Steep			Daily		
	Job Sample	Matched Sample	p-value	Job Sample	Matched Sample	p-value
Age	33.60227	31.18868	.2588292	31.64948	31.49701	.8822646
Years of education	4.852273	5.018868	.6401675	4.310881	5.197605	.0515407
Backward caste	.6818182	.7358491	.6985306	.7010309	.7005988	.9622376
Scheduled caste/tribe	.2272727	.2641509	.5086482	.2525773	.2694611	.753832
two sided travel cost (INR)	36.85227	39.23077	.7011102	37.83333	47.78443	.0221573
Time to market (hrs)	.7623106	.8413462	.977335	.8871993	1.049301	.3388932
Baseline Wage Theft	.1022727	.1698113	.2975967	.0984456	.1017964	.7966292
Have worked with firm who back-loaded payment	.4431818	.4528302	.8987636	.4300518	.4431138	.6127116
Daily consumption	193.8636	184.717	.4474201	186.1198	197.3054	.1278429
Know workers who have reneged on work	.5340909	.509434	.7989108	.5520833	.569697	.4708321
Fraction days worked	.6022727	.6556604	.7236715	.5579897	.5	.3921123
Total earnings (previous 4 days) (INR)	1239.545	1408.491	.2870022	1147.072	1096.898	.7532244

Notes: The table shows summary statistics for workers who were allotted job after the experiment and random selection (Job Sample) and the workers who were eventually matched (Matched Sample) for Daily and Steep contracts. Column 3 captures the p-value of that difference with labor stand fixed effects.

E Experiment Implementation

E.1 Worker experiment

Avoiding Duplicate offers To avoid making duplicate job offers to the same worker on different days, we uniquely identified each worker using their name, age, and village name. This data was provided to the field supervisors through an online tool which was deleted after completing the survey. The enumerator was instructed to check prior to making the offer if the worker had been previously offered a job. Enumerators were also instructed to ask the worker whether they had previously been offered a job. Based on the uniqueness of name, age and village we identify zero duplicate offers in the experiment.

Avoiding Spillovers To minimize spillovers, enumerators were instructed to make job offers a short distance away from the main labor stand. Offers were made between 7:30 a.m. and 10:30 a.m. over a span of two to four consecutive days, depending on the size of the stand. At larger stands, the field team occasionally returned after a one-week gap to continue hiring.

In total, we conducted 154 labor stand–day observations, covering 26 distinct stands—an average of approximately 5.92 days per stand. On each stand–day, enumerators made offers to an average of 8.83 workers, with the number ranging from a minimum of 5 to a maximum of 34. This implies that, on average, enumerators approached between 1.66 and 10.33 workers per hour. Larger stands were assigned more enumerators to account for higher footfall.

On average, each stand saw a daily footfall of 180 workers, with the smallest stands drawing around 50 and the largest up to 400. Thus, we made offers to roughly 5% of workers present on any given day. Many labor stands are located at road intersections, creating 3–4 natural gathering points within each stand. Given this spatial dispersion and the relatively small share of workers approached, the risk of spillover effects is likely to be minimal.

Script of Job Offer The script of the job offer for a smooth uninsured contract of 7 days is shown below:

Enumerator reads:

We want to hire a worker for daily wage work at a nearby construction site. The job will be for 7 days, with 8 hours of work each day.

You will be paid a total of INR 3,850 for the 7 days: INR 350 at the end of each of the first six days, and INR 1,750 on the seventh day. This adds up to INR 3,850 in total.

Instruction: Ask the respondent whether they have understood these terms.

Question to respondent: **Would you like to work on these terms?**

The script of the job offer for a smooth insured contract of 7 days is shown below:

Enumerator reads:

We want to hire a worker for daily wage work at a nearby construction site. The job will be for 7 days, with 8 hours of work each day.

You will be paid a total of INR 3,850 for the 7 days: INR 350 at the end of each of the first six days, and INR 1,750 on the seventh day. This adds up to INR 3,850 in total.

We guarantee that you will receive the full amount as agreed. One of our associates will visit the site to confirm that the contractor pays you. If you complete the work but the contractor fails to pay the promised amount, our organization will pay you directly.

All these conditions—including the schedule of payments and our guarantee— will be written into a formal agreement signed by our organization. You will also receive our phone number and office address, so that you can contact us if there is any problem.

Enumerator instruction: Show the organization's card to the respondent. **Instruction:** Ask the respondent whether they have understood these terms. *Question to respondent:* **Would you like to work on these terms?**

E.2 Firm experiment

Script of Offer to Firms The script of the offer for a steep contract of 7 days is shown below:

We will provide you with a worker for 7 days for your construction work. The worker will be paid INR 3500 for 7 days; INR 300 on the first day, and INR 3200 on the last day. Additionally, we will pay the worker INR 50 at the end of each day which will be paid according to the structure of the contract. So 50 INR will be paid on the first day and 300 INR (50*6) will be paid on the last day by us. Payment due to the worker must be made at the end of the day (300 on the first day and 3200 on the last day).

Question to respondent: **Do you want to hire the worker on these terms?**

The script of the offer for a **guarantor** contract of 7 days is shown below:

We will provide you with a worker for 7 days for your construction work. You will have to pay the worker Rs. 3,500 for seven days; Rs. 500 per day. In addition, at the end of each day we will pay the worker Rs. 50. The daily wage will be paid at the end of the day. If on any of the seven days the worker does not come, or leaves the job in the middle, we will give you Rs. 200 as compensation for your loss. Along with this, if you wish, we will also provide another worker in place of the worker who has left.

We will give you all these terms in writing, with our organisation's signature on it.

Instruction: Ask the respondent whether they have understood these terms.

Would you like to employ a worker on these terms?

The script of the offer for a **credit** contract of 7 days is shown below:

We will provide you with a worker for 7 days for your construction work. You will have to pay the worker Rs. 3,500 for seven days; Rs. 500 per day. In addition, at the end of each day we will pay the worker Rs. 50. The daily wage will be paid at the end of the day.

If you have difficulty making daily cash payments to the worker (because you do not have cash on hand or for any other reason), we can pay the worker on your behalf. For this, we can give you the cash needed to pay the wages each day.

Also, if it is difficult for you to go to the worksite every day to pay the worker, under this agreement one of our team members can go to the site each day and make the agreed payment to the worker. For example, if you are supposed to pay the worker Rs. 500 per day, we can give you up to Rs. 500 per day or pay the worker directly.

Under this arrangement, you may take this assistance or, if you prefer, you may choose not to. It is entirely up to you. This money will be given to you without interest, and you can repay us within two to three weeks.

Instruction: Ask the respondent whether they have understood these terms.

Would you like to employ a worker on these terms?

E.3 Matching stage

Matching workers with firms who accepted the job offers Workers who accepted a job offer were placed with firms that had agreed to the same contract type during the firm-side experiment. When a firm's demand arose, workers were contacted either by phone or directly at the labor stand. At the end of the survey and experiment, all workers were given a contact card with the address and phone number of our field team. This enabled even those without personal phones—but who could access one—to reach out and stay in touch.